Characteristics of Animal Cells

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Contents

Introduction 4
Animal cell 4
Cell growth and cycle 5
Metabolism 14
Animal Cell Culture and Biotechnology 29
Basic characterstics of Tissue Culture 29
Growth Kinetics 30
Culture Media For Animal Cell Culture 30
Culture Media 30
Culturing Animal Cells 31
Basic techniques in Animal Cell Culture 33
Quantitation of cells in cell culture 38
Kinetics of Cell Growth and Product formation and Effect of Shear Force 41
Invitro Fertilization and Embryo Transfer 51
Product and Substrate Transport 55
Micro & Macro-carrier culture 65
Livestock Improvement 70
Cloning in Animals 74
Gentic Engineering in Animal cell culture 77
Animal Cell Preservation 79

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INTRODUCTION

ANIMAL CELL

Animal cells are typical of the eukaryotic cell, enclosed by a plasma membrane and containing a membrane-bound nucleus and organelles. Unlike the eukaryotic cells of plants and fungi, animal cells do not have a cell wall. Most cells, both animal and plant, range in size between 1 and 100 micrometers and are thus visible only with the aid of a microscope.

The lack of a rigid cell wall allowed animals to develop a greater diversity of cell types, tissues, and organs. Specialized cells that formed nerves and muscles—tissues impossible for plants to evolve—gave these organisms mobility. The ability to move about by the use of specialized muscle tissues is a hallmark of the animal world. The animal kingdom is unique among eukaryotic organisms because most animal tissues are bound together in an extracellular matrix by a triple helix of protein known as collagen. Most animal cells are diploid, meaning that their chromosomes exist in homologous pairs. Different chromosomal ploidies are also, however, known to occasionally occur. The proliferation of animal cells occurs in a variety of ways. In instances of sexual reproduction, the cellular process of meiosis is first necessary so that haploid daughter cells, orgametes, can be produced. Two haploid cells then fuse to form a diploid zygote, which develops into a new organism as its cells divide and multiply.

Components found in animal cells:

- Centrioles - Centrioles are self-replicating organelles made up of nine bundles of microtubules and are found only in animal cells. They appear to help in organizing cell division, but aren't essential to the process.
- Cilia and Flagella - For single-celled eukaryotes, cilia and flagella are essential for the locomotion of individual organisms. In multicellular organisms, cilia function to move fluid or materials past an immobile cell as well as moving a cell or group of cells.
- Endoplasmic Reticulum - The endoplasmic reticulum is a network of sacs that manufactures, processes, and transports chemical compounds for use inside and outside of the cell. It is connected to the double-layered nuclear envelope, providing a pipeline between the nucleus and the cytoplasm.
- Endosomes and Endocytosis - Endosomes are membrane-bound vesicles, formed via a complex family of processes collectively known as endocytosis, and found in the cytoplasm of virtually every animal cell. The basic mechanism of endocytosis is the reverse of what occurs during exocytosis or cellular secretion. It involves the invagination (folding inward) of a cell's plasma membrane to surround macromolecules or other matter diffusing through the extracellular fluid.
- Golgi Apparatus - The Golgi apparatus is the distribution and shipping department for the cell's chemical products. It modifies proteins and fats built in the endoplasmic reticulum and prepares them for export to the outside of the cell.

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- Intermediate Filaments - Intermediate filaments are a very broad class of fibrous proteins that play an important role as both structural and functional elements of the cytoskeleton. Ranging in size from 8 to 12 nanometers, intermediate filaments function as tension-bearing elements to help maintain cell shape and rigidity.

- Lysosomes - The main function of these microbodies is digestion. Lysosomes break down cellular waste products and debris from outside the cell into simple compounds, which are transferred to the cytoplasm as new cell-building materials.

- Microfilaments - Microfilaments are solid rods made of globular proteins called actin. These filaments are primarily structural in function and are an important component of the cytoskeleton.

- Microtubules - These straight, hollow cylinders are found throughout the cytoplasm of all eukaryotic cells (prokaryotes don't have them) and carry out a variety of functions, ranging from transport to structural support.

- Mitochondria - Mitochondria are oblong shaped organelles that are found in the cytoplasm of every eukaryotic cell. In the animal cell, they are the main power generators, converting oxygen and nutrients into energy.

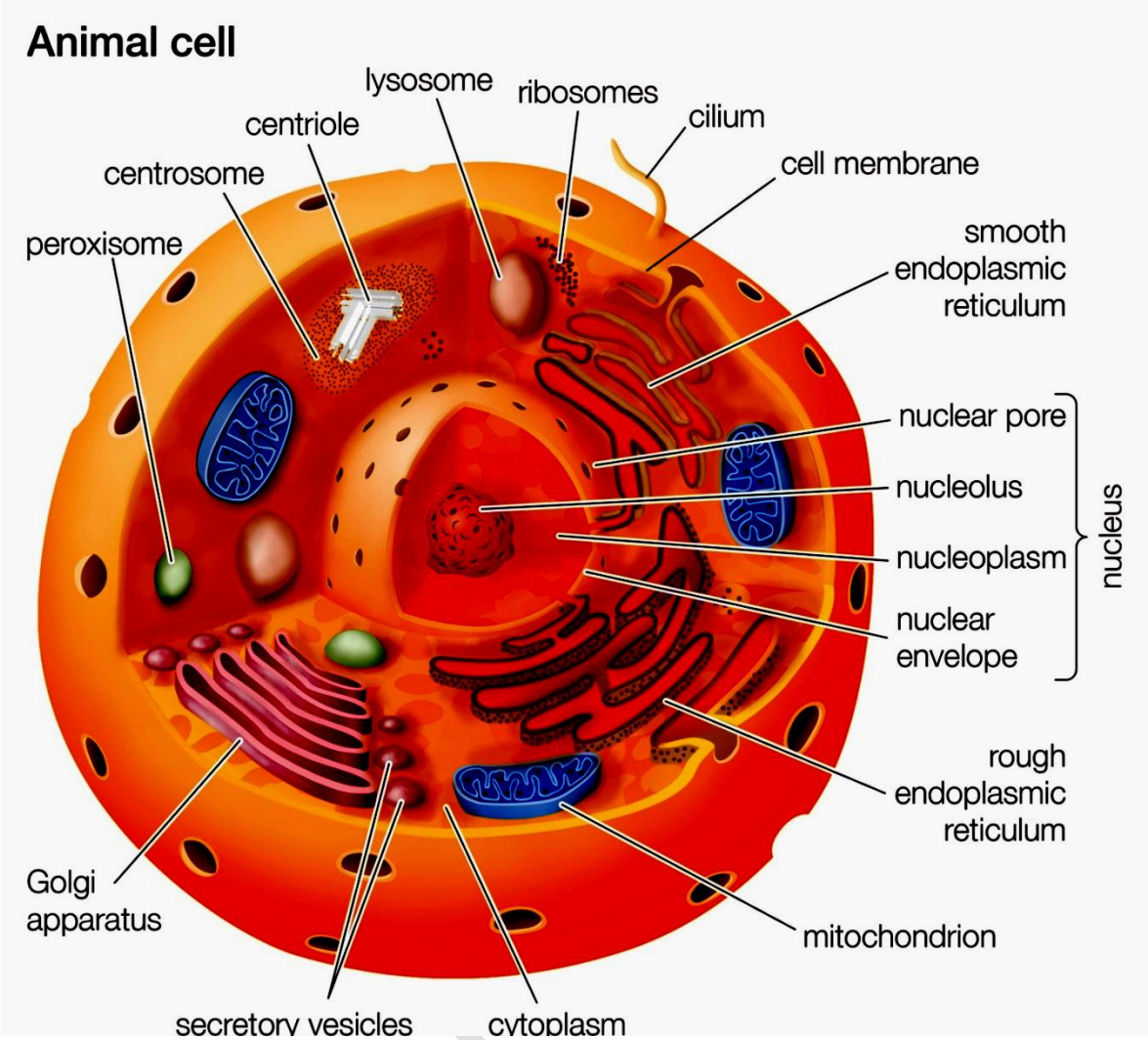
- Nucleus - The nucleus is a highly specialized organelle that serves as the information processing and administrative center of the cell. This organelle has two major functions: it stores the cell's hereditary material, or DNA, and it coordinates the cell's activities, which include growth, intermediary metabolism, protein synthesis, and reproduction (cell division).

- Peroxisomes - Microbodies are a diverse group of organelles that are found in the cytoplasm, roughly spherical and bound by a single membrane. There are several types of microbodies but peroxisomes are the most common.

- Plasma Membrane - All living cells have a plasma membrane that encloses their contents. In prokaryotes, the membrane is the inner layer of protection surrounded by a rigid cell wall. Eukaryotic animal cells have only the membrane to contain and protect their contents. These membranes also regulate the passage of molecules in and out of the cells.

- Ribosomes - All living cells contain ribosomes, tiny organelles composed of approximately 60 percent RNA and 40 percent protein. In eukaryotes, ribosomes are made of four strands of RNA. In prokaryotes, they consist of three strands of RNA.

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Animal cell
lysosome
ribosomes
centriole
cilium
centrosome
cell membrane
peroxisome
smooth endoplasmic reticulum
nuclear pore
nucleolus
nucleoplasm
nuclear envelope
nucleus
rough endoplasmic reticulum
Golgi apparatus
mitochondrion
secretory vesicles
cytoplasm

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Differences between Animal cells and Plant cells

Cell Size	Plant Cell	Animal Cell
Cell size	*Large*	Smaller than plant cells
Cell Shape	*Rectangular*	Circular
Vacuoles	*A single centrally located vacuole. It takes up almost 90% of the cell volume. The vacuole stores water and maintains turgidity of the cell.*	If any, there are a number of small vacuoles spread throughout the cytoplasm that store water, irons and waste materials.
Cell Wall	*A rigid cell wall, made of cellulose, is present around a plant cell that helps it maintain its shape.*	Cell wall is absent. This allows animal cells to adopt different shapes.
Chloroplast	*Present. Chlorophyll is the pigment that traps sun's energy which is used by plants to make food through the process of photosynthesis. This pigment is present in the chloroplasts.*	Absent. As animals lack this pigment, they cannot make their own food.
Cell Division	*Cell division takes place by the formation of cell place in the center of the diving cell. This becomes the cell wall between the two daughter cells.*	Animal cells divide with the formation of a cleavage furrow. This is formed as the chromosomes move to the ends of the microtube spindle formed by the centrioles.
Centrioles	*Present only in lower forms. Plants instead have microtubule organizing centers that produce the microtubules.*	Present. Centrioles help in division of animal cells by creating microtubule spindles that pull the chromosomes to opposite ends for cell division to occur.
Centrosome	*Absent. Instead, two small clear areas called polar caps are present.*	Present.
Lysosomes	*Absent*	Present. Lysosomes are vesicles that contain enzymes that destroy dead cell organelles and other cells debis.
Golgi Bodies	*In place of golgi bodies, its sub units known as dicysomes are present.*	Complex golgi bodies are present close to the nucleus.

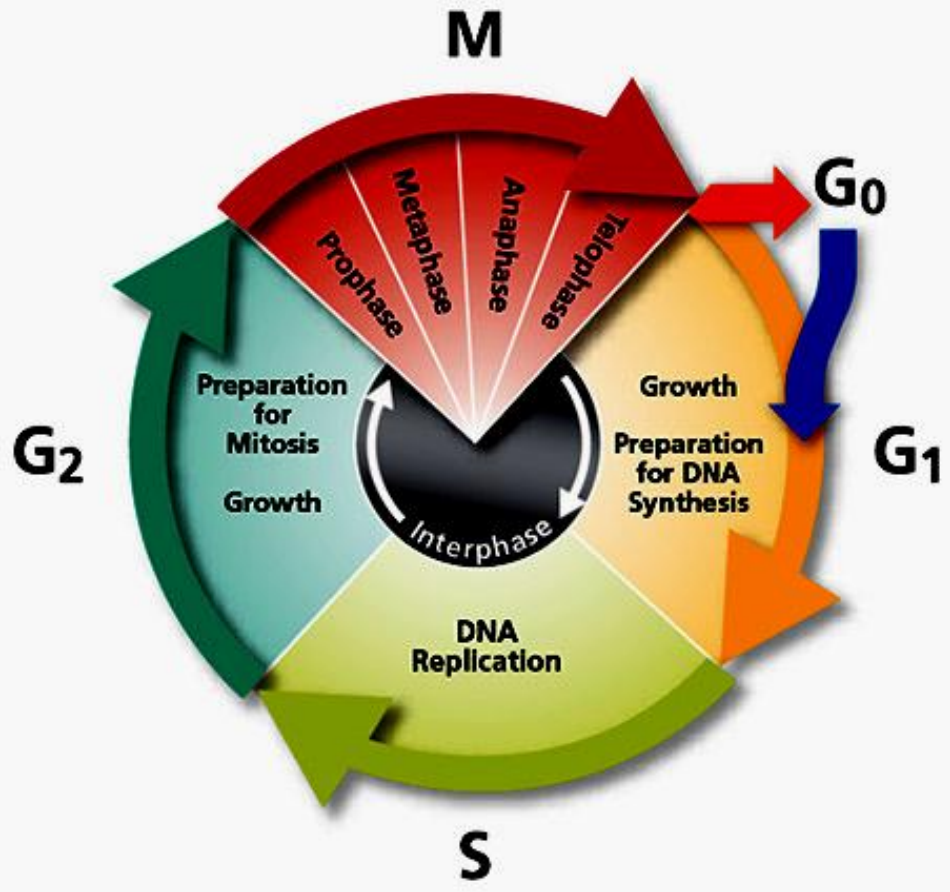
CELL GROWTH AND CELL CYCLE

The cell theory states that cells can only arise by the division of existing cells. Cell growth and division is therefore a critical process in the life of cells and organisms. The reproduction of cells, from when the cell is produced by division of the mother cell until the new cell itself divides, is known as the cell cycle. The length of this cycle varies depending on the cell type. Bacterial cells can divide every 20 minutes when growth conditions are favourable, whereas human liver cells only divide about once a year. Mammalian cells in tissue culture have a cell cycle time of about 20 hours; frog embryo cells divide much more frequently with a cycle time of around 30 minutes.

Despite the variation in the duration of the cell cycle, there are certain basic requirements for cell division. If two new cells are to be produced, clearly the amount of cell material must double if the daughter cells are to remain the same size as the parent. This is particularly critical for the DNA of the cell, which must be copied exactly if the genetic integrity of the cell is to be maintained. This is achieved by the process of DNA replication, which is an important marker during the cell cycle.

How has the cell cycle been studied? If you were to look at a mammalian cell under the microscope for a complete cell cycle, you would see very little evidence of cytological changes until the last hour of the process, when suddenly there is a flurry of activity as the chromosomes move apart and two new cells are formed. Thus early descriptions of the cycle separated the process into two parts, called interphase and division. The division process itself is more accurately called mitosis or the M phase of the cycle. Mitosis refers to the separation of the chromosomes, and this is followed by division of the cell during cytokinesis (CK). Interphase is sometimes called the resting phase, but in biochemical terms this is in fact a very active period of growth. A more accurate description of the cell cycle splits interphase into three parts. The period of DNA replication is known as the synthesis or S phase, and there are 'gaps' known as G1 and G2. As our knowledge has increased, it has become clear that 'gap' is perhaps not the best way to describe these two periods, as this suggests little activity, as does the term interphase. A diagram of the cell cycle is shown in the figure below.

The cell cycle. Interphase is made up of phases G1, S and G2. During S phase the DNA is replicated, although no cytological changes are distinguished in the light microscope. The cell grows during interphase and enters the M phase (mitosis). The four sub-stages of M phase are prophase (P), metaphase (M), anaphase (A) and telophase(T).This is followed by cytokinesis (CK) which generates two new daughter cells, each of which enters G1 to begin the cycle again. For animal cells in tissue culture, cycle duration is around 20 hours.



MITOSIS

The actual division of the cell requires a series of co-ordinated events that separate the replicated chromosomes and then split the cell into two, each half having received a complete set of chromosomes. After DNA replication the chromosomes are composed of two **chromatids**, held together at the **centromere**. Following separation during mitosis, the newly separated sister chromatids are known as **daughter chromosomes**. Although mitosis is a dynamic process, there are four stages that can be recognised easily in the light microscope. These are **prophase**,**metaphase**, **anaphase** and **telophase**. Movement of chromosomes is achieved by **spindle fibres**, which are **microtubules**. These are composed of alternating dimers of α and β **tubulin** (a protein). The spindle begins to form at prophase, and is organised by **spindle poles** or **centrosomes** at the two poles of the cell. During mitosis the spindle serves to guide the daughter chromosomes and pull them apart to opposite ends of the cell. Figure below shows the events of mitosis

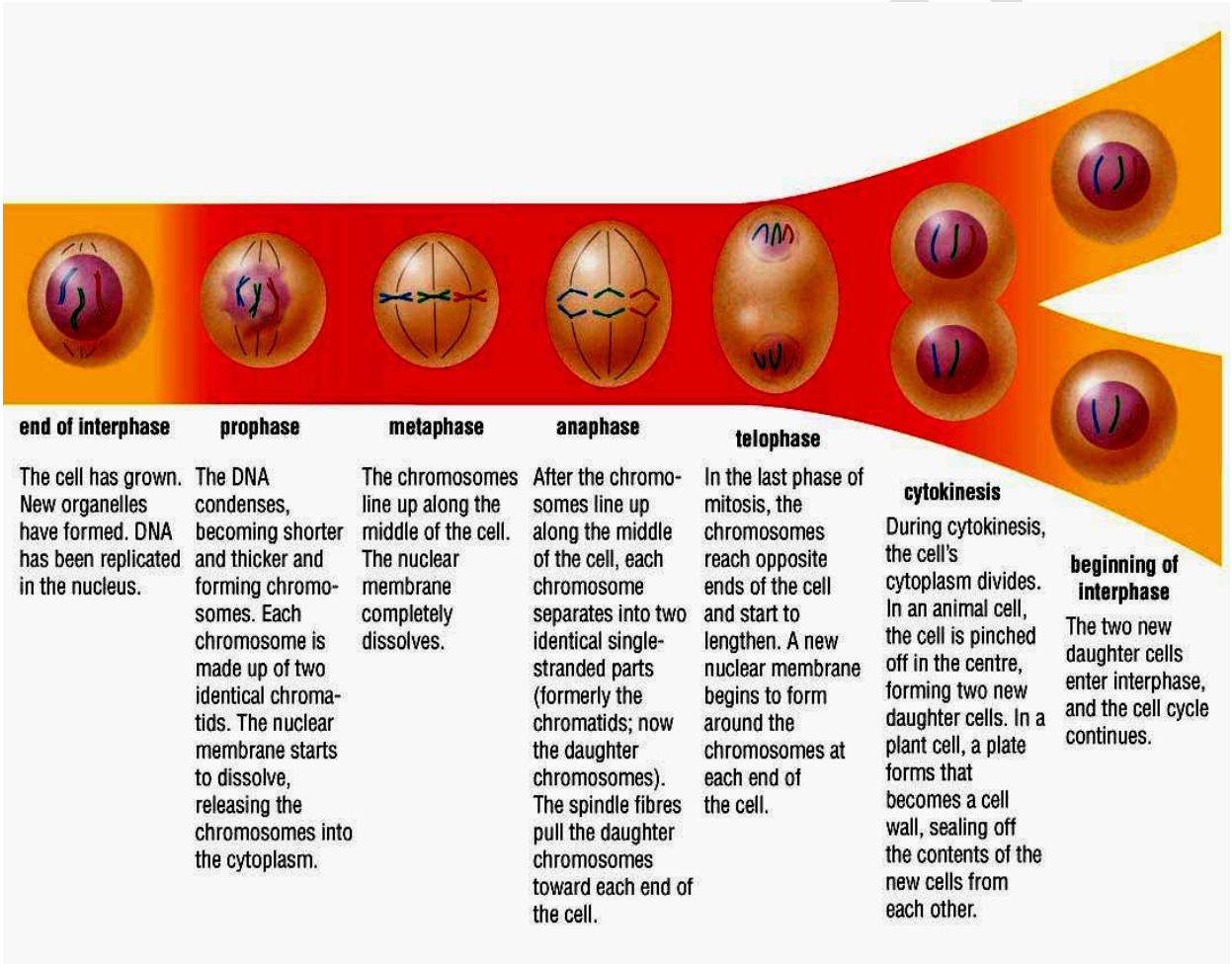


Figure shows Mitosis. The diagram and explanation describe the events in each stage.

Due to the structural differences between plant and animal cells, **cytokinesis** is achieved by different mechanisms in these two cell types. In animal cells, constriction of the cytoplasm by a contractile ring of actin and myosin produces a cleavage furrow, which essentially pinches the cell into two pieces. In plants the cell wall makes this impossible, and cytokinesis is achieved by the formation of a new cell wall to separate the daughter cells.

CONTROL OF THE CELL CYCLE

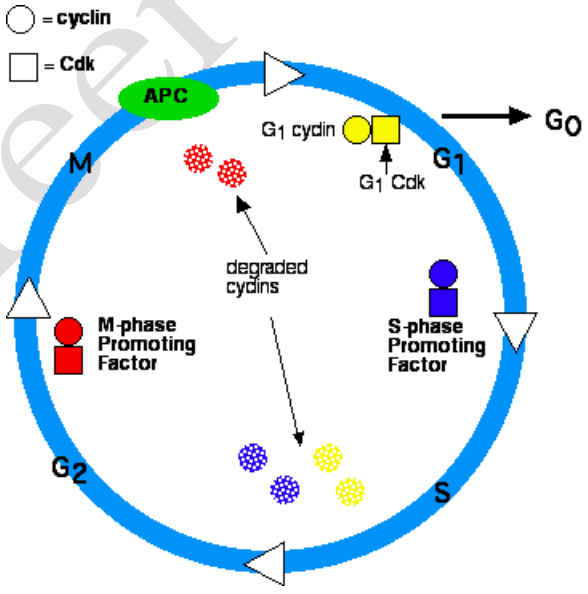
What controls cell division? This is a question that has occupied biologists for many years, and a wide variety of different organisms and cell types has been used to try to answer some of the main questions. Obviously it is important that DNA replication is complete before mitosis, and that the cell mass has increased sufficiently to enable two daughter cells to be formed. Also, the division processes of mitosis and cytokinesis have to be controlled in both temporal and spatial terms if success is to be achieved. In multicellular eukaryotes, additional constraints on cell division are required if cell numbers are to be controlled. To divide, these cells require specific **growth factors**, of which over 50 types have been isolated. In the absence of enough growth factor, cells stop at the G1 checkpoint and enter a non-growing phase called G0. Our current understanding of cell cycle control is that there is a central mechanism that is used to assess the status of the cell as it progresses through the cycle. This mechanism works through a series of three main checkpoints:

• G1 checkpoint – towards the end of G1 phase, cell size is assessed. If sufficiently large to allow division, the cell enters S phase and DNA replication begins.

• G2 checkpoint – the success of DNA replication is monitored, and if all is well entry into mitosis is triggered.

• M checkpoint – this occurs during metaphase and triggers the exit from mitosis and cytokinesis, and entry into the next G1 phase in the daughter cells.

The molecular mechanisms that control the cell cycle involve the interactions of many different genes and proteins to 'trigger' the events of the cell cycle in their proper sequence. For example, a critical part of cell division is the entry into mitosis. This is triggered by a complex called **mitosis promoting factor (MPF)**, which in turn is controlled by other intracellular cell cycle signals.



DIFFERENTIATION OF CELLS INTO TISSUES AND ORGANS

Although the cell is essentially a complete living system in its own right, in multicellular organisms cells are organised into tissues and systems for specific functions such as support,

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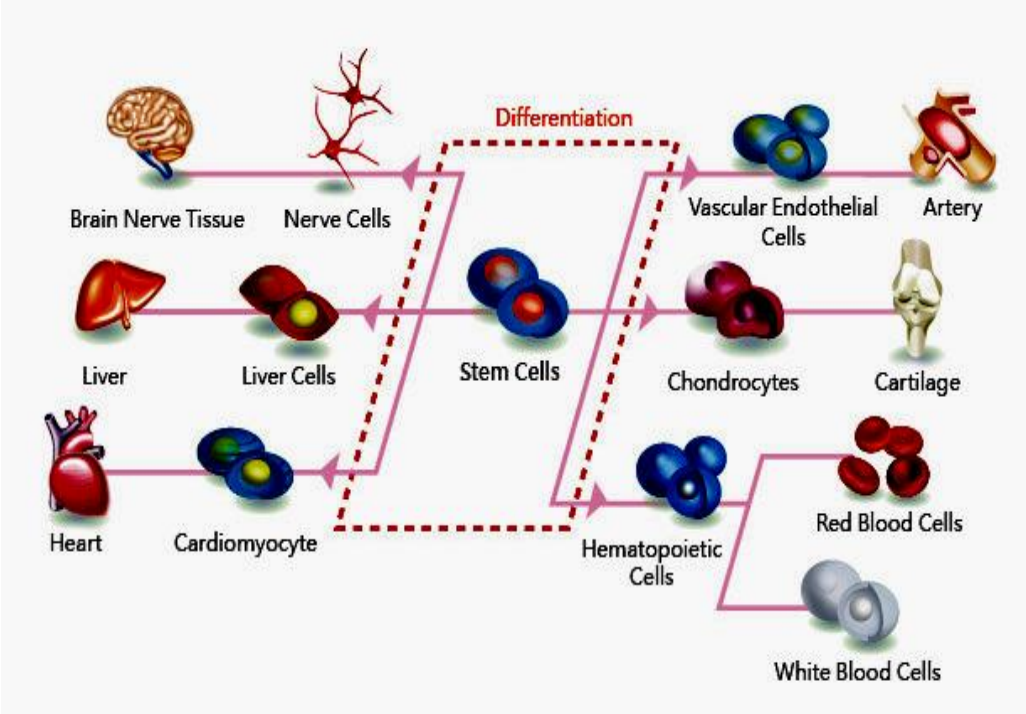
movement, nutrition, coordination and control. Much useful information about tissues and systems has been obtained by studying the final product – liver, kidney, nerves, glands and so on – but the key question is concerned with how these tissues and systems arise in the developing embryo. This is the area of **developmental biology**. When scientists began to isolate and study genes, a logical step was to try to understand how genes function in the control of developmental processes. Thus the field of developmental genetics has emerged as one of the central areas of modern biology. Despite great advances in our knowledge, development remains one of the most complex and astonishing branches of science. One way of looking at complex processes is to study simple organisms, to see if any functions correlate with those in higher organisms. Although there are not many developmental processes evident in bacteria, some aspects of how genes are expressed have become clearer by looking at adaptive responses such as expression of the *lac* operon.

Multicellular plants provide relatively well-ordered systems, with much simpler developmental patterns than animals, yet development in plants is not as well understood as in animals. Development in animals begins with a single fertilised egg undergoing successive divisions. Cell proliferation is of course needed to make up the cell number required as the embryo develops – an adult human has more than 1,000,000,000,000 cells! However, all these cells divide mitotically, and so have identical genomes. So how do different cells arise? And how do they end up in the correct place as the embryo grows?

DIFFERENTIAL GENE EXPRESSION IN DEVELOPMENT

To generate different cells from the same genetic information, there must be some sort of control over gene expression. This control can be both **temporal** (different genes expressed at certain specific times in development) and **spatial** (cells in different places in the embryo expressing different genes). Development is not a fixed series of genetic events, but rather a 'conversation' between cells as the embryo develops. This 'conversation' involves gene products, which influence cellular events, which in turn create patterns due to cell movement and differentiation. Unravelling all this complexity is an immense task, but it is becoming clear that similar processes operate in all animals. Stem cells are the body's master cells and have self-renewal and multipotency capacity. They are able to differentiate into different kinds of functional cells to renew and repair organs, tissues, blood and immune systems.

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MEIOSIS

Meiosis is the formation of gametes, performed by reproductive cells only. This will result in a reduction of the chromosome number, forming haploid cells (n). So, unlike mitosis, which produces two daughter cells with identical chromosomes, meiosis produces 4 daughter cells, each with half the number of chromosomes, that are not identical to each other. This is again important because it keeps the chromosome number constant over generations, in the adult organism. When the egg (n) and the sperm (n) combine (n + n) in fertilization, the offspring is a diploid cell (2n).

Meiosis I – Separates Homologous Chromosomes

Prophase I

During early prophase I:

• Chromosomes begin to condense, and homologs loosely pair along their lengths, aligned gene by gene.
• Paired homologs become physically connected to each other along their lengths by a zipper-like protein structure, the *synaptonemal complex;* this state is called **synapsis.**
• **Crossing over**, a genetic rearrangement between nonsister chromatids involving the exchange of corresponding segments of DNA molecules, begins during pairing and synaptonemal complex formation, and is completed while homologs are in synapsis.

At the Prophase stage:

• Synapsis has ended with the disassembly of the synaptonemal complex in mid-prophase, and the chromosomes in each pair have moved apart slightly.

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• Each homologous pair has one or more X-shaped regions called **chiasmata** (singular, *chiasma*). A chiasma exists at the point where a crossover has occurred. It appears as a cross because sister chromatid cohesion still holds the two original sister chromatids together, even in regions beyond the crossover point, where one chromatid is now part of the other homolog.
• Centrosome movement, spindle formation, and nuclear envelope breakdown occur as in mitosis.

Later in prophase I:

• Microtubules from one pole or the other attach to the two kinetochores, protein structures at the centromeres of the two homologs. The homologous pairs then move toward the metaphase plate.

Metaphase I

• Pairs of homologous chromosomes are now arranged at the metaphase plate, with one chromosome in each pair facing each pole.
• Both chromatids of one homolog are attached to kinetochore microtubules from one pole; those of the other homolog are attached to microtubules from the opposite pole.

Anaphase I

• Breakdown of proteins responsible for sister chromatid cohesion along chromatid arms allows homologs to separate.
• The homologs move toward opposite poles, guided by the spindle apparatus.
• Sister chromatid cohesion persists at the centromere, causing chromatids to move as a unit toward the same pole.

Telophase I and Cytokinesis

• At the beginning of telophase I, each half of the cell has a complete haploid set of duplicated chromosomes. Each chromosome is composed of two sister chromatids; one or both chromatids include regions of nonsister chromatid DNA.
• Cytokinesis (division of the cytoplasm) usually occurs simultaneously with telophase I, forming two haploid daughter cells.
• In animal cells like these, a cleavage furrow forms. (In plant cells, a cell plate forms.)
• In some species, chromosomes decondense and nuclear envelopes form.
• No chromosome duplication occurs between meiosis I and meiosis II.

Meiosis II – Separates Sister Chromatids

Prophase II

• A spindle apparatus forms.
• In late prophase II, chromosomes, each still composed of two chromatids associated at the centromere, move toward the metaphase II plate.

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Metaphase II

• The chromosomes are positioned at the metaphase plate as in mitosis.
• Because of crossing over in meiosis I, the two sister chromatids of each chromosome are *not* genetically identical.
• The kinetochores of sister chromatids are attached to microtubules extending from opposite poles.

Anaphase II

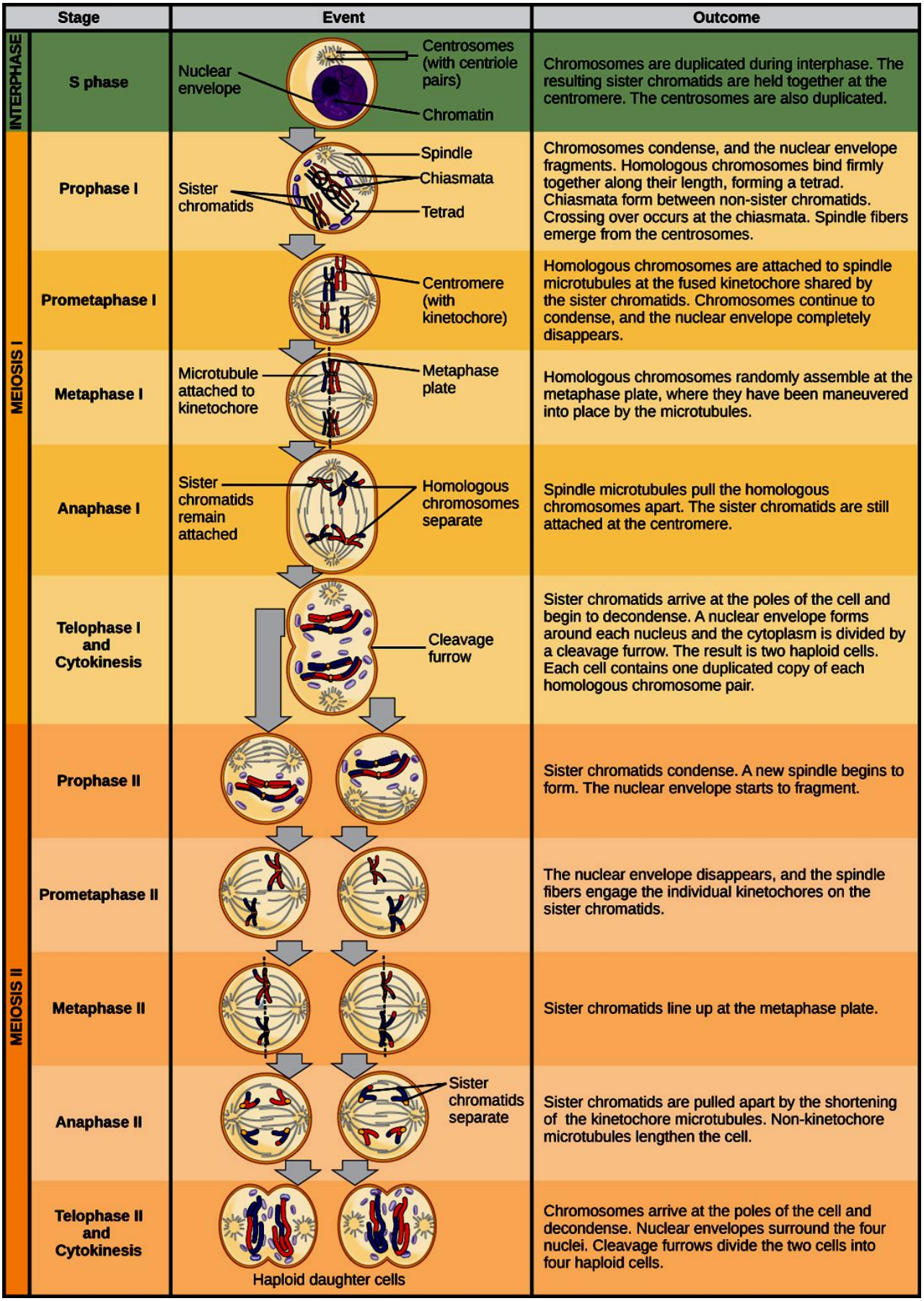
• Breakdown of proteins holding the sister chromatids together at the centromere allows the chromatids to separate. The chromatids move toward opposite poles as individual chromosomes.

Telophase II and Cytokinesis

• Nuclei form, the chromosomes begin decondensing, and cytokinesis occurs.
• The meiotic division of one parent cell produces four daughter cells, each with a haploid set of (unduplicated) chromosomes.
• The four daughter cells are genetically distinct from one another and from the parent cell.

DIFFERENCES BETWEEN MITOSIS AND MEIOSIS		
EVENTS	**MITOSIS**	**MEIOSIS**
Occurrence	In all the body cells including germ cells.	Only in the germ (reproductive) cells.
Definition	It is an equational division.	It is a reductional division.
Number of daughter cells	Only two	Four
Prophase	Involves relatively few changes.	Involves a series of changes in chromosomes distinguished into 5 substages.
Chromom-eres	Not visible in prophase.	Visible in the leptotene stage of prophase - I
Synapsis	Does not occur.	Occurs in zygotene of prophase-I.
Crossing over	Does not occur.	Occurs in pachytene stage of prophase - I.
Metaphase	Chromosomes arrange along the equator.	Chromosomes arrange equally on either side of the equator in metaphase-I.
Centromeres in Anaphase	Each centromere splits into two.	Centromeres do not split in metaphase-I.
Centrome- res in Metaphase	Orient towards the equator while chromatids orient towards poles.	Orient towards poles while chromatids orient towards the equator in metaphase-I.
Telophase	Results in the formation of two daughter nuclei having the same no. of chromosomes as that of parent cell.	Telophase-II results in the formation of two daughter nuclei, each having half the no. of chromosomes as that of parent cell.
Cytokine- sis	Follows immediately after karyokinesis.	May or may not occur at the end of first karyokinesis.

Unlike Mitosis, Meiosis is not a cycle, and the daughter cells end up with haploid number of chromosomes



Diagrammatic representation of Meiosis



GATE QUESTIONS

GATE 2011

Q.1 A cell in G_1 of interphase has 12 chromosomes. How many chromatids will be found per cell during metaphase II of meiosis?

(A) 6 (B) 12 (C) 18 (D) 24

Q.2 In balanced growth phase of a cell

P. all components of a cell grow at the same rate

Q. specific growth determined by cell number or cell mass would be the same

R. the growth rate is independent of substrate concentration.

S. the growth rate decreases with decreasing substrate concentration

(A) P,Q and S only (B) Q, R and S only

(C) P,Q and R only (D) P only

GATE 2012

Q.3 Which one of the following is **NOT** a protoplast fusion inducing agent??

(A) Inactivated Sendai virus (B)Ca^{2+} at alkaline pH

(C) Polyethylene glycol (D) Colchicine

Answers:

1-B 2-A 3-C

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METABOLISM

Metabolism is the set of life-sustaining chemical transformations within the cells of living organisms. These enzyme-catalyzed reactions allow organisms to grow and reproduce, maintain their structures, and respond to their environments. Metabolism is usually divided into two categories. Catabolism, that breaks down organic matter and harvests energy by way of cellular respiration, and anabolism that uses energy to construct components of cells such as proteins and nucleic acids.

Cellular Respiration

1. Cellular respiration involves various metabolic pathways that break down carbohydrates and other metabolites with the concomitant buildup of ATP.
2. Cellular respiration consumes oxygen and produces CO_2; because oxygen is required, cellular respiration is aerobic.
3. Cellular respiration usually involves the complete breakdown of glucose into CO_2 and H_2O.
4. The net equation for glucose breakdown is: $C_6H_{12}O_6 + 6\ O_2 = 6\ CO_2 + 6\ H_2O +$ energy
5. Glucose is a high-energy molecule; CO_2 and H_2O are low-energy molecules; cellular respiration is thus *exergonic* because it releases energy.
6. Electrons are removed from substrates and received by oxygen, which combines with H^+ to become water.
7. Glucose is oxidized and O_2 is reduced.
8. The reactions of cellular respiration allow energy in glucose to be released slowly; therefore ATP is produced gradually.
9. In contrast, if glucose were broken down rapidly, most of its energy would be lost as non-usable heat.
10. The breakdown of glucose yields synthesis of 36 or 38 ATP (depending on certain conditions); this preserves about 39% of the energy available in glucose.
11. This is relatively efficient compared to, for example, the 25% efficiency of a car burning gasoline.

A. NAD^+ and FAD

1. Each metabolic reaction in cellular respiration is catalyzed by a specific enzyme.
2. As a metabolite is oxidized, NAD^+ (nicotinamide adenine dinucleotide) accepts two electrons and a hydrogen ion (H^+); this results in $NADH + H^+$.
3. Electrons received by NAD^+ and FAD are high-energy electrons and are usually carried to the electron transport chain.
4. NAD^+ is a coenzyme of oxidation-reduction since it both accepts and gives up electrons; thus, NAD^+ is sometimes called a *redox coenzyme*
5. Only a small amount of NAD^+ is needed in cells because each NAD^+ molecule is used repeatedly.
6. FAD coenzyme of oxidation-reduction can replace NAD^+; FAD accepts two electrons and two hydrogen ions to become $FADH_2$.

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B. Phases of Cellular Respiration

1. Cellular respiration includes four phases:
   a. Glycolysis is the breakdown of glucose in the cytoplasm into two molecules of pyruvate.
      1) Enough energy is released for an immediate yield of two ATP.
      2) Glycolysis takes place outside the mitochondria and does not utilize oxygen; it is therefore an anaerobic process.
   b. In the preparatory (prep) reaction, pyruvate enters a mitochondrion and is oxidized to a two-carbon acetyl group and CO_2 is removed; this reaction occurs twice per glucose molecule.
   c. The citric acid cycle:
      1) occurs in the matrix of the mitochondrion and produces NADH and $FADH_2$;
      2) is a series of reactions that gives off CO_2 and produces one ATP;
      3) turns twice because two acetyl-CoA molecules enter the cycle per glucose molecule;
      4) produces two immediate ATP molecules per glucose molecule.
   d. The electron transport chain:
      1) is a series of carriers in the inner mitochondrial membrane that accept electrons from glucose--electrons are passed from carrier to carrier until received by oxygen;
      2) passes electrons from higher to lower energy states, allowing energy to be released and stored for ATP production.

Outside the Mitochondria: Glycolysis

1. Glycolysis occurs in the cytoplasm outside the mitochondria.
2. Glycolysis is the breakdown of glucose into two pyruvate molecules.
3. Glycolysis is universally found in organisms; therefore, it likely evolved before the citric acid cycle and electron transport chain.
4. Glycolosis can be divided into the energy-investment steps where ATP is used to "jump start" glycolosis, and the energy-harvesting steps, where more ATP is made than used.

A. Energy-Investment Steps

1. Glycolysis begins with the activation of glucose with two ATP; the glucose splits into two C_3 molecules known as G3P, each of which carries a phosphate group.
2.

B. Energy-Harvesting Steps

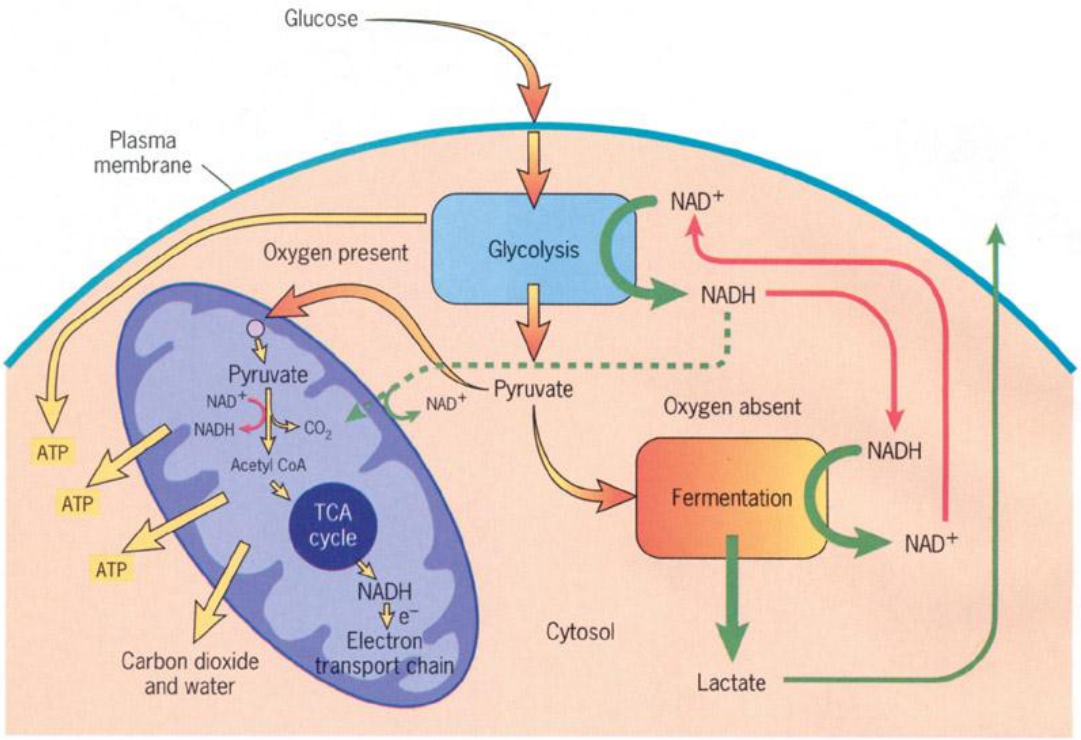
1. Oxidation of G3P occurs by removal of electrons and hydrogen ions.
2. Two electrons and one hydrogen ion are accepted by NAD^+, resulting in two NADH; later, when the NADH molecules pass two electrons to the electron transport chain, they become NAD^+ again.
3. The oxidation of G3P and subsequent substrates results in four high-energy phosphate groups, which are used to synthesize four ATP molecules; this process is called substrate-level phosphorylation.

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4. Two of four ATP molecules produced are required to replace two ATP molecules used in the initial phosphorylation of glucose; therefore there is a *net* gain of two ATP from glycolysis.
5. Pyruvate enters a mitochondrion (if oxygen is available) and cellular respiration ensues.
6. If oxygen is not available, *fermentation occurs* and pyruvate undergoes reduction.

Fermentation

1. Fermentation is an *anaerobic* (i.e., occurs in the absence of oxygen) process which consists of glycolysis plus reduction of pyruvate to either lactate or to alcohol and CO_2 (depending on the organism).
2. Animal cell fermentation results in lactate.
3. Bacteria can produce an organic acid like lactate, or an alcohol and CO_2.
4. Yeasts produce ethyl alcohol and CO_2.
5. NADH passes its electrons to pyruvate instead of to an electron transport chain; NAD^+ is then free to return and pick up more electrons during earlier reactions of glycolysis.



A. Advantages and Disadvantages of Fermentation

1. Despite a low yield of two ATP molecules, fermentation provides a quick burst of ATP energy for muscular activity.
2. Fermentation products are toxic to cells.
   a. When blood cannot remove all lactate from muscles, lactate changes pH and causes muscles to fatigue.

b. The individual is in oxygen debt because oxygen is needed to restore ATP levels and rid the body of lactate.
c. Recovery occurs after lactate is sent to the liver where it is converted into pyruvate; some pyruvate is then respired or converted back into glucose.

B. Efficiency of Fermentation

1. Two ATP produced per glucose molecule during fermentation is equivalent to 14.6 kcal.
2. Complete glucose breakdown to CO_2 and H_2O during cellular respiration represents a potential yield of 686 kcal per molecule.
3. Efficiency of fermentation is 14.6/686 or about 2.1%, far less efficient than complete breakdown of glucose.

C. Fermentation Helps Produce Numerous Food Products

1. Yeast Fermentation
a. Baker's yeast, *Saccharomyces cerevisiae*, is added to bread for leavening. The dough rises when yeasts give off CO_2.
b. Yeasts ferment the carbohydrates of fruit to produce ethyl alcohol in wine, and ferment grains to produce ethyl alcohol in beer.
c. The acetic acid bacteria, *Acetobacter aceti*, spoil wine, and convert the alcohol to acetic acide to produce vinegar.
2. Bacterial Fermentation
   a. Lactic acid bacteria cause milk to sour and produce yogurt, sour cream, and cheese
   b. Brine cucumber pickles, sauerkraut, and kimchi are pickled vegetables produced by acid producing fermenting bacteria.
3. Soy Sauce Production
   a. Yeast and fermenting bacteria are added to soy beans and wheat to produce soy sauce.

Inside the Mitochondria

1. The next reactions of cellular respiration involve the preparatory reaction, the citric acid cycle, and the electron transport chain.
2. These reactions occur in the mitochondria.
3. A mitochondrion has a double membrane with an intermembrane space (between the outer and inner membrane).
4. *Cristae* are the inner folds of membrane that jut into the *matrix*, the innermost compartment of a mitochondrion that is filled with a gel-like fluid.
5. The prep reaction and citric acid cycle enzymes are in the matrix; the electron transport chain is in the cristae.
6. Most of the ATP produced in cellular respiration is produced in the mitochondria; therefore, mitochondria are often called the "powerhouses" of the cell.

A. Preparatory Reaction

1. The preparatory reaction connects glycolysis to the citric acid cycle.
2. In this reaction, pyruvate is converted to a two-carbon acetyl group, and is attached to *coenzyme A*, resulting in the compound acetyl-CoA.

Career Avenues

3. This redox reaction removes electrons from pyruvate by a dehydrogenase enzyme, using NAD^+ as a coenzyme.
4. This reaction occurs twice for each glucose molecule.
5. CoA carries the acetyl group to the citric acid cycle.
6. The two NADH carry to electrons to the electron transport chain.
7. The CO_2 diffuses out of animal cells into blood, transported to lungs, and exhaled.

B. Citric Acid Cycle

1. The citric acid cycle occurs in the matrix of mitochondria.
2. The cycle is sometimes called the Krebs cycle, named for Sir Hans Krebs, who described the fundamentals of the reactions in the 1930s.
3. The cycle begins by the addition of a two-carbon acetyl group to a four-carbon molecule, forming a six-carbon citrate (citric acid) molecule.
4. In the subsequent reactions, at three different times two electrons and one hydrogen ion are accepted by NAD^+, forming NADH.
5. At one time, two electrons and one hydrogen ion are accepted by FAD, forming $FADH_2$.
6. NADH and $FADH_2$ carry these electrons to the electron transport chain.
7. Some energy is released and is used to synthesize ATP by *substrate-level phosphorylation*.
8. One high-energy metabolite accepts a phosphate group and transfers it to convert ADP to ATP.
9. The citric acid cycle turns twice for each original glucose molecule.
10. The products of the citric acid cycle (per glucose molecule) are 4 CO_2, 2 ATP, 6 NADH and 2 $FADH_2$.
11. Production of CO_2
    a. The six carbon atoms in the glucose molecule have now become the carbon atoms of six CO_2 molecules, two from the prep reaction and four from the citric acid cycle.

C. The Electron Transport Chain

1. The electron transport chain is located in the cristae of mitochondria and consists of carriers that pass electrons successively from one to another.
2. NADH and $FADH_2$ carry the electrons to the electron transport system.
3. Members of the Chain
    a. NADH gives up its electrons and becomes NAD^+; the next carrier then gains electrons and is thereby reduced.
    b. At each sequential redox reaction, energy is released to form ATP molecules.
    c. Some of the protein carriers are cytochrome molecules, complex carbon rings with a heme (iron) group in the center.
4. Cycling of Carriers
    a. By the time electrons are received by O_2, *three* ATP have been made.
    b. When $FADH_2$ delivers electrons to the electron transport system, *two* ATP are formed by the time the electrons are received by O_2.
    c. Oxygen serves as the terminal electron acceptor and combines with hydrogen ions to form water.
5. The Cristae of a Mitochondrion and Chemiosmosis

Career Avenues

a. The electron transport chain consists of three protein complexes and two protein mobile carriers that transport electrons.
b. The three protein complexes include *NADH-Q reductase complex*, the *cytochrome reductase complex*, and the *cytochrome oxidase complex*; the two protein mobile carriers are *coenzyme Q* and *cytochrome c*.
c. Energy released from the flow of electrons down the electron transport chain is used to pump H^+ ions, which are carried by NADH and $FADH_2$, into intermembrane space.
d. Accumulation of H^+ ions in this intermembrane space creates a strong electrochemical gradient.
e. ATP synthase complexes are channel proteins that serve as enzymes for ATP synthesis.
f. As H^+ ions flow from high to low concentration, ATP synthase synthesizes ATP by the reaction: ADP + P = ATP.
g. Chemiosmosis is the term used for ATP production tied to an electrochemical (H^+) gradient across a membrane.
h. Once formed, ATP molecules diffuse out of the mitochondrial matrix through channel proteins.
i. ATP is the energy currency for all living things; all organisms must continuously produce high levels of ATP to survive.

D. Energy Yield From Glucose Metabolism

1. Substrate-Level Phosphorylation
   a. Per glucose molecule, there is a net gain of two ATP from glycolysis in cytoplasm.
   b. The citric acid cycle in the matrix of the mitochondria produces two ATP per glucose.
   c. Thus, a total of four ATP are formed by substrate-level phosphorylation outside of the electron transport chain.
2. ETC and Chemiosmosis
   a. Most of the ATP is produced by the electron transport chain and chemiosmosis.
   b. Per glucose, ten NADH and two $FADH_2$ molecules provide electrons and H^+ ions to the electron transport chain.
   c. For each NADH formed within the mitochondrion, three ATP are produced.
   d. For each $FADH_2$ formed by the citric acid cycle, two ATP are produced.
   e. For each NADH formed outside mitochondria by glycolysis, two ATP are produced as electrons are shuttled across the mitochondrial membrane by an organic molecule and delivered to FAD.
3. Efficiency of Cellular Respiration
   a. The energy difference between total reactants (glucose and O_2) and products (CO_2 and H_2O) is 686 kcal.
   b. An ATP phosphate bond has an energy of 7.3 kcal; 36 to 38 ATP are produced during glucose breakdown for a total of at least 263 kcal.
   c. This efficiency is 263/686, or 39% of the available energy in glucose is transferred to ATP; the rest of the energy is lost as heat.

Career Avenues

Metabolic Pool

1. In a metabolic pool, substrates serve as entry points for degeneration or synthesis of larger molecules.
2. Degradative reactions (catabolism) break down molecules; they tend to be exergonic.
3. Synthetic reactions (anabolism) build molecules; they tend to be endergonic.

A. Catabolism

1. Just as glucose is broken down in cellular respiration, other molecules in the cell undergo catabolism.
2. Fat breaks down into glycerol and three fatty acids.
   a. Glycerol is converted to G3P, a metabolite in glycolysis.
   b. An 18-carbon fatty acid is converted to nine acetyl-CoA molecules that enter the citric acid cycle.
   c. Respiration of fat products can produce 108 kcal in ATP molecules; fats are an efficient form of stored energy.
3. Amino acids break down into carbon chains and amino groups.
   a. Hydrolysis of proteins results in amino acids.
   b. *R*-group size determines whether carbon chain is oxidized in glycolysis or the citric acid cycle.
   c. A carbon skeleton is produced in the liver by removal of the amino group, by the process of deamination.
   d. The amino group becomes ammonia (NH_3), which enters the urea cycle and ultimately becomes part of excreted urea.
   e. The size of the *R*egroup determines the number of carbons left after deamination.

B. Anabolism

1. ATP produced during catabolism drives anabolism.
2. Substrates making up pathways can be used as starting materials for synthetic reactions.
3. The molecules used for biosynthesis constitute the cell's metabolic pool.
4. Carbohydrates can result in fat synthesis: G3P converts to glycerol, acetyl groups join to form fatty acids.
5. Some metabolites can be converted to amino acids by *transamination*, the transfer of an amino acid group to an organic acid.
6. Plants synthesize all the amino acids they need; animals lack some enzymes needed to make some amino acids.
7. Humans synthesize 11 of 20 amino acids; the remaining 9 essential amino acids must be provided by the diet.

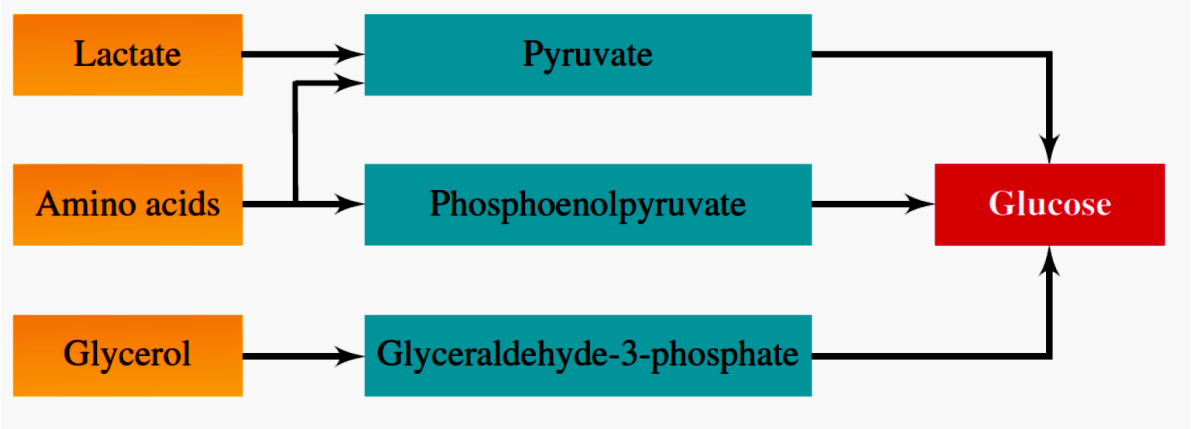
Gluconeogenesis

A continuous supply of glucose is needed by the body, especially for the brain and the nervous system. But the amount of glucose and glycogen present in the body will last for only about four hours at normal metabolic rates. Thus, there is a metabolic need for a

Career Avenues

pathway that produces glucose from non-carbohydrate sources. The formation of glucose from non-carbohydrate sources is called gluconeogenesis. Most of the glucose formed during gluconeogenesis comes from lactate, certain amino acids, and the glycerol of fats. In each case, these molecules are converted to Embden–Meyerhof pathway intermediates.

Lactate is converted to pyruvate. Amino acids first undergo deamination (loss of amino groups) and then are converted to pyruvate or phosphoenolpyruvate. Glycerol is converted to glyceraldehyde-3-phosphate. Most of the steps in the Embden–Meyerhof pathway are reversed to transform these pathway intermediates into glucose.
Gluconeogenesis takes place primarily in the liver and also in the kidneys. These organs have the enzymes that catalyze reversal of the Embden–Meyerhof pathway. Because of this capability, the liver is the organ that is primarily responsible for maintaining normal blood-sugar levels.



An overview of gluconeogenesis. All transformations except lactate to pyruvate require a series of reactions. (Detailed gluconeogenesis pathway is given under carbohydrate metabolism in Biochemistry section)

Glycogenesis and glycogenolysis

Glycogenesis and **Glycogenolysis** another two important metabolic pathways that can be seen in animals are also dealt in detail under biochemistry section.

LIPID METABOLISM

Fats (triacylglycerols) are stored primarily in adipose tissue, which is widely distributed in the body. Fat tends to accumulate under the skin (subcutaneous fat), in the abdominal region, and around some internal organs, especially the kidneys. Fat deposited around internal organs acts as a shock absorber, or cushion. Subcutaneous fat acts as an insulating blanket. It is developed to an extreme degree in mammals such as seals, walruses, and whales, which live in cold water.

Fat is the major reserve of potential energy. It is metabolized continuously. Stored fat does not remain in the body unchanged: There is a rapid exchange between the triacylglycerols of the plasma lipoproteins and the triacylglycerols in the adipose tissue. The plasma lipoprotein–

bound triacylglycerols are broken down by the enzyme lipoprotein lipase that is found on the walls of all capillaries, and the resulting free fatty acids are transported into the adipose cells. The fatty acids are then converted back to triacylglycerols. When the body needs energy from fat, adipose cell enzymes hydrolyze triacylglycerols, and the fatty acids are exported to other body tissues. This vital process is under careful hormonal control. For example, a part of the "fight or flight" response caused by the hormone epinephrine (adrenaline) is an increased fatty acid output from the adipose tissue. Conversely, when there is more energy available in the diet than the body needs, the excess energy is used to make body fat. Continued eating of more food than the body can use results in obesity

Lipogenesis or biosynthesis of Fatty acids

The biosynthesis of fatty acids from acetyl-CoA is called lipogenesis. Acetyl-CoA can be obtained from the catabolism of carbohydrates, fats, or proteins. After they are synthesized, fatty acids combine with glycerol to form triacylglycerols, which are stored in adipose tissue. Consequently, lipogenesis is the pathway by which all three of the major classes of nutrients are ultimately converted to body fat.

Is lipogenesis just the reverse of fatty acid oxidation (beta oxidation)? By analogy with carbohydrate metabolism, we might expect this to be the case However, fatty acid biosynthesis is not simply a reversal of fatty acid oxidation. The following are the major differences between the two pathways:

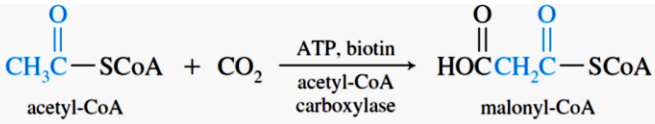
1. Fatty acid catabolism occurs in the mitochondria, but fatty acid anabolism (lipogenesis) occurs in the cytoplasm.

2. Lipogenesis requires a set of enzymes that are different from the enzymes used in the catabolism of fats.

3. In the anabolic pathway (lipogenesis), the growing fatty acid chain bonds to a special Acyl Carrier Protein, ACP SH, which acts as a "handle" to transfer the growing chain from one enzyme to another through the series of enzyme-catalyzed reactions in the pathway. Coenzyme A is the carrier in fatty acid catabolism.

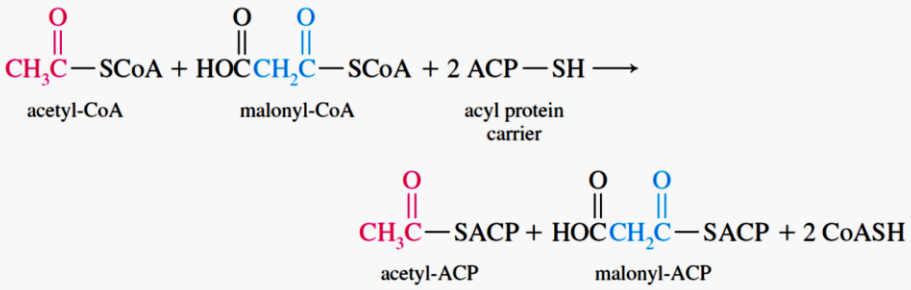
4. A preliminary set of reactions, involving malonyl-CoA, occurs for each two-carbon addition cycle in the synthesis. "Malonyl-" refers to a three-carbon group and has no counterpart in the catabolic pathway. Malonyl-CoA is synthesized from acetyl-CoA and carbon dioxide in the presence of the enzyme acetyl-CoA carboxylase, ATP, and the vitamin biotin:

Career Avenues

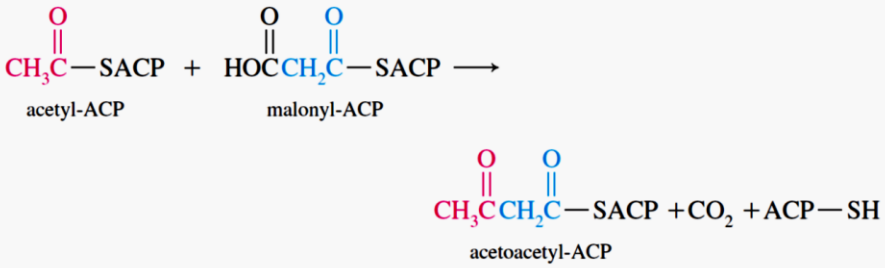


The biosynthesis of a fatty acid occurs by the addition of successive two-carbon increments to a lengthening chain starting with acetyl-CoA. Each incremental addition follows this five-step pathway or reaction sequence:

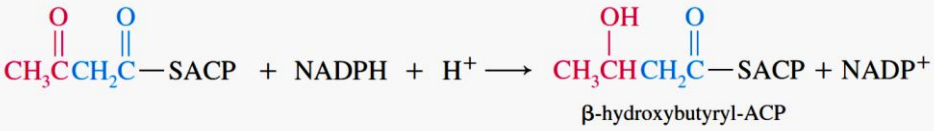
1. Acetyl-CoA and malonyl-CoA bond to separate acyl carrier proteins:



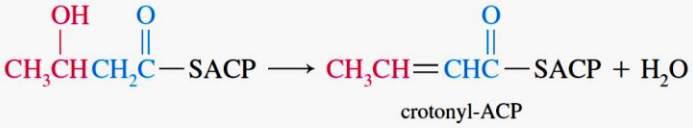
2. Acetyl-ACP and malonyl-ACP condense, with loss of carbon dioxide (decarboxylation):



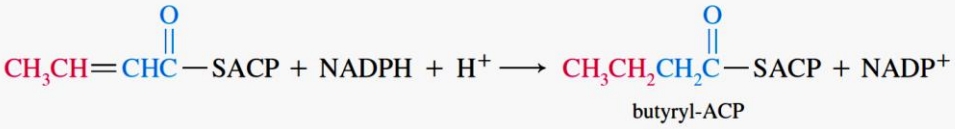
3. Reduction [hydrogenation of carbon 3 (β-keto group)]:



4. Dehydration (formation of a double bond between carbons 2 and 3):

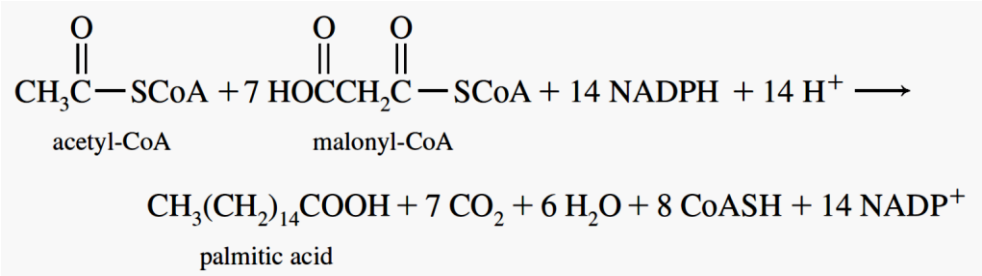


5. Reduction (hydrogenation of carbons 2 and 3):



This completes the first cycle of the synthesis; the chain has been lengthened by 2 carbon atoms. The biosynthesis of longer-chain fatty acids proceeds by a series of such cycles, each

lengthening the carbon chain by an increment of 2 carbon atoms. The next cycle begins with the reaction of butyryl-ACP and malonyl-ACP, leading to a 6-carbon chain, and so on. This synthesis commonly produces palmitic acid (16 carbons) as its end product. The synthesis of palmitic acid from acetyl-CoA and malonyl-CoA requires cycling through the series of steps seven times. The net equation for the formation of palmitic acid is



Nearly all naturally occurring fatty acids have even numbers of carbon atoms. A sound reason for this fact is that both the catabolism and the synthesis proceed by two-carbon increments.

In conclusion, it should be noted that the metabolism of fats has some features in common with that of carbohydrates. The acetyl-CoA produced in the catabolism of both carbohydrates and fatty acids can be used as a raw material for making other substances and as an energy source. When acetyl-CoA is oxidized via the citric acid cycle, more potential energy can be trapped in ATP. The ATP in turn serves as the source of energy needed for the production of other substances, including the synthesis of carbohydrates and fats.

Lipogenolysis or Fatty acid degradation or β-Oxidation has been dealt in detail under Biochemistry section

Amino acid Metabolism has also been dealt in detail under Biochemistry section

Career Avenues

ANIMAL CELL CULTURE AND ANIMAL BIOTECHNOLOGY

INTRODUCTION

One of the characteristics of most animals is that they are multicellular−in other words− they are composed of many cells. With this multicellularity comes the specialization of cells. In a multicellular organism, each cell does not have to carry out all the activities necessary for the life of the organism. Although most cells of any higher organism have many organelles and metabolic pathways in common, each cell is also unique in expressing some of these components to an enhanced degree in order to fulfill a specific function within the organism. Each cell type has its own role−to secrete a specific product, to contract, to transmit an electrical impulse, and so on. The result of this cellular specialization is that animals consist of a number of different types of cells – each with a characteristic size, shape, structure and function. Such cells are said to have differentiated. A vertebrate has more than 100 different types of cells. These cells associated in very organized patterns to perform specialized functions. Many animal cells can, with special care, be induced to grow outside of their organ or tissue of origin. Isolated cells, tissues or organs can be grown in plastic dishes when they are kept at defined temperatures using an incubator and supplemented with a medium containing cell nutrients and growth factors. The *in vitro* cultivation of organs, tissues and cells is collectively known as tissue culture, and is used in many areas of science. The partial list of different cell types which can be grown in culture includes connective tissue elements such as fibroblasts, skeletal tissue (bone and cartilage), skeletal, cardiac and smooth muscle, epithelial tissue (liver, lung, breast, skin, bladder and kidney), neural cells (glial cells and neurons, although neurons do not proliferate *in vitro*), endocrine cells (adrenal, pituitary, pancreatic islet cells), melanocytes and many different types of tumor cells. The development of these tissue culture techniques owes much to two major branches of medical research: cancer research and virology.

BASIC CHARACTERISTICS OF TISSUE CULTURE

TISSUE CULTURE

Tissue culture is used as a generic term to include the *in vitro* cultivation of organs, tissues and cells. Originally, the term is not limited to animal cells, but includes the *in vitro* cultivation of plant cells. Tissue culture can be subdivided into three major categories; organ culture, explant culture, and cell culture.

ORGAN CULTURE

Organ culture refers to a three-dimensional culture of tissue retaining some or all of the histological features of the tissue *in vivo*. The whole organ or part of the organ is maintained in a way that allows differentiation and preservation of architecture, usually by culturing the tissue at the liquid-gas interface on a grid or gel. There are disadvantages to organ cultures. Organs cannot be propagated so each piece of tissue can only be used once, which makes it difficult to assess the reproducibility of a response. And, of course, the particular cells of interest may be very small in number in a given piece of tissue so the response produced may be difficult to detect and quantify. It may not be possible to supply adequate oxygen and

Career Avenues

nutrients throughout the tissue because of the absence of a functioning vascular system, so necrosis of some cells occurs fairly rapidly. This problem may be ameliorated to some extent by keeping the organ in stirred cultures or in roller bottles which alternately provide air and soluble nutrients.

EXPLANT (OR ORGANOTYPIC) CULTURE

In explant culture, small pieces of the tissue of interest are simply allowed to attach to an appropriate substrate, usually one that has been coated with collagen, and are cultured in a rich medium, usually one containing serum. Following attachment, cell migration is promoted in the plane of the solid substrate. Traditionally, explants have been maintained in Maximov chambers in which cells are grown on cover slips sealed over a depression in a thick glass slide, and this approach is still in use. More recently, it has become common to use regular culture dishes, which are much more convenient since they do not need to be disassembled and reassembled at each feeding. As with dissociated cell culture, immature tissue grows best, and explants are generally prepared from embryonic or neonatal tissue. Typically, the tissue is cut with scalpels into slices 0.5 to 1.0 mm thick, but in some cases it is simply fragmented by passing through a nylon mesh. The need for diffusion of nutrients and oxygen to the center of the explant limits thickness to about a millimeter.

In experienced hands, explant cultures can be maintained for months, and cells within the explant continue their development more or less appropriately. One of the principal advantages of this method is that some aspects of the tissue's architecture can be preserved within the explant.

CELL CULTURE (DISSOCIATED)

Cell culture refers to cultures derived from dissociated cells taken from the original tissue ('primary cell culture'). Cells are dispersed (mechanically and/or enzymatically) into a cell suspension which may then be cultured as a monolayer on a solid substrate, or as a suspension in the culture medium. These cultures have lost their histotypic architecture and often some of the biochemical properties associated with it. However, they can be propagated and hence expanded and divided to give rise to replicate cultures. Cell cultures can be characterized and a defined population can be preserved by freezing. The most obvious advantage of cell culture, and of dissociated cell culture in particular, is that it makes individual living cells accessible. All in all, primary dissociated cell cultures are particularly amenable to study using morphological and physiological techniques, which can be applied on a cell by cell basis. They are obviously less well suited to traditional biochemical approaches because the quantity of material obtainable from these cultures is usually limited and they contain a heterogeneous population of cells.

One final drawback of working with primary cell cultures is that success is not automatic. Finding the conditions that permit good cell growth and maturation, getting culture to grow reproducibly, and documenting that you have accomplished all of this entails plenty of hard work.

Career Avenues

GROWTH KINETICS

The standard format of a culture cycle beginning with a lag phase, proceeding through the logarithmic phase to a stationary phase and finally to the decline and death of cells, is well documented. Although cell growth usually implies increase in cell numbers, increase in cell mass can occur without any replication. The difference in mean cell mass between cell populations is considerable, as would be expected, but so is the variation within the same population. Growth (increase in cell numbers or mass) can be defined in the following terms:

(a) Specific growth rate, u (i.e. the rate of growth per unit amount (weight/ numbers) of biomass):

$$\mu = (1/x)(dx/dt)h^{-1}$$

where dx is the increase in cell mass, dt is the time interval, and x is the cell mass. If the growth rate is constant (e.g. during logarithmic growth), then

$$\ln x = \ln x_o + \mu t$$

where X_0 is the biomass at time t_0

(b) Doubling time, t_d (i.e. the time for a population to double in number/mass):

$$t_d = \frac{\ln 2}{\mu} = \frac{0.693}{\mu}$$

(c) Degree of multiplication, n or number of doublings (i.e. the number of times the inoculum has replicated):

$$n = 3.32 \log(x/x_o)$$

CULTURE MEDIA FOR ANIMAL CELL CULTURE

CULTURE MEDIA

Today, with standardized media and sophisticated incubation conditions, culturing animal cells is considerably easier than it used to be. Since 1950s, tissue culture media were developed and conditions were worked out which closely simulate the situation *in vivo*. In particular, the environment is regulated with regard to the temperature, osmotic pressure, pH, essential metabolites (such as carbohydrates, amino acids, vitamins, proteins and peptides), inorganic ions, hormones and extracellular matrix. Among the biological fluids that proved successful for culturing cells, serum is the most significant. Five to twenty percents of serum is usually added to media for optimal cell growth. Serum is an extremely complicated mixture of compounds including undefined components, therefore much work has gone towards creating a chemically defined alternative to serum.

A wide variety of culture media is currently available. The choice of culture media is dependent on the requirements of cells. The components of suitable culture media include:

Career Avenues

BASIC MEDIA

The most basic media are balanced salt solutions (BSS), e.g., phosphate-buffered saline (PBS), which may be used for washing cells and for short incubations in suspension. More complex defined media are used for growth and maintenance. Defined media can also vary in complexity, by the addition of a number of constituents, e.g., from Eagle's minimum essential medium (MEM) which contains essential amino acids, vitamins and salts, to McCoy's medium, which contains a larger number of different amino acids, vitamins, minerals and other extra metabolites (such as nucleosides).

BUFFERING CAPACITY

A number of supplements to the basic media are necessary to enable them to be used for culturing cells. Cell cultures have an optimum pH for growth, generally between pH 7.4- 7.7. The type of buffering that is used for the media depends on the growth conditions. When cells are incubated in a CO_2 atmosphere an equilibrium is maintained between the medium and the gas phase. A bicarbonate−CO_2 buffering system is most often used due to its low toxicity towards the cells. HEPES, a much stronger buffer, may also be used, however, in this case much greater concentrations of HEPES than bicarbonate are required when used in a CO_2 atmosphere. Each type of media has a recommended bicarbonate concentration and CO_2 tension to achieve the correct pH and osmolarity. Nevertheless, this may vary slightly among laboratories, therefore, a sample of media should be left under the normal incubating conditions and monitored overnight, the buffering can then be adjusted accordingly. HEPES buffer should normally be used in conjunction with bicarbonate for which a relationship between the HEPES and bicarbonate exists for differing CO_2 levels, although, HEPES alone can maintain pH in the absence of exogenous CO_2. The addition of 1-5 mM pyruvate to the medium increases the endogenous cellular production of CO_2 and limits the need for a CO_2 atmosphere. Some defined media have been devised for this purpose, e.g., Leibovitz L-15 medium. Cells which produce large amounts of endogenous CO_2 under certain incubation conditions may require HEPES to buffer this CO_2 product. The density of the culture may affect the CO_2 requirement, however, in general phenol red in the medium will indicate the state of the pH at any given time.

GLUTAMINE AND AMINO ACIDS

In addition to buffering the medium, there are other growth requirements including amino acids, the requirement for which may vary with cell culture type. Commonly the necessary amino acids include cysteine and tyrosine, but some non-essential amino acids may be needed. Glutamine is also required by most cell lines and it has been suggested that cultured cells use glutamine as an energy and carbon source in preference to glucose, although glucose is present in most defined media. Glutamine is usually added at a final concentration of 2 mM, however, once added to the medium the glutamine is only stable for about 3 weeks at 4°C.

SERUM

Although there is much research aimed at attempting to reduce the requirement of cells for serum, by alternative supplementation of the media, it is apparent that most cell lines still require serum for adequate growth. Various sources of serum may be used such as calf, fetal

Career Avenues

calf, and horse. Many continuous cultures utilize calf serum, but often fetal calf serum provides the best growing conditions. The level of serum used depends on the particular cell line and should be determined empirically. Batches of serum may vary considerably in their ability to support cellular growth. It is therefore important to test batches of serum and have sufficient quantities of a batch that shows suitable growth supporting characteristics stored at -20 °C. To check these properties, cloning efficiency and growth characteristics (morphology, growth patterns) should be carried out over a concentration range of 2% to 20% serum. This wide concentration range will disclose if the alteration in the concentration of serum is possible to give optimal growth characteristics for a particular cell line.

ANTIBIOTICS AND ANTIMYCOTICS

Unless good sterile conditions can be maintained (e.g., using laminar flow hoods) it is necessary to incorporate antibiotics and antimycotics into the media. A wide range of suitable preparations are available from relatively specific antibiotics, e.g., penicillin/ streptomycin solutions, to broader spectrum antibacterial/antimycotic agents such as kanamycin or amphotericin B. The antibiotics chosen should clearly not to be toxic to the cells in culture and may depend on the type of contamination experienced in the individual laboratory.

SUPPLY AND PREPARATION OF CULTURE MEDIA

The choice of culture media used will depend on the type of primary cell, cell line, and the incubation conditions. However, it is best to start with the medium recommended by the original supplier of the cells. Changing the medium necessitates the investigation on growth characteristics such as growth curves and cloning efficiency. To render a change in culture conditions from one medium to another it is advisable to 'condition' the cells, by increasing the ratio of new to old medium with successive passages. Culture media have a limited storage life and the recommendations indicated by the supplier should be followed. Liquid defined media may have a storage life at 4 °C of up to one year, while glutamine lasts only 3 weeks at 4 °C. Serum lasts for about one year at -20 °C. Culture media can be supplied in powdered form which requires dissolving and filter sterilizing, as a 10x concentrate liquid which requires dilution prior to use, or as a 1x liquid media. Media preparation, in all cases, requires high quality water. Bottles of media should be prepared in small batches, for instance two weeks supply at a time. This will ensure that constituents such as glutamine do not have time to deteriorate and also that if contamination should occur it is confined to a few bottles.

PRACTICAL CONSIDERATIONS

i. Temperature of Medium

Always pre-warm the medium to the operating temperature (usually 37°C) and stabilize the pH before adding the cells. Shifts in pH during the initial stages of a culture create many problems, including a long lag phase and reduced yield.

ii. Growth Phase of Cells

Avoid using stationary-phase cells as an inoculum since this will mean a long lag phase, or no growth at all. Ideally, cells in the late logarithmic phase should be used.

Career Avenues

iii. Inoculation Density

Always inoculate at a high enough cell density. There is no set rule as to the minimum inoculum level below which cells will not grow, as this varies between cell lines and depends on the complexity of the medium being used. As a guide, it may be between 5×10^4 and 2×10^5 cells/ml, or 5×10^3 and 2×10^4 cells/cm^2.

iv. Stirring Rate

Find empirically the optimum stirring rate for a given culture vessel and cell line. This could vary between 100 and 500 r.p.m. for suspension cells, but is usually in the range 200-350 r.p.m., and between 20 and 100 r.p.m. for micro carrier cultures.

v. Medium and Surface Area

The productivity of the system depends upon the quality and quantity of the medium and, for anchorage-dependent cells, the surface area for cell growth. A unit volume of medium is only capable of giving a finite yield of cells. Factors which affect the yield are: pH, oxygen limitation, accumulation of toxic products (e.g. NH4), nutrient limitation (e.g. glutamine), spatial restrictions, and mechanical/shear stress. As soon as one of these factors comes into effect, the culture is finished and the remaining resources of the system are wasted. The aim is, therefore, to delay the onset of any one factor until the accumulated effect causes cessation of growth: at which point the system has been maximally utilized. Simple ways of achieving this are: a better buffering system (e.g. Hepes instead of bicarbonate), continuous gassing, generous headspace volume, enriched rather than basal media, with nutrient-sparing supplements such as lactalbumin hydrolysate or peptone, perfusion loops through ultrafiltration membranes or dialysis tubing for detoxification and oxygenation, and attention to culture and process design.

CULTURING ANIMAL CELLS

SELECTING SOURCES OF TISSUE FOR CULTURE

Adult or embryonic tissue

Cultures can be derived from adult tissue or from embryonic tissue. Cultures derived from embryonic tissue generally survive and grow better than those taken from adult tissue. Tissues from almost all parts of the embryo are easy to culture, whereas tissues from adult are often difficult or even impossible to culture. This presumably reflects the lower level of specialization and presence of replicating precursor or stem cells in the embryo. Adult tissues will usually have a lower growth fraction and a high proportion of non-replicating specialized cells, often within a more structured, and less readily disaggregated, extracellular matrix. Initiation and propagation are more difficult, and the lifespan of the culture is often shorter. Embryonic or fetal tissue has many practical advantages, but it must always be remembered that in some instances the cells will be different from adult cells and it cannot be assumed that they will mature into adult-type cells unless this can be confirmed by appropriate characterization. Examples of widely used embryonic cell lines are the various 3T3 lines (mouse embryo fibroblasts) and MRC-5 and other human fetal lung fibroblasts. Mesodermally derived cells (fibroblasts, endothelium, myoblasts) are also easier to culture

Career Avenues

than epithelium, neurons or endocrine tissue but this may reflect the extensive use of fibroblast cultures during the early years of the development of culture media together with the response of mesodermally derived cells to mitogenic factors present in serum. A number of new selective media have now been designed for epithelial and other cell types and with some of these it has been shown that serum is inhibitory to growth and may promote differentiation.

Embryonic stem cells

A more recent development has been the removal of embryonic stem cells (ES-cells) from the embryo during the blastocyst stage of development. These cells can be grown in culture for many generations and are of particular interest because they can be manipulated in culture and then re-introduced into embryos.

Normal or neoplastic tissue

Normal tissue usually gives rise to cultures with a finite lifespan while cultures from tumors can give continuous cell lines, although there are several examples of continuous cell lines (MDCK dog kidney, 3T3 fibroblasts) which are non-tomorigenic. Normal cells will generally grow as an undifferentiated stem cell or precursor cell and the onset of differentiation is accompanied by a cessation in cell proliferation which may be permanent. Some normal cells, e.g., fibrocytes or endothelium, are able to differentiate and still dedifferentiate and resume proliferation and in turn redifferentiate, while others, e.g., squamous epithelium and many hemopoietic cells, once initiated into differentiation are incapable of resuming proliferation. Cells cultured from neoplasms, however, can express at least partial differentiation, e.g., B16 mouse melanoma, while retaining the capacity to divide. Many studies of differentiation have taken advantage of this fact and used differentiated tumors such as the minimal deviation hepatomas of the rat and human and rodent neuroblastomas, although whether this can be taken as normal differentiation is always in doubt. Tumor tissue can often be passaged in the syngeneic host, providing a cheap and simple method of producing large numbers of cells, albeit with lower purity. Where the natural host is not available, tumors can also be propagated in athymic mice with greater difficulty but similar advantages.

Many other differences between normal and neoplastic cells are similar to those between finite and continuous cell lines and indeed the importance of immortalization in neoplastic transformation has been recognized. Selecting types of animal cell culture.

ORGAN CULTURE OR CELL CULTURE

Early attempts at culturing tissues relied upon the explantation of whole tissue or organ which could be maintained *in vitro* for only very short periods. Nowadays it is more usual to grow specific cell types from tissues, although there are still some situations where it is necessary to grow a whole organ (or a part of it). In adopting a particular type of culture the following points should be taken into account. Organ culture will preserve cell interaction, retain histological and biochemical differentiation for longer, and, after the initial trauma of explantation and some central necrosis will generally remain in a non-growing steady state for a period of several days and even weeks. They cannot be propagated, generally incur

Career Avenues

greater experimental variation between replicates, and tend to be more difficult to use for quantitative determinations due to minor variations in geometry and constitution.

ADHERENT OR SUSPENSION CULTURE

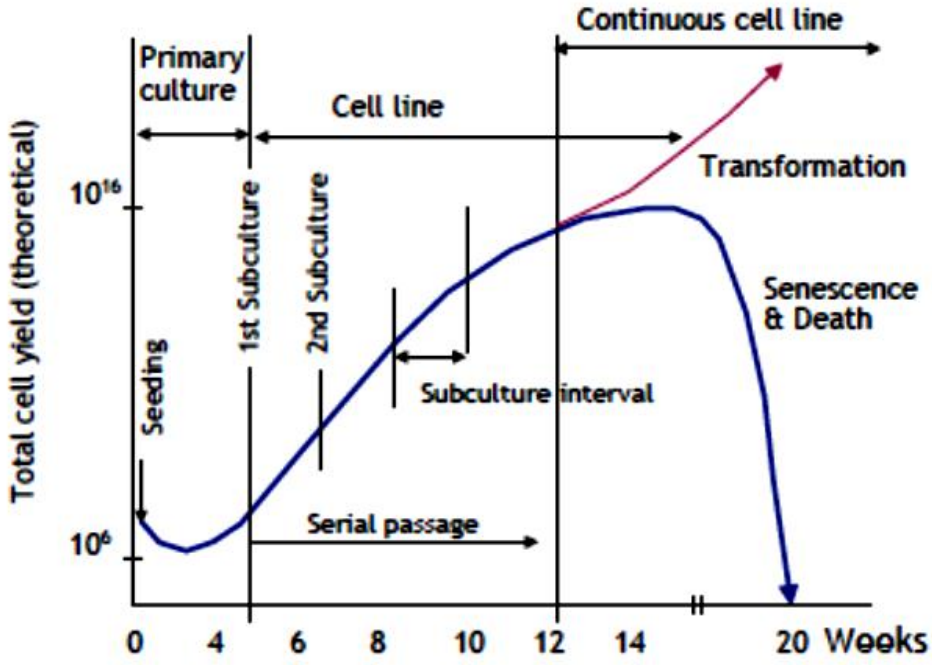
Cells may grow as an adherent monolayer or in suspension. Adherent cells are said to be anchorage-dependent and attachment to a substratum is a prerequisite for proliferation. They are generally subjected to contact inhibition, which means they grow as an adherent monolayer and stop dividing when they reach such a density that they touch each other. Most cells, with the exception of mature hemopoietic cells and transformed cells, grow in this way. In contrast to anchorage-dependent cells, cells cultured from blood, spleen or bone marrow adhere poorly if at all to the culture dish. In the body, these cells are held in suspension or are only loosely adherent. It is important to realize this if you are working with this category of cells, since the methods used to propagate these cells are very different to those for adherent cells.

Suspension cultures are easier to propagate, since subculture only requires dilution with medium. Cultures in which cells grow attached to each other or to a substratum have to be treated by a proteolytic enzyme to break the bond between cells and substratum. The most commonly used enzyme is trypsin (other enzymes, e.g., collagenase, papain, dispase, and pronase, are also used). Clearly, freely suspended cultures do not require trypsinization. They are, therefore, also easier to harvest.

PRIMARY CULTURES OR CONTINUOUS CELL LINES

If you remove tissue from an embryo, dissociate it into a suspension of single cells, and plate them out onto a culture dish, a series of characteristic events occurs. Firstly, cells are in a lag phase, usually no more than 1-2 days in length, during which there is little or no increase in cell number. During this time, cells are "conditioning" the medium, undergoing internal cytoskeletal and enzyme changes and adjusting to the new medium. Secondly, the cells undergo a period of rapid division, so-called log phase growth. Then, as they approach confluency and form contacts with one another, their rate of division slows and they begin to express a program of differentiation characteristic of their tissue of origin. Muscle cells fuse and acquire cross-striation, epithelial cells from the kidney or gut become linked by junctional complexes and transport ions from one surface to another, heart cells begin to beat spontaneously.

Career Avenues



Growth of animal cells in culture

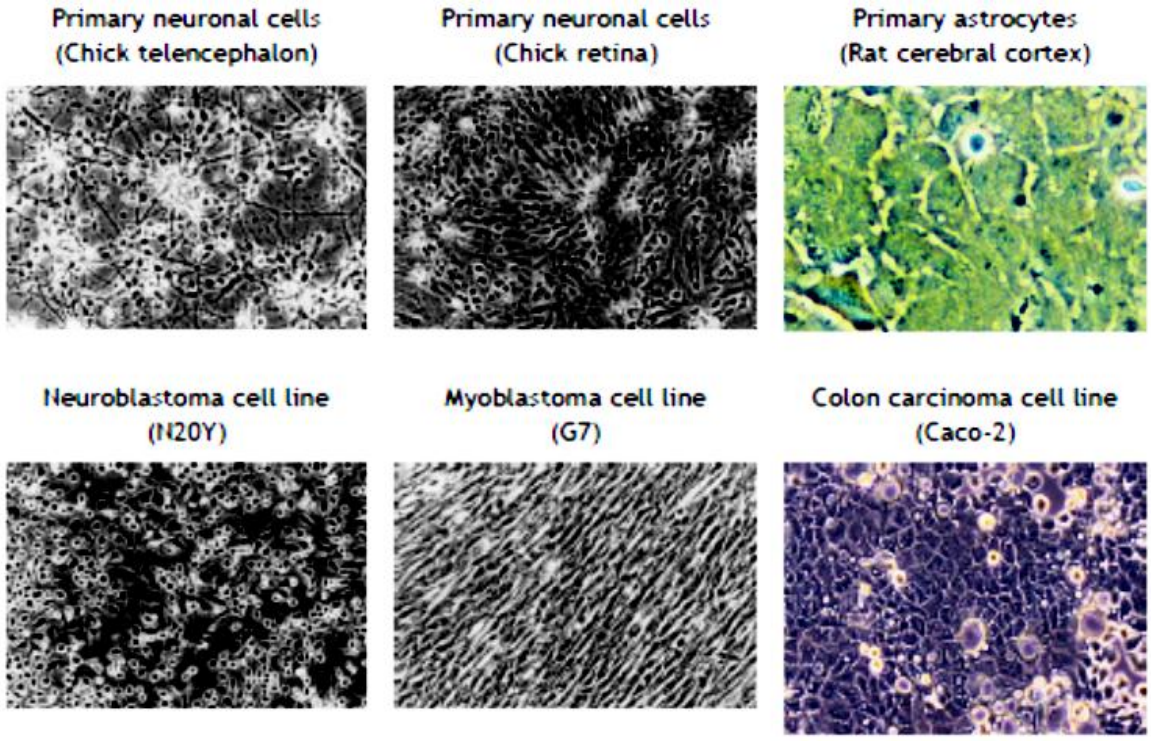
Cultures such as those just described are referred to as *primary cultures*, because they are prepared from cells taken directly from the animal. The cells divide or not (depending on what they are accustomed to), acquire differentiated characteristics, and ultimately die. For the next experiment, it's back to the animal again to obtain new tissue and prepare new culture.

Alternately, in the case of cells that divide in culture, it is possible to 'passage' or 'subculture' them by inducing them to detach from the substrate, 'splitting' them (i.e., diluting them several-fold in medium and replating them into new dishes), and allowing them to reenter log phase growth. However, the properties of the cultured cells often change gradually with passaging, as more rapidly dividing cell populations come to predominate and more 'differentiated' cells, which divide more slowly, are lost. When cells are repeatedly subcultured, most cease division after a finite number of generations, typically between 20 and 80. This is thought to reflect the same process of senescence that occurs in cells *in situ*. However, it is possible to develop populations of cells that can be passaged indefinitely and that express a reasonably stable phenotype. These are referred to as *established or continuous cell lines*.

Some cell lines have arisen spontaneously in normal cells being passaged in culture, but the majority has been obtained by culturing tumor cells. In addition to their infinite life span (their 'immortality'), such cell lines frequently share several additional properties that distinguish them from 'normal' cells in culture. They divide more rapidly, they do not require attachment to the substratum for growth, and when reintroduced into animals, they form tumors. Cell lines with these properties are sometimes referred to as *transformed cell lines*.

Once a continuous cell line has been established, it is customary to clone the cells in order to obtain a genetically homogeneous population. Many separate clonal cell lines, each

exhibiting a unique set of properties, can be obtained from a single tumor, or tumor cell line. For example, dozens have been derived from the original mouse neuroblastoma, glioma, and pheochromocytoma cells have been derived from neural tumors, some from tumors that arose spontaneously, others from tumors induced by chemical carcinogens.



Examples of primary cultures and continuous cell line cultures

Primary cultures or clonal cell lines? This is the first and most important choice that must be made after deciding to culture cells. But both approaches have advantages as well as drawbacks; both have their ardent advocates and their opponents.

PRIMARY CULTURES: ADVANTAGES AND LIMITATIONS

As previously described, a primary culture is defined as one 'started from cells, tissues or organs taken directly from organisms'. The major advantages of primary cultures are the retention of : (1) the capacity for biotransformation; and (2) tissue-specific functions.

In many cases, the metabolic profile generated by a primary cell culture such as hepatocytes has greater similarity to *in vivo* than does the pattern seen with subcellular fractions used as an exogenous source for biotransformation. Hepatocytes possess the ability for conjugation reactions that may be absent in subcellular fractions. In fact, metabolic studies in hepatocytes have resulted in the identification of previously undetected *in vivo* products. Studies in hepatocytes lead to the discovery of ketoethinimate as a metabolite of ethinimate. Subsequent investigation revealed this metabolite was also formed *in vivo*.

The second advantage of primary cultures is the retention of tissue specific functions. For example, primary cultures of rat myocardial cells consisting of synchronously beating cells

Career

can be prepared. When these cultures were exposed to tricyclic antidepressants that are cardiotoxic, both arrhythmias and cessation of beating were observed.

One limitation of primary cultures is the necessity to isolate cells for each experiment. Procedures to isolate cells require the disruption of the tissue, often with proteolytic enzymes. This may result in the loss or damage of specific membrane receptors, damage to the integrity of the membrane, and loss of cellular products. During the interval necessary to establish monolayer cultures, damage is often repaired. An example of this phenomenon is the loss of Ca++ during the isolation of hepatocytes. There is a 60% decrease in the concentration of Ca++ in freshly isolated cells; however, following 24 hours in culture, the level is restored to that of the intact liver.

Primary cultures have a limited life span and changes in metabolism and tissue specific functions will occur with time in culture. Many alterations can be postponed depending upon the culture conditions. It has been shown using hepatocytes that P-450 levels can be maintained with appropriate medium, and inducibility can be altered by the substrate upon which the cells are cultured.

In summary, primary cultures offer a cell system which can often provide a better approximation of the intact tissue than can cell lines. However, care must be taken to minimize the changes that occur with increasing time in culture.

CONTINUOUS (CLONAL) CELL LINES: ADVANTAGES AND LIMITATIONS

The following important considerations should be taken into account before undertaking any experimental study with a cell line:

1. Many cell lines have been available for extended time periods, having undergone many *in vitro* passages. Before use, cultures should be carefully examined for homogeneity of cell type, and if necessary, must be re-cloned. Cell lines are usually more resistant to toxic insult than freshly isolated short-term cultures.
2. Investigators should be aware of population doubling time since this may play a role in sensitivity to the test agents (i.e., cells may have to undergo mitosis to reflect drug effect). Replication time may also affect the number of cells seeded in order to obtain a required density.
3. Type of cell and culture conditions used at the time of exposure to the test agent should best serve the particular assay at hand (i.e., suspension versus adherent cultures; sparsely seeded versus dense cultures). For certain experimental procedures, sparsely seeded cultures maintained over several days might be most suitable. Other experiments might be more informative when cells are in exponential growth or are confluent. Types of seeding must be kept constant for all experiments within a given study.
4. Information concerning normal or tumorigenic characteristics should be noted. This may also be reflected in chromosome number and ability to replicate in medium with low serum concentration. (Tumor cells can grow more readily in low serum medium.)

Career Avenues

5. Some cell lines maintain or secrete tissue-specific proteins and enzymes which can be used for examination as biological markers. Specific biomarkers should be noted, such as special receptor sites or responses to specific growth factors.

6. If defined medium for a particular line is not available, serum concentrations should be kept as low as possible without affecting cell viability. Serum can adsorb xenobiotics, and thus reduce the concentration of test agent available.

7. Temperature effects may yield important information which can be obtained from cultures adapted to lower temperature or cell lines. For example, it may be more appropriate to expose keratinocyte cell lines, such as the XB-2 to the test agent at the lower temperatures found at the skin surface.

8. For assays involving test agents which are in need of metabolic activation to convert to a toxic intermediate, cell lines that are known to contain P-450 microsomal enzymes, or can be induced to make them, should be used. Alternatively, preparations of S-9 mixtures which contain the mixed functions oxygenase, can be added to the incubation medium.

9. Cell lines must be clean of contaminating mycoplasma, bacteria or viruses and sterile techniques must be strictly adhered to.

10. Frequent microscopic examination and monitoring of stock cultures is mandatory to detect changes in growth patterns and morphology.

11. It is essential that all experiments are repeated several times, using the same environmental conditions to determine variability.

12. Immortalized epithelial cells (nontumorigenic) derived from some organs, such as the lung and liver, are now available (due to transfection of viral components) for long term studies.

13. Stable lymphocytic lines (mostly of neoplastic origin) which maintain the functions of macrophages and other cells of the hematopoietic system are available. Similar stable cell lines maintaining the function of cells from other organ systems are not easily available, with some exceptions (for example, the PC12 neuroblastoma cell line).

Career Avenues

BASIC TECHNIQUES IN ANIMAL CELL CULTURE

ASEPTIC TECHNIQUES

In culturing animal cells, it is essential that all procedures are carried out using aseptic or sterile techniques. As mentioned earlier, laminar flow facilities or sterile rooms provide a suitable environment, but even then aseptic techniques should be employed. If a nonsterile environment is used, the reliance on aseptic techniques is very high. Basic aseptic technique is to ensure that the work area is clear, swabbed down regularly with 70% ethanol and that all the equipment used has been sterilized. Clean laboratory coats are also essential.

The basic rules for aseptic techniques which should be used even if an airflow cabinet is available include:

- If working on the bench, use a Bunsen flame to heat the air surrounding the Bunsen. This causes the movement of air and contaminants upwards and reduces the chance of contamination entering open vessels. Open all bottles and perform all maneuvers in this area only;
- Swab all bottle tops and necks with 70% ethanol to clean them before opening;
- Flame all bottle necks and pipettes by passing very quickly through the hottest part of the flame. This is not necessary with sterile, individually wrapped, plastic flasks and pipettes;
- Avoid placing caps and pipettes down on the bench; practice holding bottle tops with the little finger while holding the bottles for pouring or pipetting;
- Work either left to right or *vice versa*, so that all material to be used is on one side and, once finished, is placed on the other side of the Bunsen burner. (This may also stop the operator using the same reagent twice!);
- Manipulate bottles and flasks carefully. The tops of bottles and flasks must not be touched by the operator. Touching of open vessels should also be prevented when pouring. If necessary practice pouring from one container to another keeping a distance of 5 mm between the two vessels;
- Clear up spills immediately and always leave the work area clean and tidy. Dispose of glassware in appropriate bins and discard used plasticware in marked polythene bags for autoclaving or incineration. All glassware or plasticware used for infectious work must always be autoclaved before incineration. Re-usable glassware should be immersed in disinfectant whilst awaiting transfer to an autoclave.

CHECKING AND PREVENTION FOR CONTAMINATION

Perhaps the biggest pre-occupation of the cell culture biologist is how to prevent contamination of the tissue cultures. The media used to grow the cells also provide excellent nutrition for unwanted organisms. Contamination of cell cultures by bacteria, fungi and mycoplasma often results in the loss of a great deal of time and money. An additional problem is the cross contamination of cells from other cell lines. Before we examine how to ensure contamination is kept at a minimum we need to briefly consider how we would detect contamination.

Career Avenues

Regularly checking aliquots from culture flasks using a microscope can reveal some types of contamination, for example, yeast and fungi can be seen easily. Bacteria can be Gram stained or plated out on blood agar plates and general cloudiness of medium also indicates yeast or bacterial contamination. Mycoplasma are more insidious, the medium is not cloudy, the organisms cannot be seen under an ordinary microscope and most mycoplasma species are very difficult to grow on agar plates. Testing for mycoplasma is time consuming but must be done regularly because cross contamination is very easy.

DNA stains or molecular probes may be used to detect their presence in cells, those cells which turn out to contain DNA in the cytoplasm must be discarded. Much of the contamination comes from the culture medium and its components or from inadequately cleaned glassware. Routine sterility checks on the medium, serum and nutrients are recommended. To do this, a small aliquot of the medium should be incubated at room temperature and another similar aliquot simultaneously incubated at 37 °C for up to one week before the medium is used for culturing cells. If these small samples are found to be contaminated, the medium should be autoclaved and discarded.

The use of antibiotics in the medium helps to contain the problem to some extent but it should never be seen as an alternative to good aseptic technique and careful monitoring. Generally, if contamination occurs, affected cultures should be disposed of into 2.5% hyper chlorite solution. Media bottles known, or suspected, to be contaminated should be disposed of as well. Some basic preventive measures may be taken: checking sterilizing procedures (autoclave and oven procedures); checking sterility of laminar flow hoods; regular checks on cultures; each worker having his or her own set of media; and disposal of contaminated cultures rather than attempting to decontaminate them.

PREPARATION OF PRIMARY CULTURES

It is not always necessary to disaggregate tissue before culturing. Some embryonic tissues can be cultured simply by leaving the whole tissue on the flask surface and individual cells will simply grow out from the whole tissue and proliferate. After a few days the original tissue can be removed and the culture medium replenished to allow the new cells to continue growing. This method works for some tissues but it is not suitable for growing specific cells from a piece of tissue.

Tissues that do not require enzymatic disaggregation

Tissue for primary explant can be treated as follows:

- Excess blood is removed by rinsing with a sterile balanced salt solution (BSS). If the tissue is to be transported it can be kept in BSS or medium;
- Unwanted material such as fat and cartilage is cut off and the rest is chopped finely using a pair of crossed scalpels. It is important to make clean cuts, the use of tearing actions or scissors may damage the cells;
- The cell suspension is transferred to a sterile 50 ml centrifuge tube together with the buffered saline and the cells are allowed to settle out;

Career Avenues

- The medium is carefully pipetted off and the pellet is washed in fresh BSS. The cells are allowed to settle again or the suspension is gently centrifuged at about 1,000 rpm for 5 min;
- The cell pellet is resuspended in 10-15 ml of medium and the suspension is aliquoted into 2 or 3 of 25 cm2 flasks. If the pieces of tissue are still quite large the suspension may be passed through a sieve before culturing, the cultures are incubated at the appropriate temperature for 18-24 hours. (Note that in describing the cultivation of adherent cells, it is usual to describe the size of the vessels in terms of surface area rather than volume);
- If the pieces of tissue have adhered, the medium should be changed weekly until a significant outgrowth of cells has occurred;
- The original tissue pieces which show outgrowth can then be picked off and transferred to a fresh flask;

The medium in the original flask is replaced and the cells cultivated until they cover at least 50% of the surface available for growth. They can then be subcultured if required.

All steps must, of course, be done in such a way as to reduce the chances of the tissue, media and vessels becoming contaminated. Thus all dissection tools, centrifuge tubes and reagents need to be sterilized before use. This method is useful for small pieces of tissue such as skin biopsies. Fibroblasts, glial cells, epithelium and myoblasts all migrate out of the tissue very successfully. Some selection occurs because not all cells will adhere to the same extent and some cells rapidly outgrow the others.

Tissues that require enzymatic disaggregation

The tissue may be kept whole or may be cut into small pieces as described above. However, densely packed tissue is more difficult to digest and prolonged contact with digestive enzymes such as trypsin often causes destruction of viable cells. Large pieces of tissue for enzymatic disaggregation should be cut into smaller pieces as for the primary explant and digested with trypsin for about 30 min at 37 °C. Then the trypsin is neutralized by the addition of serum. The amount of trypsin that is required to achieve satisfactory release of cells is tissue-dependent and also depends upon the activity of the enzyme preparation. Trypsin suppliers often provide useful guidance for the use of their own preparations. Typically 0.25% w/v trypsin solution is satisfactory.

Trypsin is a proteolytic enzyme which hydrolyzes proteins. By adding a lot of protein (in the form of serum), the enzyme begins to hydrolyze these proteins rather than the proteins which bind cells together. What we are really doing is 'diverting' the enzyme to an alternative substrate so that it is no longer available to attack the extracellular matrix. Thus once trypsinization is complete; medium containing serum should be added to neutralize enzyme activity. If trypsin is used to disaggregate whole tissue, the dissociated cells should be harvested after about 30 min and washed to remove trypsin. The remaining whole tissue may be retrypsinized. An alternative method is to soak the piece of tissue in cold trypsin at 4 °C overnight. The cold trypsin penetrates throughout the whole tissue but will have minimal activity at this temperature. The tissue can be disaggregated by increasing the temperature to 37 °C for 30-40 min, then enzyme activity is neutralized by the addition of medium with serum as before.

Career Avenues

Collagenase and versene (phosphate buffered saline + EDTA) are also used to disaggregate tissues. Both are gentler in action than trypsin but expensive because larger amounts are required. Collagenase or versene can also be used together with trypsin to allow trypsin to be used at lower concentrations.

A variety of other hydrolytic enzymes (for example pronase and dispase) have also been employed to disaggregate tissues in some laboratories.

Once disaggregation is complete, the cells can be washed in fresh BSS or medium and seeded out in 24 cm2 flasks. The cell layer can be supplied with fresh medium after 48 hours. This culture is referred to as a primary culture.

Removing non-viable cells from the primary culture

If the primary culture is of anchorage-dependent or adherent cells, the non-viable cells can be removed by pouring off the medium and rinsing the cell layer with buffer before adding fresh medium. Cells which are not viable will not be able to adhere to the substratum. If the cells are to be cultured in suspension, the non-viable cells gradually become diluted as the viable cells proliferate. If it is necessary to remove the non-viable cells from the suspension, the cells may be layered onto Ficoll or 'lymphoprep' and be centrifuged at 2,000 rpm for 15-20 min. The non-viable cells will sink to the bottom and the viable cells can be collected from the medium-Ficoll interface.

MAINTAINING THE CULTURE

If a primary culture is not to be used as such, it may be sub-cultured to produce a cell line. Cell lines may be of a very limited lifespan or they may be passaged several times before the cells become senescent. Some cells, such as macrophages and neurons do not divide *in vitro* and can only be used as primary cultures.

Sub-culturing from primary to secondary cell culture

A primary culture contains a very heterogeneous population of cells from the original explant. Some of these cells will die, some will fail to grow, others will grow quickly and become the dominant cell type present. On sub-culturing the primary cell culture, the dominant types will become even more dominant. We can, therefore, foresee that subculturing enables us to produce more homogeneous cell populations.

We call the culture produced after sub-culture of a primary culture, a secondary culture. Sub-culturing therefore, enables us to produce cell lines from our original explant. These cell lines may be further sub-cultured, characterized and cloned.

Producing a cell line has certain obvious advantages. A homogeneous population of characterized cells can be grown to a large scale, replicates are uniform and this makes designing experiments much easier. There are, however, disadvantages. In establishing a cell line only those cells best suited to the *in vitro* conditions are selected for. These cells may lose some of the differentiated characteristics they had while growing *in vivo* and are prone to genetic instability, particularly if they divide rapidly.

Career Avenues

Producing cell lines of a particular cell type

We really have only two main options here. Firstly, of course, we must choose the appropriate tissue in the primary explant stage. It is no good attempting to produce a particular epithelial cell line if we use tissues which do not contain the appropriate cell type. Secondly, not all cells will grow equally well in the same medium. So, in principle, by selecting our medium carefully we may provide the condition most suited to our cells of interest.

Propagating a cell line

Once a cell line is established, it needs to be propagated in order to produce sufficient cells for characterization and storage, as well as for particular experiments. A cell line is given a name or code which identifies its source (for example HuT, Human T cells) and, if more than one line was developed from the same source, a cell line number is also given (for example HuT 78). If cells from this line are cloned, then a clone number need to be given, e.g., HuT 78 clone 6D5. If the cell line is likely to be viable for only a few sub-culturing or generations each generation should be noted, for example HuT 78 clone 6D5/2 or HuT 78 clone 6D5/3.

Once a culture is confluent (i.e., the cells cover 60-70% of the growth surface available) it can be transferred into a holding or maintenance medium which provides just enough nutrients to keep the cells alive and healthy but reduces the replication rate so that the cells do not overgrow. The usual method is to use the same basic medium and reduce the amount of serum it contains. If cells are required for experiments or storage the contents of the flask can be split or divided to seed 2-4 new flasks depending on the vigor of the cell growth.

The content of a flask of cells grown in suspension is very simple to split. This is done in the following way:

- Stand the flask upright for 1-5 min to allow cells to settle; " A septically remove as much medium as possible without disturbing the cells. A 10 ml graduated pipette is best. Discard used medium into a waste pot for autoclaving;
- Resuspend cells in the remaining volume, measure and divide between the required number of flasks;
- Top up each flask with fresh growth medium and incubate as normal;
- Record the cell line code, clone number and the passage number.

Cells which grow as monolayer adhering to the flask surface are a little more difficult to split. The steps are:

- Remove medium by pouring it off aseptically;
- Rinse the monolayer with pre-warmed (37 °C) phosphate buffered saline (PBS) or BSS containing 1 mM EDTA; pour the medium off into discard pot;
- Add 10 ml of a 0.25% (w/v) solution of trypsin in saline. Tilt the flask so that all the cells are covered with trypsin. Other enzymes may also be used in the same way;
- Pour off the trypsin and incubate the flask at 37 °C for 1-10 min. Cells which are very susceptible to trypsin should be checked after 1 min;

Career Avenues

- Once the cells have begun to round up and are sliding off the flask surface (it may help to tip the side of the flask gently), the trypsin must be neutralized by the addition of 5-10 ml of medium containing serum;
- Wash all the cells down from the sides of the flask and aspirate gently using a pipette. This should break up any aggregates and allow for easier cell counting if required. Avoid creating froth as this can lead to contamination and cell damage;
- A small aliquot (0.2 ml) of the cell suspension may be removed for counting if required. Otherwise, measure the total volume of the cell suspension and divide between the required number of flasks and top up with fresh medium and incubate as before;
- Remember to record the cell line code, clone number and the number of times it has been divided (the passage number).

The intervals between sub-culturing or changing medium will depend on the cell line and the rate at which it grows. Rapidly dividing cell lines such as HeLa and VERO needs to be sub-cultured at least once a week with a medium change in between. Fibroblast HEL cells may be sub-cultured once every 10-14 days. The rate of growth of fibroblasts may be increased, if more frequent sub-culturing is required, by increasing the serum concentration in the medium.

QUANTITATION OF CELLS IN CELL CULTURE

For properly run experiments, it may be necessary to count the cell numbers before, after and even during the experiment. Day to day maintenance of cell lines also requires quantitative assessment of cell growth so that optimum cell densities for sub-culturing and storing can be determined.

We can divide the methods available for determining cell growth into two sub-groups. These are:

- Direct methods;
- Indirect methods.

In the direct method, cell numbers are determined directly either by counting using a counting chamber or by using an electronic particle counter. In the indirect methods measurement of some parameters such as DNA content or protein content related to cell number is used as the method of estimating biomass.

DIRECT METHODS FOR QUANTITATION OF CELLS IN CULTURE

Counting chambers

A hemocytometer (Improved Neubaur) is the simplest and cheapest method of counting. A coverslip slightly wet at the edges is placed over the counting grid such that interference patterns ('Newtons rings') formed when glass attaches to glass indicate that it is securely in place. Introduce the medium containing cells under the coverslip from a Pasteur pipette by capillary action. Fluid will run into the grooves of the chamber to take away the excess. The cells in the central grid area can then be counted. The hemocytometer is designed such that

Career Avenues

when properly set up an area of 1 mm2 and 0.1 mm deep is filled − this is a central area of 25 smaller squares bounded by triple parallel lines. Each of the 25 squares is further subdivided into 16 to aid counting. For routine subculture about 100-300 cells should be counted, but 500-1,000 for more precise counts. For the Improved Neubaur chamber, the cell concentration, calculated from the number of cells (n) in the central 25 square area (1 mm2, depth 0.1 mm), is n x 10^4 cells/ml. Viability determinations can be easily carried out after diluting cells with dye. This counting method is open to error in a number of ways. Errors may be introduced by incorrect preparation of the chamber, incorrect sampling of cells, and aggregation of cells. However, it has the advantage of allowing visual inspection of the morphology of the cells.

Coulter counters

Electronic particle counters consist of two electrodes separated by a small orifice. If a potential is applied to the electrodes, current will pass between them through the buffer in the orifice. The amount of current will be dependent upon the conductance (dielectric constant) of the buffer. As the particle enters the orifice, the conductance of the solution between electrodes would be reduced. Thus the current flowing would be reduced and this could be detected electronically.

The size of the change in current flow depends on the size of the particle and the difference in the dielectric constant (conductivity) of the particle and the suspending buffer. This is the principle upon which cell counting using a Coulter counter is carried out.

Cells in suspension are drawn through a fine orifice in the Coulter counter and, as each cell passes through, it produces a change in the current flowing across the orifice. Each change is recorded as a pulse and it is these pulses that are sorted and counted. The size of the pulse is proportional to the volume of the particle passing through, so signals of varying size are produced. The pulse height threshold can be set, therefore, to eliminate electronic noise and weak pulses produced by debris.

Electronic counters, of which the Coulter counter is the most widely used, provide rapid results, but high cell numbers are required to give an accurate count. Another disadvantage is that they cannot distinguish between dead and viable cells and clumps of cells may register as a single pulse thus leading to inaccurate counts.

Indirect methods for determining cells in culture

Other methods of quantitation such as radioisotope labelling and estimation of total DNA or protein are used less frequently. They are useful when cells are grown in microwell plates or as hanging drop cultures. The cells can be mixed *in situ*, stained and counted by eye under a microscope. DNA and protein assays are inaccurate, particularly if cells are multinucleated. They do not distinguish viable and non-viable cells.

CELL VIABILITY DETERMINATION

When cells are freshly isolated from a tissue or confluent monolayers are subcultured, the proportion of living, or viable, cells should be determined before they are used. This is most often determined by assessment of membrane permeability, under the assumption that a cell with a permeable membrane has suffered severe, irreversible damage.

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Trypan blue exclusion

This is a rapid test for gross damage which is conveniently combined with determining cell number.

- Mix a small aliquot of cell suspension with an equal volume of 0.4% trypan blue solution.
- Then, within 1-5 min, introduce the suspension into a hemocytometer chamber.
- Non-viable cells appear blue, with the nucleus staining particularly darker.
- Count the viable (unstained) cells and the total number of cells.
- Express the number of viable cells as a percentage of the total. Calculate the cell number after multiplying by 2 to allow for the dilution with dye.

LDH leakage

Leakage of cytosolic enzymes such as lactate dehydrogenase (LDH) provides a test that is similar in sensitivity to dye exclusion and can be more accurately quantitated but takes rather longer to perform.

- Centrifuge an aliquot of cell suspension at a speed suitable for sedimenting all the cells without causing cell damage (e.g., 50 x g for 2 min)
- Transfer the supernatant medium to a clean tube and keep on ice
- Add an equal volume of solubilizing agent to the cell pellet and vortex mix.
- Samples should be kept on ice and the assays performed on the same day.
- To a cuvette, add 3 ml of substrate at 37 °C, 50 µl of NADH and 25-200 µl of sample (depending on the cell type).
- Mix rapidly and follow the increase in absorbance at 340 nm at 37 °C.
- Calculate the initial change in absorbance per min for the medium and for the solubilized cells and express the activity of the medium as a percentage of the total (i.e., to give percent leakage).

Career Avenues

KINETICS OF CELL GROWTH AND PRODUCT FORMATION AND EFFECT OF SHEAR FORCE

The growth state and productivity of animal cells in culture can be monitored and characterized by applying the appropriate stoichiometric and kinetic analytical methods. These methods use knowledge of the stoichiometric requirements of cells at various growth states as well as familiarity with mathematical modeling principles to describe kinetics of cell growth, metabolic activity, and product formation. Stoichiometric and kinetic analyses are critical tools for developing efficient processes for the production of high-value cell culture-based products such as therapeutic proteins or vaccines. These analyses are necessary for cell culture scientists to determine the culture environment for optimum cell growth and nutrient utilization without performing numerous cycles of empirical experimentation.

Although the principles are similar, the stoichiometric and kinetic analyses of cell culture processes are somewhat different than for microbial cultures. The typical animal cell volume is a few pL (~1000 times larger than bacteria), and cell sizes exhibit a distribution within the culture that changes with the growth stage (1). The size of cells varies with cell line, medium, growth stage, pH, osmolality, etc. For anchorage-dependent cells, the size is also dependent on cell density. The size distribution can be obtained along with cell number using instruments that automatically image, count, and size viable and nonviable cells (such as the Cedex from Innovatis or the ViCell from Beckman-Coulter). In general, the kinetic parameters to characterize cell properties are calculated based on cell number, rather than on dry cell weight. Expressing the kinetic parameters this way does not take into account the differences in cell size among different cell lines or the same cell line under different culture conditions.

QUANTITATIVE DESCRIPTION OF CELL GROWTH AND PRODUCT FORMATION

Typically a cell culture process is characterized by its cell growth curve, nutrient consumption curves, and product concentration profile. Three classes of quantities contribute to a complete description of the process: concentrations (cell, nutrient, metabolite, or product), kinetic parameters (specific rates and stoichiometric ratios of growth, nutrient consumption, and product formation), and the stage of the culture (lag, exponential, stationary, or declining phase). This section includes how the concentrations and calculated kinetic parameters interact and influence each other at different stages of the culture. A typical batch mode cell growth curve is shown in Fig. 1 on the next page. The different stages of the culture are annotated on the figure.

Stages of Cell Growth

- Lag phase: Newly inoculated cells may experience hour-long or even 1- to 2-d long periods of very slow or no growth. This phase can be caused by inoculation of the culture with cells that were in stationary or decline phase or if the cells are being inoculated into medium with significant changes in factors such as pH, temperature, osmolality, or dissolved oxygen concentration. However, in most cases this lag phase is not experienced by the culture.

Career Avenues

- Exponential phase: This phase is the classic, first-order growth stage for cells. This phase is marked by the greatest and steadiest specific cell growth rate (defined below). It is during this phase that cell-doubling times are typically measured and reported.
- Stationary phase: After the period of rapid cell growth during the exponential phase, cultures exhibit reductions in cell growth rate. This may be induced by exhaustion (or reduction to rate-limiting levels) of key nutrients in the culture and/or an increase in the concentration of metabolites. The latter part of this stage is marked by the flat cell growth curve, indicating a balance between cell growth and death rates.
- Decline or death phase: The cell death rate is higher than the growth rate during the death phase and the viable and total cell concentration declines because of exhaustion of key nutrients or accumulation of inhibitory levels of metabolites or other toxic species.

Figure 1 also shows the integral of viable cell concentration (represented by the shaded area underneath the viable cell curve) over the culture time to that time point, called integral of viable cell density (IVCD), or cumulative cell time. In comparing two cultures of equal specific productivity, the one with the greater IVCD will produce more product. There are many other ways to describe the state of cultures, including physiological state (e.g., a quiescent state vs actively growing state), cell morphology (e.g., well-spread-out state on a flat surface vs a cuboidal shape on a high-curvature surface), clumping, or aggregation (e.g., single cell suspension or spheroid state).

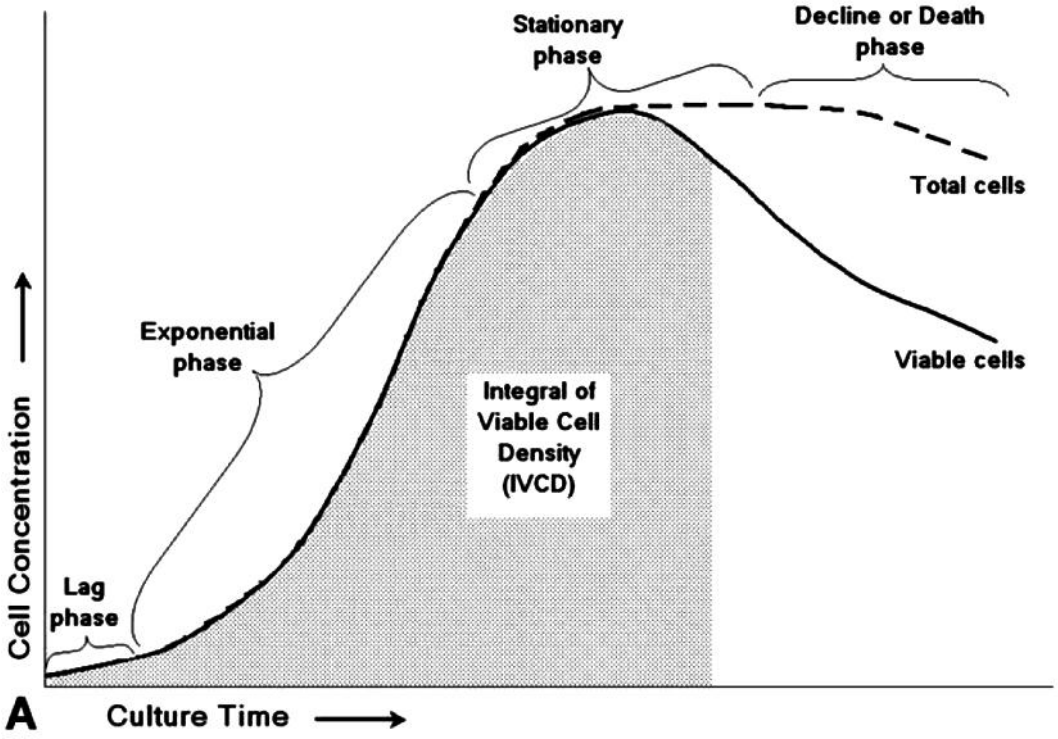
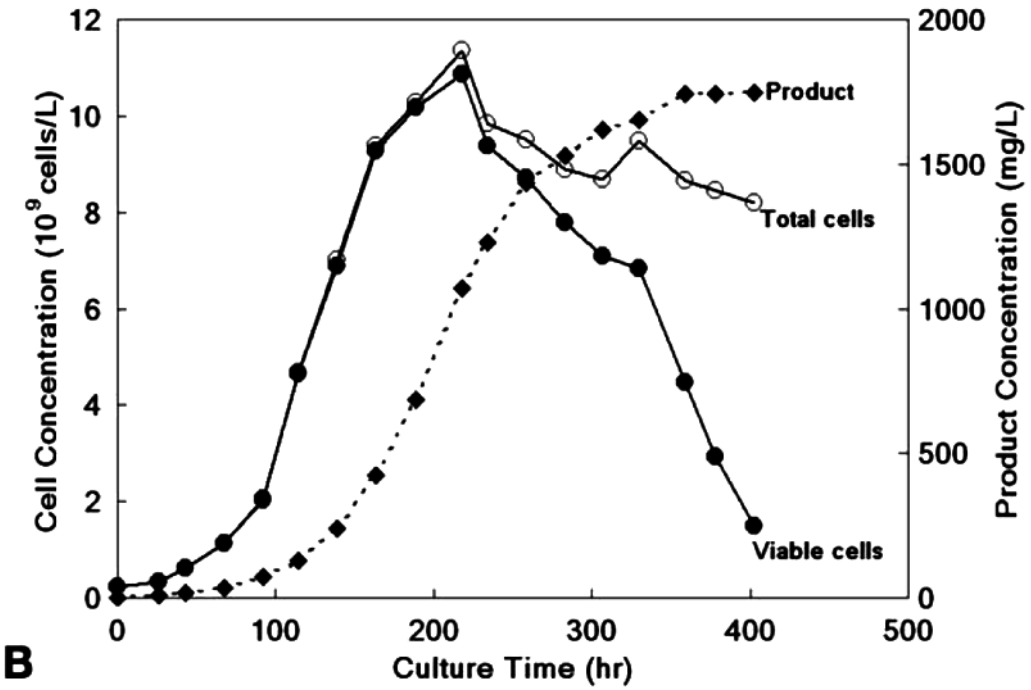


Fig. 1. Typical cell growth and productivity curves. Traces for both viable cells (solid line) and total cells (dashed line) as functions of culture time. Each phase of the culture is shown with braces along the length of the curves.



*Graph of experimental data obtained from a fed-batch culture of Chinese hamster ovary (CHO) cells producing a monoclonal antibody (previously unpublished data presented for illustration of typical growth and production rates). Each culture stage illustrated in **A** is visible in the cell growth curves in **B**.*

BALANCE GROWTH

Growth is an autocatalytic process in that the change of cell concentration over time is dependent on the cell concentration present at the time

$$\frac{dx}{dt} = \mu x$$

where *x* is the concentration of cells in terms dry biomass. For microorganisms which divide by fission the increase in cell number can be described by

$$\frac{dn}{dt} = \mu n$$

where *n* is the cell number per unit volume. In both equations, μ (or μn) is the specific growth rate. It is a measure of how fast cells grow. Basically both equations state that the rate of increase in cell concentration is proportional to its own concentration. The interval between two time points, t1 and t2, that cell number or cell biomass concentration increases by two fold ($n2=2n1$) is called doubling time (t_d).

GROWTH STAGES AND COMPOSITION CHANGES

As cells grow they increase in number, mass and size. Their growth in culture can be divided into stages in response to the concentration of nutrients (Figure 4).

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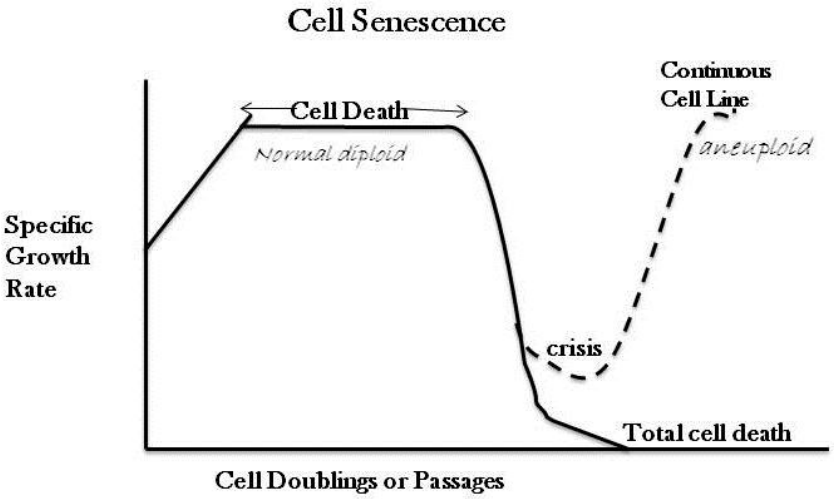
Although cells derived from higher organisms, especially those derived from animals, often also require growth stimulators (mitogens) or even attachment to the surface in order to multiply. After inoculating into a nutrient medium, they may experience a short period of lag phase. During the lag phase they may grow in mass, synthesizing the machinery required for them to double, but not divide until they are ready. Subsequently, they enter a period of rapid growth, called exponential growth phase or log phase. During the log phase they have suffi cient amount of nutrients to grow at the maximum rate achievable in the chemical and physical conditions provided. After the nutrient concentration decreases to a level too low to sustain the maximum growth rate, cell growth slows down. Eventually cell concentration reaches a plateau and the culture enters a stationary phase. Some cells may enter a decline phase in which they lose viability and the cell concentration begins to decrease.

For most microbial cells and many animal cells capable of growing in suspension, this process of growth can be repeated by transferring a small portion of the culture to a new culture medium to let cell growth resume and the cycle of culture stages starts over. Some animal cells do not grow when suspended in liquid phase and require surface adhesion in order to multiply (these are anchorage-dependent cells). These cells are isolated from tissues by enzymatic dissociation. Subsequent to dissociation, they are plated on a compatible surface overlaid with medium for them to attach and grow. They enter an exponential phase similar to microbial cells. The growth stops when they occupied the entire surface area to form a "monolayer". These cells grow to confluence experience contact inhibition. These cells, in addition to nutrients, often also require growth factors or other growth regulators (such as insulin) in order to growth. However, such "**contact inhibition**" of growth at high cell density is independent of nutrient and growth factor concentrations. Even with abundance of nutrient and growth factors, their growth is "contact inhibited". Their growth resumes only after cells are detached from the surface using a proteolytic enzyme and are placed on a larger surface area. Therein they grow to confluence again.

The process can be repeated to allow more cells to be grown. This growth is therefore different traditional microbiological growth. These cells can be kept in contact inhibited or "confluent" state for some time if nutrient medium is continued to be provided. Most of such cells possess two sets of chromosomes, and are diploid cells. However, they cannot be grown in culture indefinitely by simply repeating the "attachment-growth confluence-detachment" cycle, or frequently referred to as "passage" in culture. Normal diploid cells from animals (except stem cells) have limited life span in culture. Fibroblasts (a cell type from connective tissue) isolated from a mouse embryo can be cultured in vitro for about 60 doublings.

As that limit approaches the cells begin to fail to reach confluence and eventually cease growth. This is referred to as having reached "crisis". The observation that normal diploid cells having such a limit of proliferate potential is called "**Hayflick phenomenon**". This phenomenon is common for all normal diploid cells obtained from vertebrates (Figure).

Career Avenues



Normal diploid cells have limited life span in culture, after a number of population doubling, they will lose the capability to grow. Even if they are frozen and kept for years, upon thawing and returning to culture, they "remember" how many more doublings they can go through.

On some occasions a small fraction of cells survive the crisis and began to grow. Those then become "continuous cell line" with "unlimited" growth potential. They also reach confluence, and cease growth until being passaged. As long as the passaging continues, they grow in culture. However, those cells are aneuploid, i.e. their chromosome number is not normal two sets, but varies from cell to cell. Many cancerous cells can also be adapted to grow in culture "indefinitely", but they are also not normal diploid. Most are not contact inhibitors after reading confluence. Rather, they grow as mulitlayers.

Change in a population is caused by growth, death and migration.

$$\frac{dN}{dt} = \frac{d(nV)}{dt} = V(\mu_n - \mu_d + \mu_m)n$$

where μ_n : specific growth rate based on cell number and concentration

μ_d : specific death rate (based on cell number)

μ_m : specific net migration rate.

MONAD MODEL FOR GROWTH KINETICS

The Monod model is commonly used to describe growth rate dependence on the "limiting" substrate.

$$\boldsymbol{\mu} = \frac{\mu_{max} S}{K_s + S}$$

where s is the limiting substrate, concentration, μ_{max} is the maximum specific growth rate under the culture conditions. With the monod model, the growth rate can be largely viewed in two regions of dependence on substrate concentration. At a very high substrate concentration, $s>>K_s$ It is zero order region, the growth rate is μ_{max} and is not affected by substrate concentration. At low substrate concentrations, $K_s>>s$:

$$\boldsymbol{\mu} \approx \frac{\mu_{max} S}{K_s}$$

The growth rate is linearly dependent on the substrate concentration, and is said to be in the first order kinetics region. Basically, such saturation type dependence of growth rate on substrate concentration (i.e. μ approaches a maximum value asymptotically as 's' continues to increase) depicts an almost universal strately of microbial growth in nature: when the nutrient resources are abundant, cells maximize their growth rate to expand the population of their species. When the resources are scarce, they conserve and minimize expansion of biomass. In a batch culture in which μ_d and μ_m terms are considered to be negligible. The growth can be described by the balance equation,

$$\frac{dx}{dt} = \mu x = \frac{\mu_{max} S}{K_s + S} x$$

The substrate utilization is related to the amount required to make a given amount of biomass and can be described as

$$\frac{ds}{dt} = -\frac{\mu x}{Y_{x/s}} = -\frac{\mu_{max} S}{K_s + S} \frac{x}{Y_{x/s}}$$

In the analysis of growth kinetic data, it is useful to calculate the rate of nutrient consumption. It is often referred to the specific rate of substrate utilization.

$$\frac{ds}{dt} = \frac{-\mu x}{Y_{x/s}} = -q_s x$$

So, q_s is defined analogously to the specific growth rate.

$$q_s = \frac{-1}{x} \frac{ds}{dt}$$

$$\mu = \frac{1}{x} \frac{dx}{dt}$$

With this model one assumes the specific substrate utilization rate is proportional to the specific growth rate.

$$q = \frac{\mu}{Y_{x/s}}$$

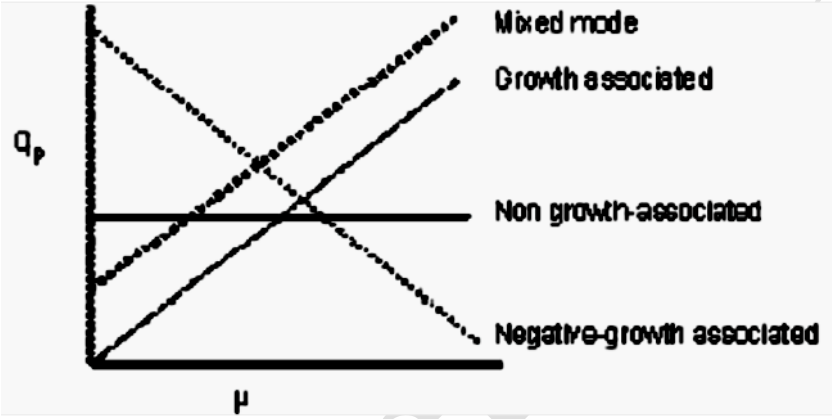
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PRODUCT FORMATION

To describe product formation quantitatively one can define a term analogous to substrate utilization, called specific product formation rate. The change of product concentration in culture is described as

$$\frac{dp}{dt} = q_p x$$

The specific product formation rate, qp, is thus a measure of the activity of cells in producing the product. The productivity may be affected by many factors, including the physical and chemical environment as well as the biological nature of the product and the organism. For many natural products, extensive strain improvement program is employed to obtain one that produce more, or even one that has different production kinetic's. The biosynthesis of many products is affected by the growth of microorganism. Their kinetic behavior is frequently classified according to its relation to growth rate (Figure).



For some product, the productivity is largely proportional to the specific growth rate. For example, ethanol production by yeast provides the energy for cell growth, and is largely growth associated. Some are not affected by growth rate. Yet others are negatively related to the growth rate. A simple equation with a growth dependent term and a growth independent term is used to represent those cases. Obviously, α can be of positive or negative value.

It is also clear that the above expression can only serve as a first approximation of specific product formation rate. The dependence on growth may not be linear or maybe not be as simple as it implies. For most products, the productivity is the result of the interplay of many factors in addition to growth rate. Factors which have strong effect on product formation include the concentration of substrate, precursor, and even product itself or side-product. With the recombinant DNA technology, a preferred mechanism for product formation may be an absolute control of the time profile of its production by manipulating the expression of the gene or genes for product formation using inducible promoter. Upon reaching the desired stage for product formation one can "turn on" the product formation gene(s) for the production will commence. In many batch culture processes, the volume taken up by biomass is rather small. For those processes, the amount of water incorporated into biomass is small. The system is then treated as homogeneous system having a volume V, rather than separated

Career Avenues

into two phases of Vw and Vx for volume of medium and biomass. The material balance equations for system are:

$$\frac{dx}{dt} = \mu x$$

$$\frac{ds_i}{dt} = -\frac{\mu x}{Y_{x/s_i}}$$

$$\frac{dp_i}{dt} = q_{p_i} x$$

This is particularly suitable for animal cell culture processes for which the cell concentration rarely exceeds 1010/L. At such a concentration the biovolume constitutes only about 1~3% of the total volume.

STOICHIOMETRIC ANALYSIS

Culture state can also be characterized by yield on amount of substrate used and the stoichiometric relationships among different nutrients and products (metabolites). The yield coefficient on cells, i.e., cells produced per unit mass of substrate consumed is expressed as

$$Y_{x/s} = \frac{\Delta x}{\Delta s} = \frac{dx}{ds}$$

and mass of protein product produced per mass of nutrients consumed (e.g. total amino acids) gives a yield coefficient for product,

$$Y_{p/s} = \frac{\Delta p}{\Delta s} = \frac{dp}{ds}$$

These yields can be used to measure process efficiency. Stoichiometric ratios are also used to characterize process efficiency cell physiological state. Important stoichiometric ratios for cell culture include the mass of metabolites (e.g., lactic acid or ammonia) produced per mass of substrate (e.g., glucose, glutamine, or oxygen) consumed. Changes in stoichiometric ratios can signal changes in cell physiological state during a culture run or between culture runs. Stoichiometric ratios calculated over the course of a culture are important metabolic indicators because combinations of stoichiometric ratios can reflect the metabolic state of cells. They help determine if the culture is in a favorable metabolic regime. If not, the analyses and mathematical models described in this chapter can help make decisions to drive the culture to a favorable regime. Medium design is an integral part of developing an optimal process for animal cell culture, and stoichiometric analyses play a key role. Key nutrients must be maintained at an optimal range of concentration and at an acceptable ratio to other nutrients that will support their consumption at a desired rate to sustain growth and production. The stoichiometric analyses help determine the demand, i.e., specific uptake rate, and how it varies as a function of time, stage of culture, and concentration as well as interaction between the nutrients. Hence, these analyses can help determine optimal concentration of medium components.

Career Avenues

EFFECT OF SHEAR STRESS

Animal cell culture has been used extensively for the production of a wide variety of substances with biological and therapeutic activity. In spite of being a mature area of development, no universally accepted method of design or scale-up of animal cell culture has been developed. When designing or scaling-up a new process, mixing and aeration are among the most important operations to be considered since they achieve some of the most fundamental objectives carried out in a bioreactor such us keeping certain level of homogeneity in the physicochemical parameters of the culture and providing oxygen to the cells.

Another most recent and exciting advances in modern medicine is the emerging field of regenerative medicine, where live cell-based therapies are used to restore function to ailing tissues and organs. The key challenge still faced for successful commercialization of cell therapies is their production on a large scale. Consequently, the bioprocessing steps involved in the manufacture of cell therapies have to be scalable and capable of producing cellular products of a reproducibly high quality. It is important that the cells maintain their integrity throughout the whole bioprocess and that the resulting cell product has the appropriate functional identity that is required for therapy. In order to achieve this, it is essential to have a detailed understanding of how the bioprocessing environment impacts on the cells at both the laboratory and industrial manufacture scales.

Cell behavior is influenced and affected by the microenvironment in which they reside. In vivo, cells are exposed to a combination of both biochemical and physical cues that regulate their function. Biochemical stimuli such as growth factors and chemokines stimulate a wide range of responses by binding to their appropriate receptors and initiating intracellular signaling pathways. Physical cues can be provided by other cells or matrix components and be passive in nature or the result of applied force.

Bioprocessing of cells for therapy involves ex vivo expansion and differentiation in an artificial microenvironment. Therefore, careful consideration has to be given to the impact that this artificial environment will have on cells that will be transplanted into a patient. It is important that nutrients and growth factors are delivered in suitably formulated growth medium, providing the cells with the appropriate nutritional support. It is also important to monitor and control the physicochemical environment, as subtle changes in temperature, O2, CO2, and pH will all potentially alter cell behaviour. A further consideration that is often overlooked in basic science research is the impact of bioprocess forces on those cells. Bioprocess forces encountered during cell culture include hydrodynamic shear caused by shaking of the flasks to aid their detachment following trypsinisation, forces produced during centrifugation prior to resuspending and shear stresses resulting from transfer through capillaries or by pipetting to resuspend cell pellets

Shear forces on cells in a Suspended culture

Suspension adapted mammalian cells are commonly used in the biopharmaceutical industry for the production of monoclonal antibodies and recombinant proteins. Stirred tank and air-lift bioreactors are extensively used in the culturing of these cells and whilst increasing

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agitation improves mixing and nutrient transport, which is necessary to achieve high purity and yield of therapeutic product, they can also cause harm to the cells. Stirring and sparging of the suspension culture results in the formation of spatial differences in the pattern of liquid flow, leading to cell damage caused by hydrodynamic shear stress. Undesired cell lysis can affect the stability of therapeutic protein products due to the large amounts of proteases that are released from the dying cells, making purification of the target protein difficult.

Many studies have been carried out to investigate the sensitivity of suspension-adapted mammalian cells to shear forces, and there is an overwhelming consensus that shear forces are damaging to these cells. Turbulent shear stress is generally thought to be more damaging to cells than laminar shear stress of the same magnitude. Cell lysis at high shear is caused by the applied shear force exceeding the cell bursting force, whereas the damage caused at low shear stress is mainly due to the papillated state of the cell, something that occurs when the turgor pressure pushes the cytoplasm out through transient pores formed in the cell. At high levels of hydrodynamic shear stress (induced by high intensity agitation of 1500 rpm in a turbine impeller baffled fermenter), the viability of murine hybridoma cells is affected.

In 1992, Born and colleagues proposed a cell-damage model. The model proposed that cell loss is independent of duration of exposure to shear stress as a cell exposed to laminar shear stress is either disrupted or unaffected. However, this model may apply specifically to suspension-adapted cells, because for adherent cells in suspension, increased shear stress exposure time results in an increase in cell damage

The large variation in susceptibility of cells in suspension to shear damage is dependent not just on the specific cell type but also on inherent variation within a single population of cells. For instance, the susceptibility of cells to hydrodynamic damage can vary as they move through the cell cycle, with larger cells in S and G2 phase being more prone to damage than G1 cells, presumably due to their size

Shear forces on Adherent cells in suspension

Bioprocess forces experienced during the isolation of primary cells from tissues for ex vivo expansion have to be considered carefully, as they may introduce a source of poorly characterized stimuli to the cells that can alter their phenotype even before the onset of expansion and/or differentiation. Harvesting of bone marrow requires physical removal from the marrow cavity, introducing low-level shear stress (due to the large diameter of the 11G biopsy needle ~3 mm) during capillary transfer. However, when obtaining adipose tissue as a source of adipose-derived stem cells, procedures often involve mechanically abrasive and high shear stress during suction of often-fibrous adipose tissue. For other tissue types, such as retrieval of dermal fibroblasts for the production of induced pluripotent cells, the forces encountered during retrieval of primary tissue are minimal, as tissue biopsy needles have a large diameter (~6 mm), and hence, the majority of cells do not experience mechanical force. However, when liberating these cells from the collagenous matrix, shear stresses are unavoidably encountered during mechanical digestion of ECM components.

Once isolated, routine passaging of cells subjects them to shear stress caused by pipetting, capillary transfer, and centrifugation. The limited work conducted on adherent cells in suspension to date suggests that cells from different sources have differing capacities to withstand shear stress and numerous physical properties are likely to impact such as cell size,

Career Avenues

density, membrane strength, and flexibility and surface characteristics such as cohesion. Even functionally distinct subsets of a cell population from a single tissue can tolerate shear stress to different degrees

Manual processing can introduce several sources of bioprocess forces (e.g., when detaching trypsinised cells from the culture dish and resuspending cell pellets) that are highly variable between operators and the impact of different forces on cells being prepared for therapy can potentially be of detriment to those cells. Automated approaches may overcome the variability and enable the bioprocess forces experienced by cells to be controlled to a greater degree, but still, these forces will remain.

Shear forces on Adherent cells attached to surfaces

Understanding the impact of flow-induced shear stress on adherent cell responses provides useful information for production of cell-based therapies and tissue-engineered constructs. Seeding cells throughout 3D porous scaffolds using perfusion methods for even distribution exposes those cells to shear stress. Even low levels of shear stress (0.25–0.60 N·m−2) can interfere with attachment of anchorage-dependent mammalian cells to surfaces. Exposing baby hamster kidney cells to shear stress using a flow chamber demonstrated that critical shear stress of 0.75–1.0 Nm−2 applied for 24 h resulted in decreased cell viability, and hence, careful control of perfusion rate is necessary.

Expanding adherent cells in perfusion bioreactors or on microcarriers in stirred bioreactors may potentially be advantageous for producing large quantities of cells in controlled environment, where the critical culture parameters can be monitored and controlled

Calculating Shear stress

Shear stress (**τ**) as a result of simple homogenization is very limited and can be calculated by

$$\tau = FA$$

Where **τ** , is the shear stress and **F (in Newtons)**is the force applied parallel to the surface **A (m^2) .**

Units will be **Nm^{-2}**

Shear stress in a stirred tank

The specific energy dissipation rate in a stirred tank is well known to depend on the shear rate γ and the shear stress τ as follows:

$$P/V = \tau\gamma \qquad (1)$$

Where P is the power input and V is the volume of the fluid in the tank. Furthermore, for Newtonian fluids, the viscosity μ is the ratio of shear stress and shear rate, i.e.

Career Avenues

$$\mu = \tau/\gamma \quad , \qquad (2)$$

Therefore, Eq. (1) can be written as follows:

$$P/V = \tau\gamma = \tau\gamma \times \gamma/\gamma = \mu\gamma^2 \quad , \qquad (3)$$

or

$$\gamma = (1/\mu \times P/V)^{1/2} \qquad (4)$$

Eq. (4) applies to laminar, turbulent and transitional flows.

For non-Newtonian fluids obeying the power law [1], we have:

$$\tau = K\gamma^n \qquad (5)$$

Where K is the consistency index and n is the flow behavior index of the fluid. Because the apparent viscosity μ_a is given as follows [4]:

$$\mu_a = \tau/\gamma = K\gamma^{n-1} \quad , \qquad (6)$$

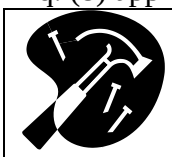
for non-Newtonian media the equation corresponding to Eq. (3) becomes the following:

$$P/V = \mu_a\gamma^2 \qquad (7)$$

and, therefore,

$$\gamma = (1/K \times P/V)^{1/(n+1)} \qquad (8)$$

Eq. (8) applies to both laminar and turbulent flow regimes



GATE QUESTIONS

GATE 2010

Q.4 Determine the correctness or otherwise of the following **Assertion(a) and Reason(r)**.
Assertion: MTT aasay is used to determine cell viability based on the principle of colour formation by DNA fragmentation.
Reason: MTT assay is used to determine cell viability based on colour development by converting tetrazolium soluble salt to insoluble salt.
(A) both (a) and (r) are true and (r) is the correct reason for (a).
(B) Both (a) and (r) are true and (r) is not the correct reason for (a).
(C) (a) is true but (r) is false.
(D) (a) is false but (r) is true.

Answers:
4-A

Career Avenues

IN VITRO FERTILIZATION AND EMBRYO TRANSFER

INTRODUCTION

In vitro fertilization and embryo transfer (IVF-ET) was first successfully used in humans over 25 years ago; since then, more than one million children have been conceived using this technology. IVF is a procedure designed to enhance the likelihood of conception in couples for whom other fertility therapies have been unsuccessful or are not possible. It is a complex process and involves multiple steps resulting in the insemination and fertilization of oocytes (eggs). The embryos created in this process are then placed into the uterus for potential implantation.

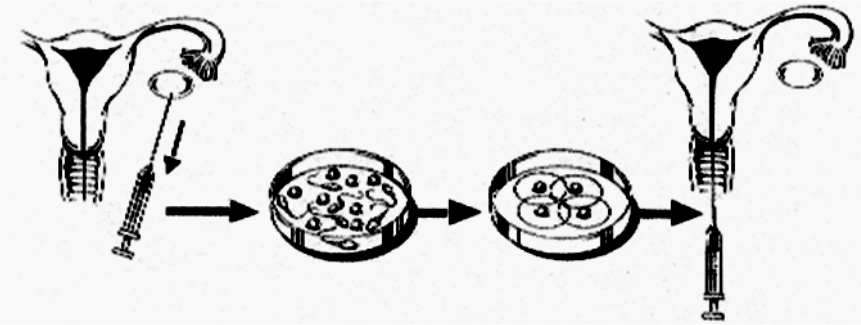
BENEFIT OF THERAPY

- IVF is an elective medical treatment. IVF may provide a couple who has been otherwise unable to conceive with a chance to establish a pregnancy.
- One of the major benefits of IVF is the ability to know before transfer takes place if the male's sperm has actually fertilized the eggs. If fertilization fails to take place (as may happen in some cases of male infertility), changes can be made in the semen processing or fertilization conditions during a future attempt to create an embryo.
- Another benefit of IVF is embryo transfer to the uterus. Because fertilized eggs are placed directly in the uterus, the woman doesn't need to have functioning fallopian tubes.

STEPS IN IN-VITRO FERTILIZATION

- **Retrieving the Oocytes (egg retrieval):** For IVF, collection of eggs is usually performed under transvaginal ultrasound guidance. To accomplish this, a needle is inserted (under IV sedation) through the vaginal wall into the ovaries using ultrasound to locate each follicle. The follicular fluid is drawn up into a test tube to obtain the eggs. Although patients are given pain medications intravenously and are carefully monitored by an anesthesiology staff, some women may experience some discomfort during the procedure. Generally, the oocyte (egg) retrieval takes 20-30 minutes.
- **Collecting and Preparing the Sperm:** A semen sample will be obtained from the partner by masturbation on the day of the oocyte (egg) retrieval. This is usually obtained while the retrieval is being performed. Abstinence from ejaculation for two to five days prior to providing this semen specimen is recommended. After the specimen is produced, the sperm will be prepared for inseminating the collected eggs in the laboratory. Because this can be a stressful time period for men, the man/partner may be unable to produce a specimen when needed. Men who feel that they may have difficulty producing a semen specimen have the opportunity to have their specimens frozen in the laboratory ahead of time for use in this situation. Testicular biopsy can also be performed as a method to extract sperm for IVF.

Career Avenues



- **Insemination of Eggs and Embryo Culture:** Following egg retrieval, the follicular fluid is immediately transferred to the laboratory for identification of eggs, evaluation, and preparation for insemination. In the process of collecting the follicular fluid, it is possible that a large number of eggs may be retrieved. It is usually strongly recommended that all of these eggs be inseminated to maximize the number of embryos available for subsequent transfer. The prepared sperm will be added to each egg and they will be allowed to incubate overnight under controlled laboratory conditions. The next day, each egg is evaluated for evidence of fertilization. However, it is possible that no eggs are fertilized. If this happens, re-insemination of the the eggs or intracytoplasmic sperm injection (ICSI) is performed in hopes of obtaining embryos for transfer. If fertilization still does not occur, the eggs will be discarded and the remainder of the procedure will be cancelled. In the case of severe male factor, the couple may be asked to consider the option of using anonymous donor sperm (obtained through a licensed sperm bank for use as a "backup" or secondary sperm source) if it is not possible to obtain sufficient sperm from the partner at the time of fertilization. The eggs that have fertilized will be allowed to develop for two or more additional days under controlled laboratory conditions before they are placed inside the woman's uterus. Depending upon the couple's wishes, some fertilized eggs/ embryos may be frozen and stored for future use

- **Transferring Embryos to the Uterus:** Embryos are transferred on either day three or day five of development. The embryologists identify "healthy" embryos and in some cases will recommend that a patient extend embryo development to day five, known as the blastocyst stage. Blastocyst transfer has become quite common in IVF cycles as it can increase chances for success while decreasing the likelihood of multiples. Embryologists determine if a day three or day five transfer would be ideal for your cycle. Embryos are transferred to the uterus through a small tube (catheter). This procedure is much like a pap smear and does not require any anesthesia and is usually painless. The embryos are placed in a small amount of fluid inside the catheter, which is passed through the cervix at the time of a speculum examination. The embryos are placed in a manner so they reach the top part of the uterus. The number of embryos transferred depends on individual circumstances of the couple, and this decision will be made collectively by patients, physicians and the embryologist. Typically, two to four embryos are be transferred in one treatment cycle.

Career Avenues

PRODUCT AND SUBSTRATE TRANSPORT

All cells acquire the molecules and ions they need from their surrounding extracellular fluid (ECF). There is an unceasing traffic of molecules and ions in eukaryotic cells, there is also transport in and out of membrane-bounded intracellular compartments such as the nucleus, endoplasmic reticulum, and mitochondria. Examples: proteins, mRNA, Ca^{2+}, ATP, glucose, Na^{+}, Ca^{2+}. Two problems to be considered:

1. Relative concentrations: Molecules and ions move spontaneously down their concentration gradient (i.e., from a region of higher to a region of lower concentration) by diffusion. Molecules and ions can be moved against their concentration gradient, but this process, called active transport, requires the expenditure of energy (usually from ATP).

2. Lipid bilayers are impermeable to most essential molecules and ions. The lipid bilayer is permeable to water molecules and a few other small, uncharged, molecules like oxygen (O_2) and carbon dioxide (CO_2). These diffuse freely in and out of the cell. Lipid bilayers are not permeable to:

- ions such as
    - K^{+}, Na^{+}, Ca^{2+} (called cations because when subjected to an electric field they migrate toward the cathode [the negatively-charged electrode])
    - Cl^{-}, HCO_3^{-} (called anions because they migrate toward the anode [the positively-charged electrode])
- small hydrophilic molecules like glucose
- macromolecules like proteins and RNA.

Transport of these vital substances is carried out by certain classes of intrinsic proteins that form a variety of transport systems: some are open channels, which allow ions to diffuse directly into the cell; others are "facilitators," which, through a little-understood chemical transformation, help solutes diffuse past the lipid screen; yet others are "pumps," which force solutes through the membrane when they are not concentrated enough to diffuse spontaneously. Particles too large to be diffused or pumped are often swallowed or disgorged whole by an opening and closing of the membrane.

Behind this movement of solutes across the cell membrane is the principle of diffusion. According to this principle, a dissolved substance diffuses down a concentration gradient; that is, given no energy from an outside source, it moves from a place where its concentration is high to a place where its concentration is low. Diffusion continues down this gradually decreasing gradient until a state of equilibrium is reached, at which point there is an equal concentration in both places and an equal, random diffusion in both directions.
A solute at high concentration is at high free energy; that is, it is capable of doing more "work" (the work being that of diffusion) than a solute at low concentration. In performing the work of diffusion, the solute loses free energy, so that, when it reaches equilibrium at a lower concentration, it is unable to return spontaneously (under its own energy) to its former high concentration. However, by the addition of energy from an outside source (through the work of an ion pump, for example), the solute may be returned to its former concentration and state of high free energy. This "coupling" of work processes is, in effect, a transferal of free energy from the pump to the solute, which is then able to repeat the work of diffusion.

Career Avenues

For most substances of biological interest, the concentrations inside and outside the cell are different, creating concentration gradients down which the solutes spontaneously diffuse, provided they can permeate the lipid bilayer. Membrane channels and diffusion facilitators bring them through the membrane by passive transport; that is, the changes that the proteins undergo in order to facilitate diffusion are powered by the diffusing solutes themselves. For the healthy functioning of the cell, certain solutes must remain at different concentrations on each side of the membrane; if through diffusion they approach equilibrium, they must be pumped back up their gradients by the process of active transport. Those membrane proteins serving as pumps accomplish this by coupling the energy required for transport to the energy produced by cell metabolism or by the diffusion of other solutes.

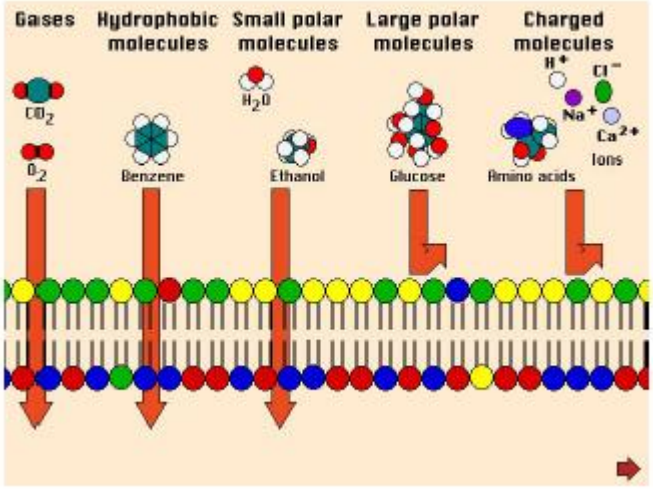
SUBSTRATE TRANSPORT

PASSIVE TRANSPORT

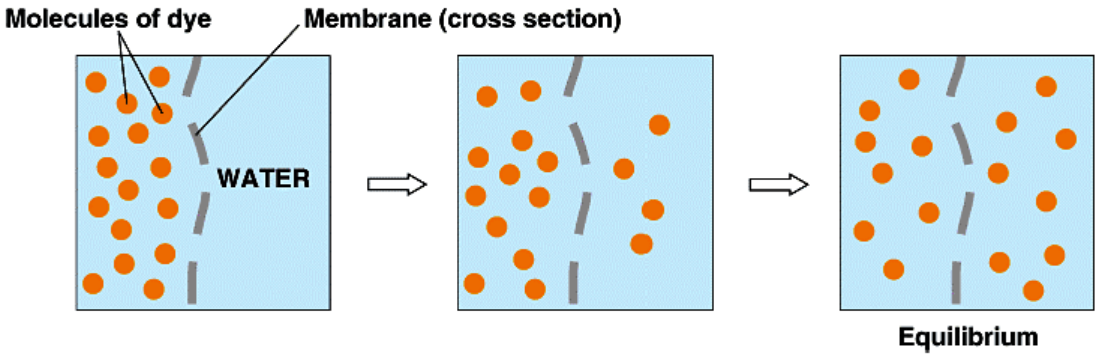
Cell membranes help organisms maintain homeostasis by controlling what substances may enter or leave cells. Some substances can cross the cell membrane without any input of energy by the cell. The movement of such substances across the membrane is known as passive transport. The activities of a cell depend on the materials that enter and leave the cell. To stay alive, a cell must exchange materials such as food, oxygen and waste with its surroundings. These materials must cross the cell membrane. Small molecules like water, oxygen and carbon dioxide can move in and out freely, since they can squeeze between the molecules of the membrane. Large or charged molecules like proteins, sugars and ions cannot. The cell membrane is thus said to be partially permeable.A selectively permeable membrane only allows certain molecules to pass through.

SIMPLE DIFFUSION

The simplest type of passive transport, diffusion does not require the cell to use energy. Only small molecules can cross the cell membrane by simple diffusion. Diffusion is the movement of molecules from an area of high concentration to one of low concentration.
This difference in the concentration of molecules across a space is called the concentration gradient. Diffusion is driven by the kinetic energy of the molecules. Because of their KE, molecules are in constant motion.



Diffusion occurs when molecules move randomly away from each other in a liquid or gas. The rate of diffusion depends on the temperature, size and the type of molecules that are diffusing. Molecules diffuse faster at higher temperatures than at lower temperatures, and smaller molecules diffuse faster than large molecules. Most transport of materials into and out of cells occurs by diffusion. The concentration gradient is the difference between the concentration of a solute in one place and its concentration in an adjacent area. Diffusion always occurs down a concentration gradient, i.e. from the area of higher concentration to the area of lower concentration. When molecules are dispersed evenly, there is no longer any diffusion because there is no longer a concentration gradient.
Diffusion will eventually cause the concentration of molecules to be the same throughout the space the molecules occupy, when they will be in equilibrium.



OSMOSIS

Osmosis is "The process by which water molecules diffuse across a partially permeable membrane from a region of higher water potential to a region of lower water potential." The symbol for water potential is the Greek letter Ψ (psi) or Ψ_w. Water moves by diffusion, like any other molecule, from a region of high concentration to one of low concentration, i.e. down its concentration gradient. Confusion occurs because 'concentration' normally refers to the solute concentration, whereas, in this case, we are referring to the solvent concentration. For this reason, the use of the term 'water potential' is essential; water then simply moves 'down the water potential gradient' – easy! The water potential of pure water is zero (0), so, since a solution must always be less than 100% pure water, all solutions (and cells) have a negative (-) water potential.
The strength of a solution is given by its solute potential (Ψ_s). The stronger the solution, the more negative this value will be. One large and one small molecule will both have the same osmotic effect (i.e. lower Ψ_s by the same amount). So, a single molecule of protein will have about 100 times less effect than the individual amino-acid molecules that make it up. Insoluble molecules do not affect the solute potential (obviously), so have no osmotic effect. Such molecules are used for storage – starch, lipids, and very large proteins (albumin). Water potential is actually a pressure, and is measured in Pascals (Pa), but since these are so small the normal unit is the kilo Pascal (kPa). Typical plant values are -200 to -2000kPa. Since animals have no cell wall, and so no pressure potential, the solute potential and the water potential must be the same:

$$(\Psi_w) = (\Psi_p)$$

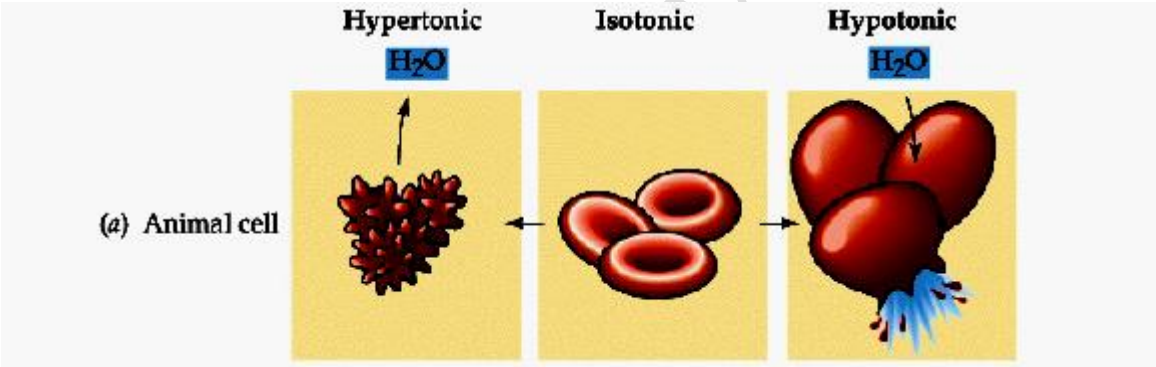
Similar logic shows that the water potential of every cell in an animals' body must be the same, since they are all in contact with blood, and thus in equilibrium ('two values that are equal to a third value, must be equal to each other') When a plant cell is turgid (its normal state), the net movement of water into and out of the cell is zero. Thus:

$$(\Psi_w) = (\Psi_s) + (\Psi_p)$$

The net direction of osmosis depends on the relative concentration of solutes on the two sides of the cell membrane. In a hypertonic solution, the concentration of solutes in the solution is higher and so it has a lower (i.e. more negative) water potential (Ψ_w). Therefore, when placed in a hypertonic solution, water leaves the cell by osmosis, until equilibrium is established. If the cell loses too much water, the cell will shrivel and shrink. Eventually they die, as their metabolism is disrupted i.e. badly wilted plants never recover fully.

Conversely, cells in a hypotonic (or weaker) solution will absorb water by osmosis until equilibrium is reached, since the cell has the lower water potential, and water 'flows downhill'. This flow of water into a cell causes it to swell.

Animal cells placed in a hypotonic solution will swell and often burst because of osmosis. The bursting of cells is called cytolysis. In an isotonic solution, the concentration of solutes on both sides of the membrane is the same and so the net movement of water is zero. This is the normal position inside an animal's body.



PERMEATION

Permeation is the diffusion, through a barrier, of a substance in solution. The rates at which biologically important molecules cross the cell membrane through permeation vary over an enormous range. Proteins and sugar polymers do not permeate at all; in contrast, water and alcohols permeate most membranes in less than a second. This variation, caused by the lipid bilayer, gives the membrane its characteristic permeability. Permeability is measured as the rate at which a particular substance in solution crosses the membrane.

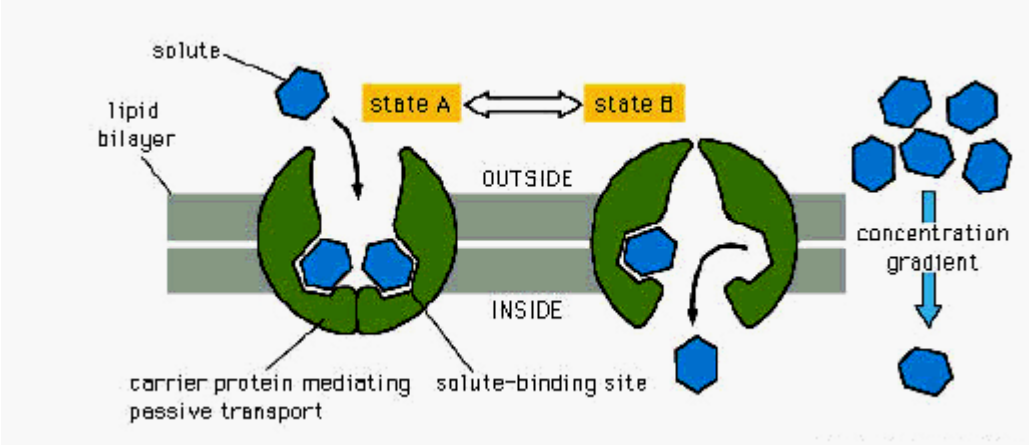
MEMBRANE CHANNELS

Many substances do not actually cross the cell membrane through permeation of the lipid bilayer. Some electrically charged ions, for example, are repelled by organic solvents and therefore cross cell membranes with great difficulty, if at all. In these cases special holes in the membrane, called channels, allow specific ions and small molecules to diffuse directly past the bilayer. A rich variety of channels has been isolated and analyzed from a wide range of cell membranes. Invariably intrinsic proteins, they appear to contain numerous amino acid sequences that traverse the membrane, clearly forming a specific hole, or pore. Certain

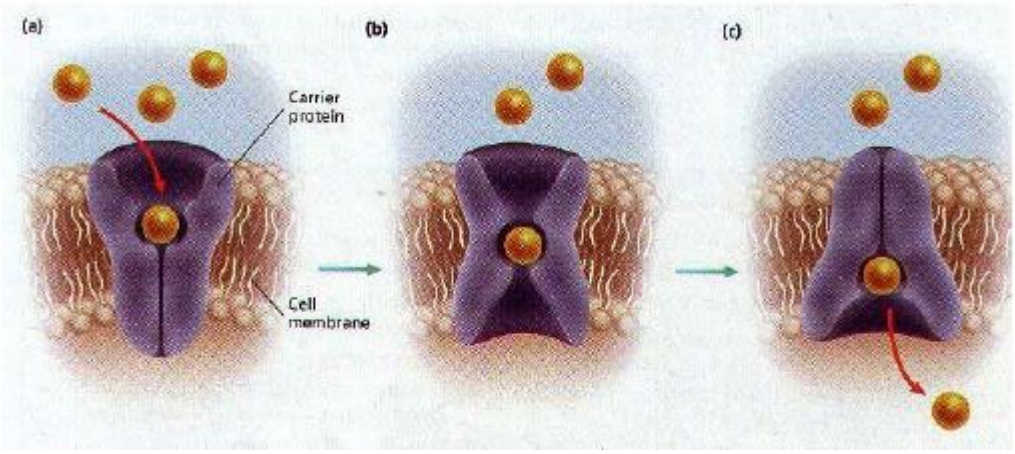
channels open and close spontaneously. Some are gated, or opened, by the chemical action of a signaling substance such as calcium, acetylcholine, or glycine, while others are gated by changes in the electrical potential across the membrane. Channels may possess a narrow specificity, allowing passage of only potassium or sodium, or a broad specificity, allowing passage of all positively charged ions (cations) or of all negatively charged ions (anions). There are channels called gap junctions that allow the passage of molecules between pairs of cells.

FACILITATED DIFFUSION

Passive transport across a membrane requires no energy input from the cell and always goes down the concentration gradient. Simple diffusion and osmosis are examples of passive transport. However, most molecules cannot cross the membrane by simple diffusion; to do so, the molecule must either be very small (H_2O, CO_2) or be soluble in both water and lipid (ethanol). Some molecules are carried across the membrane by carrier proteins which are embedded in the cell membrane.



Carrier proteins often change shape when molecules attach to them, and this change in shape enables the molecule to cross the membrane. Because the carrier protein has to fit around the molecule, it is specific to one molecule, or related class of molecules.This use of carrier proteins to cross the membrane is known as facilitated diffusion, and can be used by those molecules to cross the membrane in either direction – into or out of the cell. Like simple diffusion, facilitated diffusion always goes down the concentration gradient, and therefore continues until equilibrium is reached, for that molecule.



A good example of facilitated diffusion is the transport of glucose into the cell. Once inside the cell, the glucose is immediately turned into glucose phosphate, for which no carrier protein exists. Glucose will thus continue to enter the cell, since equilibrium can never be reached! Facilitated diffusion is therefore another form of passive transport, since it requires no energy input from the cell. Some molecules, mainly ions (e.g. Na^+, K^+) cross the membrane through tunnels made of protein called ion channels. Some ion channels are always open, but others (e.g. in neurones) have 'gates' that open to allow ions to pass or close to stop their passage. Gates open and close in response to conditions in the external environment, or in the cell. It is the opening and closing of the sodium and potassium gates that allows a nerve impulse to be formed and passed along a neurone.

THE GLUCOSE TRANSPORTER

This sugar-specific transport system enables half of the glucose present inside the cell to leave within four seconds at normal body temperature. The glucose transporter is clearly not a simple membrane channel. First, unlike a channel it does not select its permeants by size, as one type of glucose is observed to move through the system a thousand times faster than its identically sized optical isomer. Second, it operates much more slowly than do most channels, moving only 1,000 molecules per second while a channel moves 1,000,000 ions. The most important difference between a membrane channel and the glucose transporter is the conformational change that the transporter undergoes while moving glucose across the membrane. Alternating between two conformations, it moves its glucose-binding site from one side of the membrane to the other. By "flipping" between its two conformational states, the transporter facilitates the diffusion of glucose; that is, it enables glucose to avoid the barrier of the plasma membrane while moving spontaneously down its concentration gradient. When the concentration reaches equilibrium, net movement of glucose ceases.
A facilitated diffusion system for glucose is present in many cell types. Similar systems transporting a wide range of other substrates (*e.g.*, different sugars, amino acids, nucleosides, ions) are also present.

THE ANION TRANSPORTER

The best studied of the facilitated diffusion systems is that which catalyzes the exchange of anions across the red blood cell membrane. The exchange of hydroxyl for bicarbonate ions, each ion simultaneously being moved down its concentration gradient in opposite directions by the same transport molecule, is of great importance in enhancing the blood's capacity to carry carbon dioxide from tissues to the lungs. The exchange molecule for these anions is the major intrinsic protein of red blood cells; 1,000,000 of them are present on each cell, the polypeptide chain of each molecule traversing the membrane at least six times.

PRIMARY ACTIVE TRANSPORT

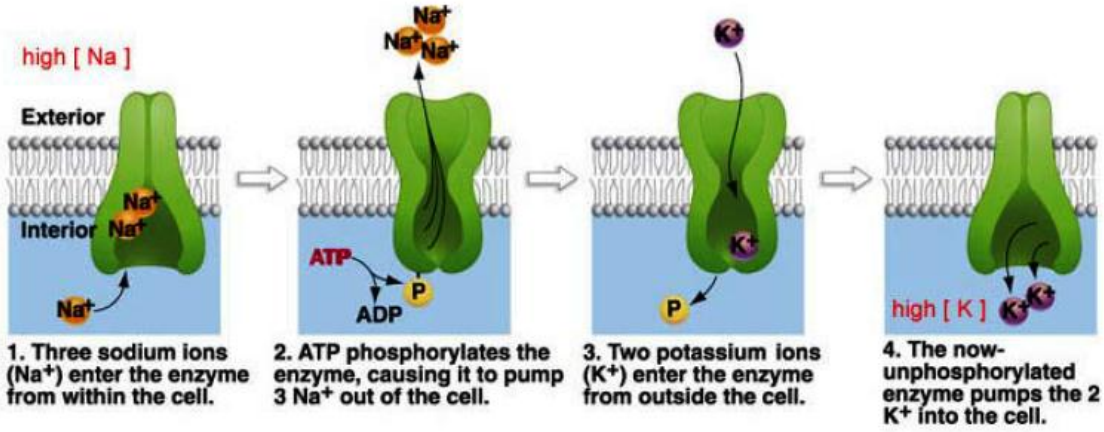
THE SODIUM PUMP

An enzyme called sodium–potassium-activated ATPase has been shown to be the sodium pump, the protein that transports the ions across the cell membrane while splitting ATP. Widely distributed in the animal kingdom and always associated with the plasma membrane, ATPase is found at high concentration in cells that pump large amounts of sodium (*e.g.*, in mammalian kidneys, in salt-secreting glands of marine birds, and in the electric organs of eels). The enzyme, an intrinsic protein, exists in two major conformations whose

Career Avenues

interconversion is driven by the splitting of ATP or by changes in the transmembrane flows of sodium and potassium. When only sodium is present in the cell, the inorganic phosphate split from ATP during hydrolysis is transferred to the enzyme. Release of the chemically bound phosphate from the enzyme is catalyzed by potassium. Thus, the complete action of ATP splitting has been demonstrated to require both sodium (to catalyze the transfer of the phosphate to the enzyme) and potassium (to catalyze the release of the phosphate and free the enzyme for a further cycle of ATP splitting).

Apparently, only after sodium has catalyzed the transferal of the phosphate to the enzyme can it be transported from the cell. Similarly, only after potassium has released the phosphate from the enzyme can it be transported into the cell. This overall reaction, completing the cycle of conformational changes in the enzyme, involves a strict coupling of the splitting of ATP with the pumping of sodium and potassium. It is this coupling that creates primary active transport. The sodium pump extrudes one net positive charge during each cycle of ATP splitting. This flow of current induces an electric potential across the membrane that adds to the potentials brought about by the diffusion of ions through gated channels. The pump's contribution to the overall potential is important in certain specialized nerve cells.



CALCIUM PUMPS

Many animal cells can perform a primary active transport of calcium out of the cell, developing a 10,000-fold gradient of that ion. Calcium-activated ATPases have been isolated and shown to be intrinsic proteins straddling the membrane and undergoing conformational changes similar to those of the sodium–potassium-activated ATPase. When a rise in the concentration of cellular calcium results from the opening of calcium-selective channels, the membrane's calcium pumps restore the low concentration.

HYDROGEN ION PUMPS

Hydrochloric acid is produced in the stomach by the active transport of hydrogen ions from the blood across the stomach lining, or gastric mucosa. Hydrogen concentration gradients of nearly 1,000,000 can be achieved by a hydrogen–potassium-activated ATP-splitting intrinsic protein in the cells lining the stomach. Apart from its specific ion requirements, the properties of this enzyme are remarkably similar to those of the sodium–potassium-activated enzyme and the calcium-activated enzyme. Other hydrogen-pumping, ATP-splitting primary active

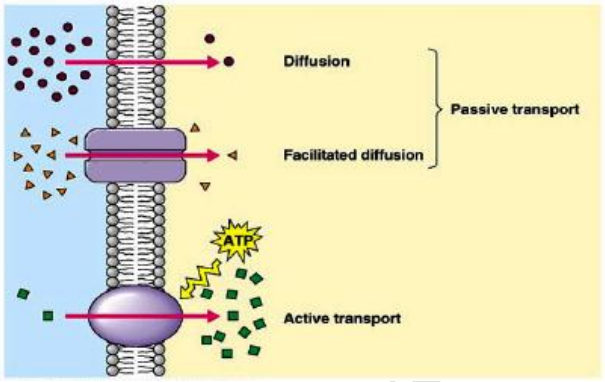
Career Avenues

transporters occur in intracellular organelles, in bacteria, and in plant cells (see below The mitochondrion and the chloroplast). The steep gradient of hydrogen ions represents a store of energy that can be harnessed to the accumulation of nutrients or, in the case of bacterial flagella, to the powering of cell movement.

SECONDARY ACTIVE TRANSPORT

In some cases the problem of forcing a substrate up its concentration gradient is solved by coupling that upward movement to the downward flow of another substrate. In this way the energy-expending diffusion of the driving substrate powers the energy-absorbing movement of the driven substrate from low concentration to high. Because this type of active transport is not powered directly by the energy released in cell metabolism, it is called secondary.

There are two kinds of secondary active transport: counter-transport(also called antiport), in which the two substrates cross the membrane in opposite directions; and co-transport(also called symport), in which they cross in the same direction.



Summary of the different methods by which molecules can enter cells.

BULK TRANSPORT

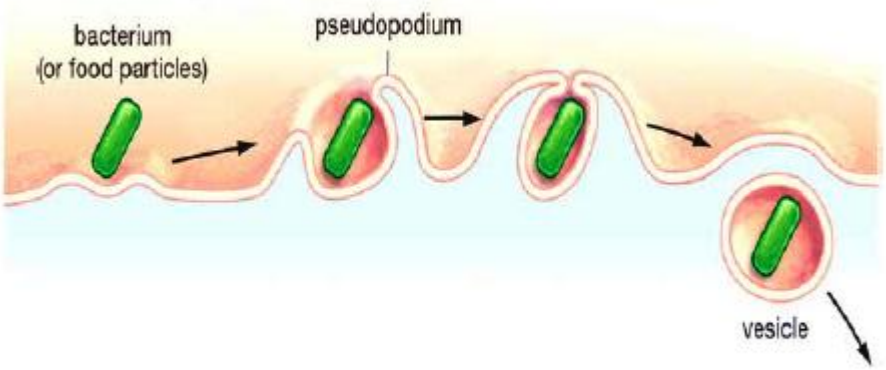
Some molecules, such as large proteins, are too large to cross the cell membrane, and enter by bulk transport. Other examples include food particles and bacteria (phagocytes). These are all packaged in membrane-bound sacs called b and moved across the membrane in this way.
Endocytosis means substances entering the cell; exocytosis means substances leaving the cell. During endocytosis the cell membrane folds into a pouch that encloses the particles, before pinching itself off to form the vesicle. The vesicle often then fuses with a lysosome, which releases its contents to digest the contents of the vesicle. This is thus a form of cell feeding – called phagocytosis. When the particles absorbed are very small (e.g. proteins), it is referred to as pinocytosis, (or 'cell drinking'). White blood cells (WBC's) known as phagocytes, destroy bacteria and other unwanted body cells by phagocytosis, including many millions of old RBC's – broken down in the liver.

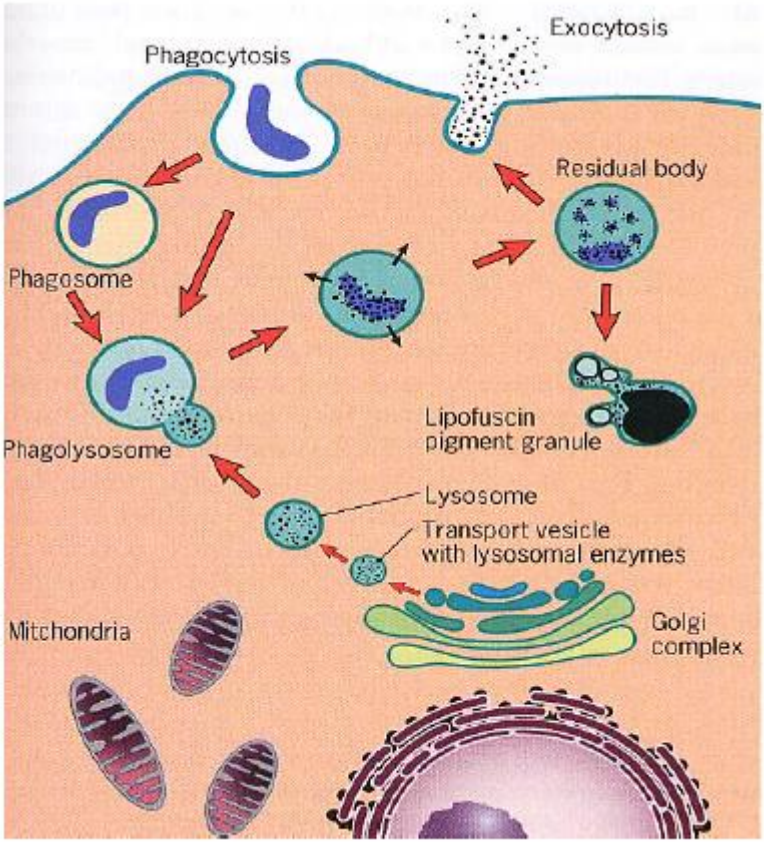
Exocytosis is the opposite of endocytosis and can be used for secretion (e.g. hormones, and enzymes, packaged in the Golgi apparatus) or for excretion of waste products (indigestible parts of food from Amoeba)

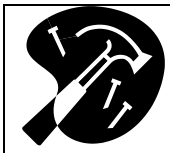
Career Avenues

PRODUCT TRANSPORT

Exocytosis is the process used for transport of products with the use of vesicles.







GATE QUESTIONS

GATE 2013

Q.5 Which one of the following is an ABC transporter?

(A) multidrug resistance protein (B) acetylcholine receptor

(C) bacteriorhodopsin (D) ATP synthase

Answers:

5-A

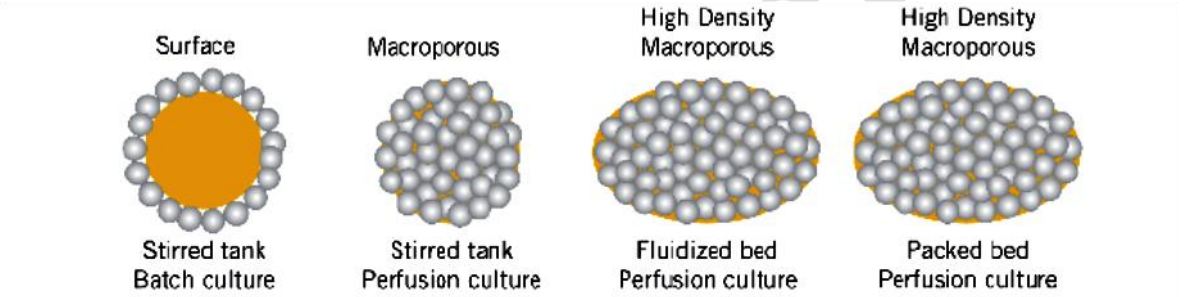
Career Avenues

MICRO & MACRO-CARRIER CULTURE

Cell culture techniques form a vital part of biotechnology, producing the raw material for a vast array of studies and the basic material for the production of many biopharmaceuticals for therapeutics and diagnostics. The use of microcarriers makes the large-scale culture of anchorage-dependent animal cells possible in a way that is efficient, practical, and economical. By using macroporous microcarriers it is possible to immobilize semi-adherent and suspension cell lines, even in protein-free media.

MICRO-CARRIER CULTURE

Microcarrier culture of anchorage-dependent or entrapped cells lowers the volume to cell density ratio and thus belongs to the second of the above alternatives. This technique has many advantages for the commercial manufacturer. It operates in batch or perfusion modes and is well suited to efficient process development and smooth scale-up (Figure). In addition, the reactors can be modified to grow other organisms.



Different microcarriers and applications.

REASONS FOR USING MICROCARRIERS

Cell culture techniques have become vital to the study of animal cell structure, function and differentiation, and for the production of many important biological materials such as vaccines, enzymes, hormones, antibodies, interferons and nucleic acids. Microcarrier culture introduces new possibilities and, for the first time, makes possible the practical high-yield culture of anchorage-dependent cells. In microcarrier culture, cells grow as monolayers on the surface of small spheres or as multilayers in the pores of macroporous structures that are usually suspended in culture medium by gentle stirring. By using microcarriers in simple suspension culture, fluidized or packed bed systems, yields of up to 200 million cells per milliliter are possible.

- **New opportunities and applications for animal cell culture:** Microcarriers provide convenient surfaces for growing animal cells or increasing the yield of cells from standard monolayer culture vessels and perfusion chambers. Applications include production of large quantities of cells, viruses and cell products, studies on differentiation and cell function, perfusion culture systems, microscopy studies, harvesting mitotic cells, isolation of cells, membrane studies, storage and transportation of cells, assays involving cell transfer and studies on uptake of labeled compounds.

Career Avenues

- **Increased production capacity:** The very large culture surface ratio offered by the microcarrier system provides high cell yields without having to resort to bulky equipment and tedious methodology. For a given quantity of cells or their products, microcarrier cultures demand much less space than other types ofmonolayer culture. Culturing cells in small compact systems is especially important when working with pathogenic organisms.

- **Separating cells from secreted product:** Microcarrier technology is one method for cell retention, which means that dilution rate(perfusion rate) becomes independent of the growth rate of the cells. Secondly, because there are fewer cells in the harvest stream, downstream processing becomes simpler.

- **Improved Control:** Microcarrier systems allow excellent control of culture parameters (e.g. pH, gastensions, etc). The technique provides a method for growing anchorage-dependent cells in a system having all the advantages of suspension culture. Improved control allows homogeneous culture systems with a wide variety of process designs. Monitoring and sampling microcarrier cultures is simpler than with any other technique for producing large numbers of anchorage-dependent cells.

- **Protection against physical and chemical stress :** Macroporous microcarriers protect cells against the tip of the stirrer, especially at large scale. Sparging with microtubules using pure oxygen is also possible due to this protection. Cells tolerate more chemical stress such as lactate, ammonia and oxygen if they create a microenvironment inside the pores.

- **Reduced requirements for culture medium:** When compared with other monolayer or suspension techniques, stirred microcarrier cultures yield up to 100-fold as many cells for a given volume of medium. Superior yields have been reported for a wide variety of systems. This reduced requirement for medium means considerable savings in cell culture costs. Macroporous microcarriers especially provide an excellent micro-environment where cells can exchange their own autocrine growth hormones, thereby allowing protein-free media to be used.

- **Reduced requirements for labor:** Because large numbers of cells can be cultured in small volumes, fewer culture vessels are required with microcarrier cultures.The simplified procedures associated with microcarriers reduce the labor needed for routine production. Separating cells from the culture medium is simple; when stirring stops, microcarriers with cells attached settle under the influence of gravity and the supernatant can be removed. Unlike true suspension culture systems, no centrifugation steps are necessary.

- **Lower risk of contamination:** In cell culture, the risk of contamination is related to the number of handling steps(e.g. opening and closing culture vessels) required to produce a given quantity of cells or products. Microcarrier culture reduces the number of handling steps. There is a much-reduced risk of contamination when producing a large quantity of cells in a single microcarrier culture than in several hundred roller botlles.

Career Avenues

The major difference created by microcarrier systems is in the mechanics of handling. Efficient stirring without grinding the beads is essential and can be achieved with a suspended rotating pendulum (Techne) or paddle (Bellco), as for suspension cell culture, rotating at 30 rpm. Technical literature is available from microcarrier suppliers to assist in setting up satisfactory cultures, and a number of protocols for microcarrier systems have been published.

***Analysis*:** Cell counting on beads can be difficult, so the growth rate of the cells should be checked by determination of DNA; protein ,if nonproteinaceous beads are used, or dehydrogenase activity, using the MTT assay on a sample of the beads.

***Variations*:** Most variations on this method arise from the choice of bead or design of the culture vessel and stirrer [Griffiths, 2000]. Bead density varies from 1.03 to 1.05 g/cc and influences the stirring speed, because a higher speed is required for higher-density beads. Composition, or coating, of the beads will also influence attachment, and more fastidious cells may prefer gelatin or collagen beads, which are also soluble when digested by protease activity. Glass beads are the easiest to recycle.

MACROCARRIERS

As well as porous microcarriers, which allow cells to grow within the interstices of the bead, there are also porous carriers that are larger with a macroscopic structure made up of a number of different materials, such as polylactic acid (PLA), polyglycolic acid (PGA), collagen, or gelatin (Gelfoam) in a variety of different geometries .These can be loaded with cells and stirred in a bioreactor or perfused in a fixed-bed or fluidized bed reactor. Fibracel is one such product designed for use in the Celligen bioreactor or a stirred bioreactor.

HYBRIDOMA TECHNOLOGY

Hybridoma technology is a technology of forming hybrid cell lines (called hybridomas) by fusing a specific antibody-producing B cell with a myeloma (B cell cancer) cell that is selected for its ability to grow in tissue culture and for an absence of antibody chain synthesis. The antibodies produced by the hybridoma are all of a single specificity and are therefore monoclonal antibodies (in contrast to polyclonal antibodies). The production of monoclonal antibodies was invented by Cesar Milstein and Georges J. F. Köhler in 1975. They shared the Nobel Prize of 1984 for Medicine and Physiology with Niels Kaj Jerne, who made other contributions to immunology.

METHOD

Laboratory animals (mammals, e.g. mice) are first exposed to an antigen to which we are interested in isolating an antibody against. Usually this is done by a series of injections of the antigen in question, over the course of several weeks. Once splenocytes are isolated from the mammal's spleen, the B cells are fused with immortalized myeloma cells. The myeloma cells are selected beforehand to ensure they are not secreting antibody themselves and that they lack the hypoxanthine-guanine phosphoribosyltransferase (HGPRT) gene, making them sensitive to the HAT medium (see below). The fusion is accomplished using polyethylene glycol or the Sendai virus. It is performed by making the cell membranes more permeable.[clarification needed]

Career Avenues

Fused cells are incubated in HAT medium (hypoxanthine-aminopterin-thymidine medium) for roughly 10 to 14 days. Aminopterin blocks the pathway that allows for nucleotide synthesis. Hence, unfused myeloma cells die, as they cannot produce nucleotides by the de novo or salvage pathways because they lack HGPRT. Removal of the unfused myeloma cells is necessary because they have the potential to outgrow other cells, especially weakly established hybridomas. Unfused B cells die as they have a short life span. In this way, only the B cell-myeloma hybrids survive, since the HGPRT gene coming from the B cells is functional. These cells produce antibodies (a property of B cells) and are immortal (a property of myeloma cells). The incubated medium is then diluted into multi-well plates to such an extent that each well contains only one cell. Since the antibodies in a well are produced by the same B cell, they will be directed towards the same epitope, and are thus monoclonal antibodies.

The next stage is a rapid primary screening process, which identifies and selects only those hybridomas that produce antibodies of appropriate specificity. The hybridoma culture supernatant, secondary enzyme labeled conjugate, and chromogenic substrate, are then incubated, and the formation of a colored product indicates a positive hybridoma. Alternatively, immunocytochemical screening can also be used.

The B cell that produces the desired antibodies can be cloned to produce many identical daughter clones. Supplemental media containing interleukin-6 (such as briclone) are essential for this step. Once a hybridoma colony is established, it will continually grow in culture medium like RPMI-1640 (with antibiotics and fetal bovine serum) and produce antibodies.

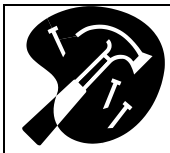
Multiwell plates are used initially to grow the hybridomas, and after selection, are changed to larger tissue culture flasks. This maintains the well-being of the hybridomas and provides enough cells for cryopreservation and supernatant for subsequent investigations. The culture supernatant can yield 1 to 60 µg/ml of monoclonal antibody, which is maintained at -20 °C or lower until required.

By using culture supernatant or a purified immunoglobulin preparation, further analysis of a potential monoclonal antibody producing hybridoma can be made in terms of reactivity, specificity, and cross-reactivity.

APPLICATIONS

The use of monoclonal antibodies is numerous and includes the prevention, diagnosis, and treatment of disease. For example, monoclonal antibodies can distinguish subsets of B cells and T cells, which is helpful in identifying different types of leukaemias.

Career Avenues



GATE QUESTIONS

GATE 2010

Q.6 Hybridoma technology is used to produce

(A) monoclonal antibodies
(B) polyclonal antibodies
(C) both monoclonal and polyclonal antibodies
(D) B cells

GATE 2011

Q.7 HAT(hypoxanthine, aminopterin and thymidine) is used for selecting the hybridomas based on the following

I. Only hybridoma will grow since it inherited the HGPRT genes from B-cells and can synthesize DNA from hypoxantine.

II. Myeloma cells will not grow in cultures since *de novo* synthesis is blocked by aminopterin and due to the lack of HGPRT enzyme.

(A) Only **I** is true
(B) only **II** is true
(C)Both I and II are true
(D) I is true and II is false

Answers:

6-A. 7-C.

Career Avenues

LIVESTOCK IMPROVEMENT

TRANSGENIC ANIMALS

A transgenic animal is an animal whose hereditary DNA has been augmented by addition of DNA from a source other than parental germplasm through recombinant DNA techniques. Transfer of genes or gene constructs allows for the manipulation of individual genes rather than entire genomes. There have been dramatic advances in gene transfer technology in the last two decades since the first successful transfer was carried out in mice in 1980. The technique has now become routine in the mouse and resulting transgenic mice are able to transmit their transgenes to their offspring thereby allowing a large number of transgenic animals to be produced. Successful production of transgenic livestock has been reported for pigs, sheep, rabbits and cattle. The majority of gene transfer studies in livestock have, however, been carried out in the pig. Although transgenic cattle and sheep have been successfully produced, the procedure is still inefficient in these species.

Transgenesis offers considerable opportunity for advances in medicine and agriculture. In livestock, the ability to insert new genes for such economically important characteristics as fecundity, resistance to or tolerance of other environmental stresses would represent a major breakthrough in the breeding of commercially superior stock. Another opportunity that transgenic technology could provide is in the production of medically important proteins such as insulin and clotting factors in the milk of domestic livestock. The genes coding for these proteins have been identified and the human factor IX construct has been successfully introduced into sheep and expression achieved in sheep milk. Moreover, the founder animal has been shown to be able to transmit the trait to its offspring. To date, the majority of genes transferred into sheep have been growth hormone encoding gene constructs. Unfortunately, in most cases the elevated growth hormone levels have resulted into a clinical diabetes situation leading to an early death of the transgenic sheep. Transgenic sheep have recently been generated which express the visna virus envelope gene.

The first reports of the production of transgenic animals created a lot of excitement among biological scientists. In the field of animal breeding, there were diverse opinions on how the technology might affect livestock genetic improvement programmes. Some believed that it would result in total reorganisation of conventional animal breeding theory while others considered the technology as an extension of current animal breeding procedures which, by broadening the gene pool, would make new and novel genotypes available for selection. Application of the technology in animal improvement is still far from being achieved. However, consideration needs to be given to its potential role in this field. Smith et al (1987) presented a comprehensive evaluation of strategies for developing, testing, breeding and disseminating transgenic livestock in the context of quantitative improvement of economic traits.

An important contribution of transgenic technology is in the area of basic research to study the role of genes in the control of physiological processes. The understanding of the molecular control of life processes has important implications for both medicine and agriculture. For example, the generation (through mutation of an endogenous gene) of an organism which lacks a specific gene is a powerful tool to investigate the function of the gene

Career Avenues

product. This type of genetic analysis has been facilitated by the availability of in vitro cultures of embryonic stem cells from mice.

Recent advances in *in vitro* technology (*in vitro* fertilisation and maturation) will increase the number of zygotes available for gene transfer purposes. This, plus the utilisation of embryonic stem cell and primodial germ cell technologies should enhance the efficiency of gene transfer in cattle and sheep considerably.

GENETIC CHARACTERISATION OF ANIMAL GENETIC RESOURCES

Developing countries are endowed with the majority of the global domestic animal diversity - landraces, strains or breeds. Some livestock breeds in these countries are in immediate danger of loss through indiscriminate crossbreeding with exotic breeds. The importance of indigenous livestock breeds lies in their adaptation to local biotic and abiotic stresses and to traditional husbandry systems. However, most of these animal genetic resources are still not characterised and boundaries between distinct populations are unclear. In such cases breeds are defined on the basis of subjective data and information obtained from local communities. Reliance on these criteria as the basis for classification for utilisation and/or conservation may be misleading. Additionally, historical evidence is not always accurate, relying as it often does on subjective judgements. Archival research can reveal much about the original type of a breed or strain but it is molecular genetic evidence which is factual and precise. It is in this sphere that biotechnology has an important role.

Genetic uniqueness of populations is measured by the relative genetic distances of such populations from each other. Polymorphism in gene products such as enzymes, blood group systems and leukocyte antigens which have traditionally been used for measuring genetic distance are being rapidly replaced by polymorphism at the level of DNA, both nuclear and mitochondria as a source of information for the estimation of genetic distances. The first DNA polymorphism to be used widely for genome characterisation and analysis were the restriction fragment length polymorphism (RFLP) which detects variations ranging from gross rearrangements to single base changes. Minisatellites sequences of 60 or so bases repeated many hundreds or thousands of times at one unique locus within the genome have been used to generate DNA fingerprints typical of individuals within species. Microsatellites repeats of simple sequences, the commonest being dinucleotide repeats are abundant in genomes of all higher organisms, including livestock. Polymorphism of microsatellites takes the form of variation in the number of repeats at any given locus and is generally revealed as fragment length variation in the products of polymerase chain reaction (PCR) amplification of genomic DNA using primers flanking the chosen repeat sequence and specific for a given locus. Ease of identification and of sequence determination and need for only small amounts of DNA, are some of the advantages of microsatellites. Additionally, because microsatellite polymorphism can be described numerically, they lend themselves to computerised data handling and analyses. Microsatellites can be used in non-PCR systems in a way similar to minisatellite probes.

Randomly amplified polymorphic DNA (RAPD) has been extensively used for genetic characterisation of a wide range of organisms. The technique uses short (up to 10 bases) primers to amplify nuclear DNA in the PCR. The procedure does not require knowledge of the sequence of DNA under study; primers are designed randomly. The basis of the polymorphism detected by this method is that products are either generated in PCR or not.

Career Avenues

Complete sequencing of the genome is the ultimate form of genetic characterisation. Sequencing has traditionally been expensive and laborious, but with the advent of automated sequencing this is changing rapidly. However, sequencing is unlikely to be used as a technique of choice for genetic characterisation.

Nei and Takezaki (1994) reviewed statistical methods for estimating genetic distances and for constructing phylogenetic trees from DNA sequence data and concluded that different analytical methods may produce different results.

CONSERVATION OF ANIMAL GENETIC RESOURCES

The terms conservation, preservation, ex situ and in situ are used here according to the definition given by FAO (1992). There are several ways, differing in efficiency, technical feasibility and costs, to conserve animal genetic resources. Developing and utilising a genetic resource is considered the most rational conservation strategy. However, there are cases where ex-situ approaches are the only alternatives. *Ex-situ* approaches include: maintenance of small populations in domestic animal zoos; cryopreservation of semen (and ova); cryopreservation of embryos; and some combinations of these. Brem et al (1989) reviewed biotechnologies for *ex-situ* conservation.

Cryopreservation of gametes, embryos or DNA segments can be quite an effective and safe approach for breeds or strains whose populations are too small to be conserved by any other means. The safety of these methods has been demonstrated by background irradiation studies. For example, studies based on irradiation of mouse embryos exposed to the equivalent of hundreds of years of background mutation showed no detectable damage.

Regeneration of offspring following transfer of frozen-thawed embryos has been successful for all major domestic species, except the buffalo. In cattle, the transfer of frozen-thawed embryos is now a commercial practice and embryo survival rate after thawing can be as high as 80% with a pregnancy rate of about 50%. Cryopreservation of oocytes followed by successful fertilisation and live births has been achieved in the mouse. Cryopreserved bovine oocytes have been successfully matured and fertilised in vitro and zygotes developed to blastocyst stage. These trends strongly suggest that long-term cryopreservation of mammalian oocytes is possible.

Respective pregnancy rates of 58 and 50% for fresh and frozen-thawed *in vitro* produced embryos have been reported. Also, calves have been produced from transfer of both split and frozen-thawed *in vitro* produced embryos.

Economic aspects of genetic conservation in farm animals have been assessed. The study concluded that costs of *ex situ* live animal conservation was moderate to high while costs of long-term cryopreservation of gametes were low.

Development in genetic engineering, cryobiology, cell biology and embryology will provide techniques that may enhance our ability to preserve germplasm *in vitro*. Techniques such as transfer of DNA within and between species and the production of viable transgenic animals are far from practical application. However, biotechnology will certainly contribute newer and cheaper methods for preservation such as storage of catalogued DNA. At present, other than live animal and embryo preservation, the other techniques do not allow preservation of

Career Avenues

genomes in a form which can be reactivated *in toto* at a later stage, but they permit the preservation of individual genes or gene combinations for possible future regeneration.

Conservation of indigenous animal genetic resources should be one of the priority livestock development activities for developing countries. The critical importance of these resources to their owners in developing countries need not be emphasised. Their importance to developed countries is also becoming evident as indicated by the increasing importation of tropical germplasm by these countries. It is highly likely that these resources will become of increasing importance to the industrialised countries either as sources of unique genes or when environmental concerns necessitate change in production systems. Developed countries should, thus, assist in the conservation and development of these resources. Technology for cryopreservation of semen and embryo is sufficiently developed to be applied in developing countries. What is missing is financial support to implement conservation programmes. Such support has been provided for world-wide conservation activities for plant germplasm. There is also a strong case for support of animal genetic resources conservation.

A wide range of applications is possible with transgenic animals and can be summarized as follows:

- • Gene Pharming: Refers to the production of recombinant biologically active human proteins in the mammary glands of transgenic animals.
- • Antibody production: Numerous monoclonal antibodies are being produced in the mammary gland of transgenic goats that can be employed both for diagnosis and treatment of diseases.
- • Xenotransplantation: Producing transgenic animals to meet the demand for organs and tissues that need to be transplanted.
- • Blood replacement: Here the whole idea is to produce functional human hemoglobin in transgenic animals.
- • Disease model: The traditional lab experimental animal mouse differs in physiology, anatomy and life span from those of humans, therefore making it an inappropriate model for many human diseases. One the other hand, farm animals, such as pigs, sheep or even cattle, may be more appropriate models to study potential therapies for human diseases which require longer observation periods.
- • Carcass composition: Transgenic animals are being produced, having significant improvements in economically important traits such as growth rate, feed conversion and body fat/muscle ratio.
- • Lactation: Emphasis on transgenic animals with improved physicochemical properties of the milk. Hypoallergenic milk, milk with modified composition of casein are a few examples.
- • Eco-friendly: Environment friendly transgenic animals like creation of transgenic animals can convert certain pollutant materials into nonpolluting products.
- • Wool production: To increase the amount and quality of fleece produced by animals.
- • Disease resistance: To create transgenic animals that can resist diseases better.

Career Avenues

- • Transgenic Sheep Some of the applications of transgenic sheep can be listed as follows:
- o Human growth hormone has been successfully introduced into sheep in order to increase their development growth and meat production. Such transgenic sheep has shown considerable improvement in their body weight, feed efficiency, meat/fat ratio and fat composition. The gene for ovine growth hormone is usually placed under control of metallothionein promoter
- o Transgenic sheep have been produced with the idea of increased wool production and improved wool quality, and this is being one of the major research efforts. Several genes have been introduced and expressed successfully for example: two bacterial genes, viz cysE (serineacetyltransferase) and cysM (O–acetylserinesulphudrolase) and mouse ultra-high-sulfur keratin promoter linked to an ovine insulin-like growth factor 1(IGF1). However, these initial works have not been that promising and much further work is needed to improve wool production and its quality.
- o Human □α1 antitrypsin deficiency results from a genetic defect. It leads to diseases associated with liver andemphysema. The human lungs are constantly challenged by the entry of dust, spores, bacteria etc and these needs to be broken down before they cause any harm. To achieve this, lungs have large numbers of neutrophils and they destroy the foreign particles by releasing anenzyme called 'elastase'. However, this elastase can harm the elastin in the lungs which maintains the elasticity of lungs. To prevent this human body release a protein called □α1 antitrypsin' 'AAT', which now referred to as'□α1 proteinase inhibitor' has been successfully expressed in sheep22, which bind to the elastase and prevent the damage caused to the lungs.
- o Clotting factor IX is an important protein produced naturally in the body. It helps the blood to form clots and stops bleeding. Injections of factor IX are used to treat hemophilia B, a condition which is sometimes called Christmas disease. If one does not have enough factor IX and become injured, the blood will not form clots as it should and may result in damage of muscles and joints.
- • Transgenic Birds Produced by injecting transgene into the germinal disc or Cells from blastoderm cells are removed and are subjected to transfection by lipofection. It is then inserted into the subgerminal space of a recipient whose blastoderm has been irradiated. Off the resulting chickens, some are chimeric and these chimeras are subsequently bred to establish transgenic lines.
- • Transgenic chickens as bioreactors The hen since from long time has been a potential candidate to produce human biopharmaceuticals at low-cost with high-yield. The reason for this is simple as: The yolk and white of the egg are sterile. The technology for fractionating egg yolk and egg white proteins is available and highly automated systems for efficiently producing and collecting thousands of eggs per day are well established. The egg white contains ~4 g of protein, more than half of which comes from the expression of a single gene i.e. ovalbumin gene (OV). Hence, the OV promoter, combined with its other expression elements, could yield significant amounts of protein with a great amount of purity and recovery of the protein is non-invasive. Added to this typically a hen lays >300 eggs per year, therefore a single hen could potentially produce 300 g of raw product annually. Above all they show post-translational modifications that can be compared with that taking place in humans.

Career Avenues

- • Transgenic Pig One of the most desired applications of transgenic technology is to get animals with increased growth rate, improved milk production and composition, improved carcass composition and improved feed utilization.
- APPLICATIONS OF TRANSGENIC SWINE
- a) Modification of milk: Some of the objectives as previously mentioned in producing transgenic swine are –
- i) Produce more amount of milk,
- ii) Produce milk with better nutritional content
- iii) More number of piglets per liter
- iv) Improved feed utilization
- v) Improved body fat
- vi) Improved carcass composition.

Career Avenues



GATE QUESTIONS

GATE 2010

Q.8 In transgenics, alterations in the sequence of nucleotide in genes are due to

P. Substitution
Q. Deletion
R. Insertion
S. Rearrangement

(A) P and Q
(B) P, Q and R
(C) Q and R
(D) R and S

Answers:

8-B.

Career Avenues

CLONING IN ANIMALS

"Clones are biological copies of normal animals."

THE CLONING PROCESS

Early methods of cloning in the 1970s involved a technology called embryo splitting, or blastomere separation. Embryos were split into several cells and then implanted into a surrogate mother for growth and development. But there were a limited number of splits that could be made, and only a few clones could be produced from one egg. The characteristics of the clone were also unpredictable because scientists were cloning from an embryo whose traits could not be predicted.

The practice of cloning took on new meaning in 1996 with the birth of Dolly the sheep, the world's first mammal cloned from an adult cell. Dolly was produced using SCNT technology. Since the cloning of Dolly, this technology has been used to clone cattle, mice, goats, pigs, rabbits, and even a cat. Unlike the embryo splitting method, in theory, SCNT can be used to make an unlimited number of copies of one animal.

The SCNT process starts with an unfertilized egg, or oocyte. Scientists remove the oocyte's nucleus, which contains the egg's genes, or hereditary "instructions." What remains after removal of the nucleus is a cell that contains nutrients essential for embryo development and other cellular machinery waiting for a new set of instructions.

A somatic cell from the animal to be cloned—or in some cases, just the cell's nucleus—is cultured in an incubator and then injected under the coating of the unfertilized oocyte. (Somatic cells are any cells of the body except sperm and eggs). Stimulated by a mild electrical pulse, the oocyte cytoplasm (everything in the cell but the nucleus) and the genetic material from the donated somatic cell combine. If fusion is successful, the resulting fused cell divides just as if it were a fertilized egg and produces an embryo. The embryo is placed in the uterus of a surrogate mother and, if development proceeds normally, an animal clone is born.

The nucleus of the adult cell is specialized, or differentiated, for a particular function. "For example, a muscle cell has a different job from a liver cell, and it has a different set of instructions available to it. The complicated part of cloning that we don't fully understand is how those instructions get reset."

The unlocking and resetting of instructions without making changes to the genetic code is called epigenetic reprogramming. This process allows the cell to develop into a new organism instead of continuing to do its old specified cellular functions. And it's the epigenetic reprogramming that scientists haven't yet mastered and that accounts for frequent cloning failures.

Career Avenues

EXAMPLES OF ANIMAL CLONING

CARP

Chinese embryologist Tong Dizhou successfully inserted the DNA from a male Asian carp into the egg of a female Asian carp to create the first fish clone in 1963. In 1973 Dizhou inserted Asian carp DNA into a European crucian carp to create the first interspecies clone.

CAT

- In December 2001, scientists at Texas A&M University created the first cloned cat, CC (CopyCat). Even though CC is an exact copy of his host, they have different personalities; i.e. CC is shy and timid, his host on the other hand is playful and curious.
- In 2004, the first commercially cloned cat, Little Nicky, was created by Genetic Savings & Clone.

CATTLE

- First World cloned calf (Gene) was born on February 7, 1997 on American Breeders Service facilities in Deforest, Wisconsin. Later it was transferred and kept to Minnesota Zoo Education Center.
- GARIMA- I: world's first buffalo calf through the "Hand guided Cloning Technique" was born on February 6, 2009 at NDRI, Karnal (India).
- GARIMA- II: NDRI, Karnal (India).
- Cloned male buffalo calf Shresth born on August 26, 2010 at National Dairy Research Institute, Karnal, India-132001

DEER

- Dewey was born on May 23, 2003 at Texas A&M University.

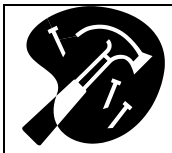
GOAT

- Downen TX 63 684 (nicknamed Megan) was cloned from a top producing Boer goat born on March 29, 2001 at Texas A&M University.

HORSE

- Prometea, female, born May 2003, Italy
- Pieraz, male, born February 2005, Italy
- Paris-Texas, male, born March 2005, USA

Career Avenues



GATE QUESTIONS

GATE 2010

Q.9 Somatic cell gene transfer is used for

P. transgenic animal production

Q. transgenic diploid cell production

R. *in-vitro* fertilization

S. classical breeding of farm animals.

(A)P, R AND S (B) P,Q AND R (C) P ONLY

Answers:

9-A.

Career Avenues

GENETIC ENGINEERING IN ANIMAL CELL CULTURE

VIRAL VECTORS AS GENETIC INFORMATION CARRIERS

The successful transfer of genetic material to target cells still represents a major experimental challenge in all aspects of molecular life sciences, including biotechnology and therapeutic initiatives. A promising delivery method involves the use of engineered animal and human viruses. Such vectors are designed to transfer a desired transgene by exploiting the efficient delivery machinery of the wild-type virus while at the same time minimizing the risk of pathogenicity. Different systems, based on animal as well as human viruses, have recently been developed and, in some cases, advanced to clinical application. Worldwide more than 400 clinical trials using viral vectors have been, or are currently being, conducted. The majority of these trials have been performed using retroviral vectors, followed by adenoviral and other viral vector systems, among them adeno-associated virus (AAV) vectors. Because of the high transfer efficiency and reasonably straightforward production protocols, viral vectors have also become increasingly popular for routine applications in various fields of life sciences. The majority of viral vector systems have been designed to retain, and even extend, the viruses' cell tropism while ensuring high biosafety characteristics.

For some viruses the ability to transduce many cell types is intrinsically present (e.g., adenovirus or AAV). Other viruses have a far narrower host range (e.g., lentiviruses, which are mainly restricted to CD4-expressing cells). Such viruses must be altered through pseudotyping with envelope or capsid proteins from other compatible virus species, thereby enabling the recombinant vector to transduce a broader variety of cell types.

To ensure biosafety, most vectors are rendered replication incompetent through selective deletion of wild-type viral genetic information, thereby preventing productive viral life cycles within the target cell. In addition to the removal of nonessential sequences from the wild-type genome, a fundamental principle underlying the production of recombinant viral particles is the redistribution of indispensable elements to either helper plasmids or producer cell lines. The resulting free space flanked by the viral inverted/long terminal repeats (ITRs/LTRs) is then employed to accommodate the heterologous DNA to be delivered. For a number of vector types only one viral open reading frame (ORF) is replaced and delivered *in trans* (e.g., first-generation adenovirus), whereas for other types the entire viral coding sequence is replaced by foreign DNA with only minimal viral elements retained (e.g., adeno-associated virus, "gutless" adenovirus).

The architecture of a viral vector has significant impact on the safety of the transduction system .Typically, the more the viral genome is segregated and lacks overlapping sequences, the smaller is the probability for recombination-based generation of replication-competent revertants during the production process. Because wild-type virus is on the whole highly optimized and adapted to its environment, altering its organization and expression patterns, as is done for the construction of viral vectors, is likely to decrease production efficiency. This often manifests in lower titers, poor encapsidation of viral genomes, or changes in transduction efficiency of recombinant viral particles. It is therefore crucial to find an optimal

Career Avenues

balance between high-safety profiles while retaining high efficiency and quality of viral particle production.

GENE THERAPY

The term gene therapy applies to approaches to disease treatment based upon the transfer of genetic material (DNA, or possibly RNA) into an individual. The concept of gene therapy has been inspired by major discoveries made in basic genetic research since the 1950s, and strategies for gene delivery were matured through the 1980s leading to the first clinical trial in 1990. By the beginning of January 2004 more than 3500 patients had received a gene therapy treatment. Today's gene therapy research may be seen as pursuing intelligent drug design through a logical extension of results of fundamental biomedical research on the molecular basis of disease. To allow transfer and proper function in a patient the therapeutic gene must be built into a vector. The most efficient gene delivery systems currently available are based upon the gene transfer machinery used in nature by animal viruses of which retroviral and adenoviral vectors are the most widely used. They will be treated in more detail in this chapter. However, in principle any virus can be modified in a way to eliminate the pathogenic/toxic genes and to maintain its capacity to transfer the gene of interest into the target cell. Since LV and AAV vectors provide several advantages in comparison to retroviral and adenoviral vectors and have thus a rather bright future in the gene therapy field, they will also be treated in more detail here. Other gene transfer systems use naked DNA or DNA coupled to chemicals that may facilitate various steps of entry of DNA into cells. Although of theoretical interest, mainly as a result of the improved safety profile in comparison to viral vectors, they are not really used today because of their inefficiency to transfer genes into the cellular genome; however, the ongoing scientific work will improve this situation in the future. Gene and cell therapy are very much related because many gene therapy approaches are ex vivo approaches. By the ex vivo gene transfer principle, genetic material is transferred to cells outside the host; following transfer the genetically modified cells are then implanted into the host (= cell therapy). By the in vivo gene transfer principle, genetic material is transferred directly to cells located within the host and no ex vivo cell cultures are necessary for cell amplification and transduction.

Career Avenues

ANIMAL CELL PRESERVATION

PRINCIPLES OF CRYOPRESERVATION

THEORETICAL BACKGROUND TO CELL FREEZING

Optimal freezing of cells for maximum viable recovery on thawing depends on minimizing intracellular ice crystal formation and reducing cryogenic damage from foci of high concentration solutes formed when intracellular water freezes.

This is achieved (a) by freezing slowly to allow water to leave the cell but not so slowly that ice crystal growth is encouraged, (b) by using a hydrophilic cryoprotectant to sequester water, (c) by storing the cells at the lowest possible temperature to minimize the effects of high salt concentrations on protein denaturation in micelles within the ice, and (d) by thawing rapidly to minimize ice crystal growth and generation of solute gradients formed as the residual intracellular ice melts.

CELL CONCENTRATION

Cells appear to survive freezing best when frozen at a high cell concentration. This is largely an empirical observation but probably is related partly to the reduced viability on thawing requiring a higher seeding concentration and partly to improved survival at a high cell concentration if cells are leaky because of cryogenic damage. A high concentration at freezing also allows sufficient dilution of the cryoprotectant at reseeding after thawing so that centrifugation is unnecessary (at least for most cells). The number of cells frozen should be sufficient to allow for 1:10 or 1:20 dilution on thawing to dilute out the cryoprotectant but still keep the cell concentration higher than at normal passage; for example, for cells sub-cultured normally at 1×10^5/mL, 1×10^7 should be frozen in 1 mL of medium, and, after thawing the cells, the whole 1 mL should be diluted to 20 mL of medium, giving 5×10^5 cells/mL (five times the normal seeding concentration). This dilutes the cryoprotectant from 10% to 0.5%, at which concentration it is less likely to be toxic. Residual cryoprotectant may be diluted out as soon as the cells start to grow (for suspension cultures) or the medium changed as soon as the cells have attached (for monolayers).

FREEZING MEDIUM

The cell suspension is frozen in the presence of a cryoprotectant such as glycerol or dimethyl sulfoxide (DMSO) [Lovelock & Bishop, 1959]. Of these two, DMSO appears to be the more effective, possibly because it penetrates the cell better than glycerol. Concentrations of between 5% and 15% have been used, but 7.5% or 10% is more usual.

There are situations in which DMSO may be toxic or induce cells to differentiate (*see* Section 17.7.2), e.g., with hematopoietic cell lines such as L5178Y or HL60 (M. Freshney, personal communication), and in these cases it is preferable either to use glycerol or to centrifuge the cells after thawing to remove the cryoprotectant. It has been claimed that cells should be kept at 4◦C after DMSO is added to the freezing medium and before freezing, but preliminary experiments by the author suggested that this did not improve survival after freezing and may

Career Avenues

even have reduced it [unpublished observations], perhaps by inhibiting intracellular penetration. However, this issue is still not fully resolved and is worth further DMSO should be colorless, and it needs to be stored in glass or polypropylene, as it is a powerful solvent and will leach impurities out of rubber and some plastics. Glycerol should be not more than one year old, as it may become toxic after prolonged storage.

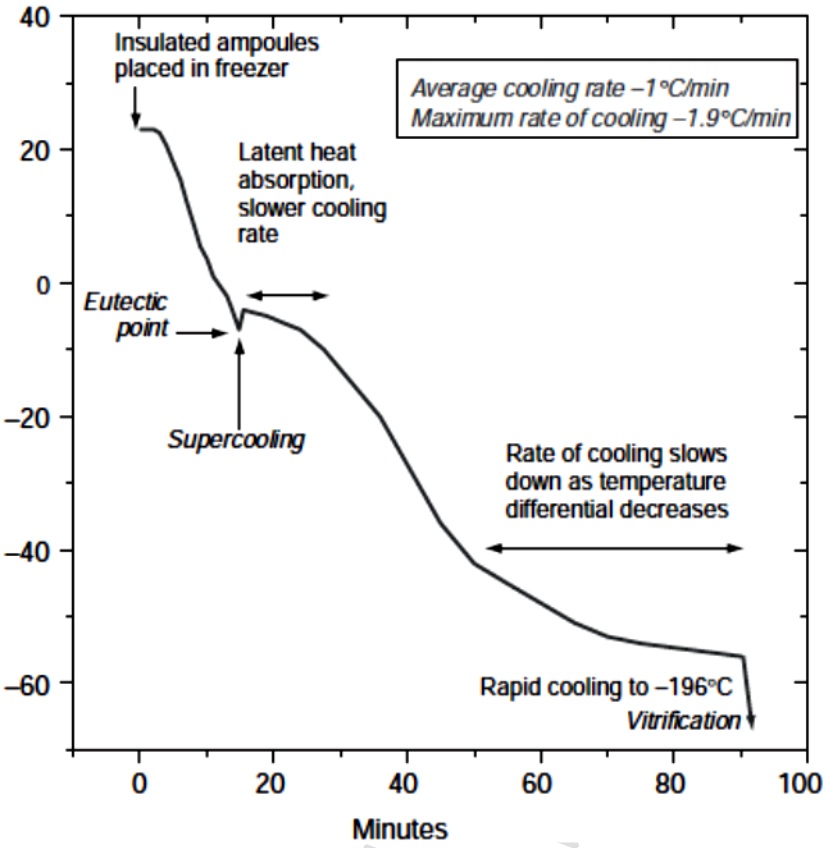
Other cryoprotectants have been suggested, such as polyvinylpyrrolidone (PVP) [Suzuki et al., 1995], polyethylene glycol (PEG) [Monroy et al., 1997], and hydroxyethyl starch (HES) [Pasch et al., 2000], but none has had the general acceptance of either DMSO or glycerol, although there may be some improvement with trehalose [Eroglu et al., 2000; Buchanan et al., 2004]. Many laboratories also increase the serum concentration in freezing medium to 40%, 50%, or even 100%.

COOLING RATE

Most cultured cells survive best if they are cooled at 1°C/min. This is probably a compromise between fast freezing minimizing ice crystal growth and slow cooling encouraging the extracellular migration of water. The shape of the cooling curve is governed by (a) the ambient temperature, (b) any insulation surrounding the cells, including the ampoule. (c) The specific heat and volume of the ampoule contents, and (d) the latent heat absorption during freezing. When cells are frozen in an insulated container placed in an ultradeep freeze, as in (1)–(4) below, this results in a curve with a rapid cooling rate at the start, when the temperature differential is greatest, down to a minimum at the eutectic point, a slight rise as freezing commences, followed by a plateau as the latent heat of freezing is absorbed, and then a more rapid fall as freezing is completed, gradually slowing down as the temperature of the freezing chamber is reached When cells are transferred to the liquid nitrogen freezer (or the end stage is reached in a programmable freezer) the temperature drops rapidly to between −180 and −196◦C. A controlled cooling rate can be achieved in several ways:

- Lay the ampoules on cotton wool in a polystyrene foam box with a wall thickness of ∼15 mm. This box, plus the cotton wool, should provide sufficient insulation such that the ampoules will cool at 1◦C/min when the box is placed at −70◦C or −90◦C in a regular deep freeze or insulated container with solid CO2.
- Insert the canes in tubular foam pipe insulation, with a wall thickness of ∼15 mm, and place the insulation at −70◦C or −90◦C in a regular deep freeze or insulated container with solid CO2.
- Place the ampoules in a Taylor Wharton freezer neck plug (Fig. 20.3) and insert the plug into the neck of the nitrogen freezer.
- Place the ampoules in a Nalge Nunc freezing container and place at −70◦C or −80◦C.
- Use a controlled-rate freezer programmed to freeze at 1◦C/min, with accelerated freezing through the eutectic point.

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FREEZING CURVE

Record of the fall in temperature in an ampoule containing medium, clipped with five other ampoules on an aluminum cane, enclosed in a cardboard tube, placed within a polyurea-foam tube and placed in a freezer at −70◦C.

ENERGY METABOLISM

Most culture media contain 4–20 mMglucose, which is used mainly as a carbon source for glycolysis, generating lactic acid as an end product. Under normal culture conditions (atmospheric oxygen and a submerged culture), oxygen is in relatively short supply. In the absence of an appropriate carrier, such as hemoglobin, raising the O_2 tension will generate free radical species that are toxic to the cell, so O_2 is usually maintained at atmospheric levels. This results in anaerobic conditions and the use of glycolysis for energy metabolism. However, the citric acid cycle remains active, and it has become apparent that amino acids—particularly glutamine—can be utilized as a carbon source by oxidation to glutamate by glutaminase and entry into the citric acid cycle by transamination to 2-oxoglutarate. Deamination of the glutamine tends to produce ammonia, which is toxic and can limit cell growth, but the use of dipeptides, such as glutamyl-alanine or glutamyl-glycine, appears to minimize the production of ammonia and has the additional advantage of being more stable in the medium (e.g., Glutamax, Invitrogen).

Career Avenues

INITIATION OF THE CULTURE

Primary culture techniques are described in detail later. Briefly, a culture is derived either by the outgrowth of migrating cells from a fragment of tissue or by enzymatic or mechanical dispersal of the tissue. Regardless of the method employed, primary culture is the first in a series of selective processes that may ultimately give rise to a relatively uniform cell line. In primary explantation, selection occurs by virtue of the cells' capacity to migrate from the explant, whereas with dispersed cells, only those cells that both survive the disaggregation technique and adhere to the substrate or survive in suspension will form the basis of a primary culture. If the primary culture is maintained for more than a few hours, a further selection step will occur. Cells that are capable of proliferation will increase, some cell types will survive but not increase, and yet others will be unable to survive under the particular conditions of the culture. Hence, the relative proportion of each cell type will change and will continue to do so until, in the case of monolayer cultures, all the available culture substrate is occupied. It should be realized that primary cultures, although suitable for some studies such as cytogenetic analysis, may be unsuitable for other studies because of their instability. Both cell population changes and adaptive modifications within the cells are occurring continuously throughout the culture, making it difficult to select a period when the culture may be regarded as homogeneous or stable.

After *confluence* is reached (i.e., all the available growth area is utilized and the cells make close contact with one another), cells whose growth is sensitive to contact inhibition and density limitation of cell proliferation will stop dividing, while any transformed cells, which are insensitive to density limitation, will tend to overgrow.

Keeping the cell density low (e.g., by frequent subculture) helps to preserve the normal phenotype in cultures such as mouse fibroblasts, in which spontaneous transformants tend to overgrow at high cell densities. Some aspects of specialized function are expressed more strongly in primary culture, particularly when the culture becomes confluent. At this stage, the culture will show its closest morphological resemblance to the parent tissue and retain some diversity in cell type.

EVOLUTION OF CELL LINES

After the first subculture, or passage, the primary culture becomes known as a *cell line* and may be propagated and subcultured several times. With each successive subculture, the component of the population with the ability to proliferate most rapidly will gradually predominate, and nonproliferating or slowly proliferating cells will be diluted out. This is most strikingly apparent after the first subculture, in which differences in proliferative capacity are compounded with varying abilities to withstand the trauma of trypsinization and transfer.

Although some selection and phenotypic drift will continue, by the third passage the culture becomes more stable and is typified by a rather hardy, rapidly proliferating cell. In the presence of serum and without specific selection conditions, mesenchymal cells derived from connective tissue fibroblasts or vascular elements frequently overgrow the culture. Although this has given rise to some very useful cell lines (e.g., WI-38 human embryonic lung fibroblasts, BHK21 baby hamster kidney fibroblasts, COS cells, CHO cells, and perhaps the most famous of all, the L-cell, a mouse subcutaneous fibroblast treated with methylcholanthrene. This overgrowth represents one of the major challenges of tissue culture

Career Avenues

since its inception — namely, how to prevent the overgrowth of the more fragile or slower-growing specialized cells such as hepatic parenchyma or epidermal keratinocytes. Inadequacy of the culture conditions is largely to blame for this problem, and considerable progress has now been made in the use of selective media and substrates for the maintenance of many specialized cell lines.

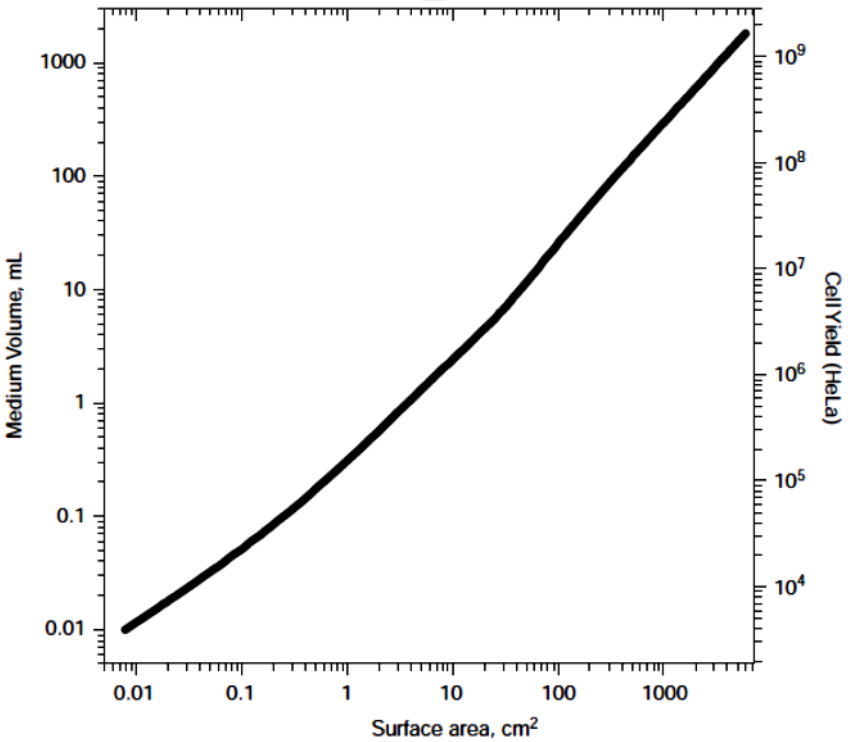
SENESCENCE

Normal cells can divide a limited number of times; hence, cell lines derived from normal tissue will die out after a fixed number of population doublings. This is a genetically determined event involving several different genes and is known as *senescence*. It is thought to be determined, in part, by the inability of terminal sequences of the DNA in the telomeres to replicate at each cell division. The result is a progressive shortening of the telomeres until, finally, the cell is unable to divide further. Exceptions to this rule are germ cells, stem cells, and transformed cells, which often express the enzyme telomerase, which is capable of replicating the terminal sequences of DNA in the telomere and extending the life span of the cells, infinitely in the case of germ cells and some tumor cells.

KINETICS OF GROWTH

CELL YIELD

For monolayer cultures, the cell yield is proportional to the available surface area of the flask (Fig. 8.1).



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CELL YIELD AND SURFACE AREA

Relationship of volume of medium and cell yield, to the surface area of a culture vessel.

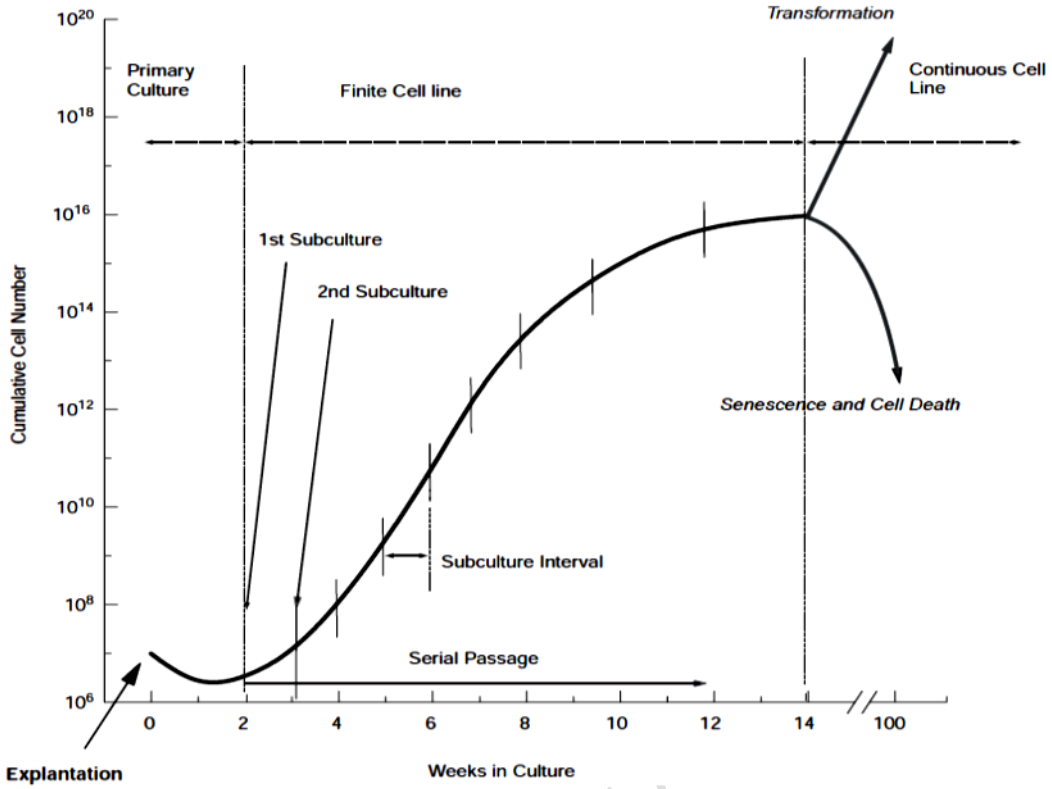
The graph is plotted on the basis of the volume of the medium for each size of vessel and is nonlinear, as smaller vessels tend to be used with proportionally more medium than is used with larger vessels. The cell yield is based on the volume of the medium and is approximate.

Small volumes and multiple replicates are best performed in multiwell plates, which can have a large number of small wells (e.g., microtitration plates with 96 or 144 wells, 0.1–0.2 mL of medium, and 0.25-cm^2 growth area or 24-well ‘‘cluster dishes’’ with 1–2 mL medium in each well, 1.75-cm2 growth area) up to 4-well plates with each well 5 cm in diameter and using 5 mL of culture medium. The middle of the size range embraces both Petri dishes and flasks ranging from 10 cm2 to 225. Flasks are usually designated by their surface area (e.g., 25 cm2 or 175 cm2, and sometimes T25 or T175, respectively), whereas Petri dishes are referred to by diameter (e.g., 3.5 cm or 9 cm). Glass bottles are more variable than plastic because they are usually drawn from standard pharmaceutical supplies. Glass bottles should have (1) one reasonably flat surface, (2) a deep screw cap with a good seal and nontoxic liner, and (3) shallow-sloping shoulders to facilitate harvesting of monolayer cells after trypsinization and to improve the efficiency of washing.

If you require large cell yields (e.g.∼1 × 109 HeLa cervical carcinoma cells or 2 × 108 MCR-5 diploid human fibroblasts), then increasing the size and number of conventional bottles becomes cumbersome, and special vessels are required.

Flasks with corrugated surfaces or multilayered flasks offer an intermediate step in increasing the surface area. Cell yields beyond that require multisurface propagators or roller bottles on special racks. Increasing the yield of cells growing in suspension requires only that the volume of the medium be increased, as long as cells in deep culture are kept agitated and sparged with 5% CO_2 in air.

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EVOLUTION OF A CELL LINE

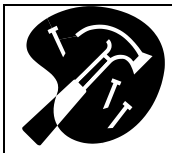
The vertical (*Y*) axis represents total cell growth (assuming no reduction at passage) for a hypothetical cell culture. Total cell number (cell yield) is represented on this axis on a log scale, and the time in culture is shown on the *X*-axis on a linear scale. Although a continuous cell line is depicted as arising at 14 weeks, with different cells it could arise at any time. Likewise, senescence may occur at any time, but for human diploid fibroblasts it is most likely to occur between 30 and 60 cell doublings, or 10 to 20 weeks, depending on the doubling time.

Career Avenues

Cell Culture Based Vaccines

Table 1 Some viral vaccines currently available for human and veterinary use.

Human vaccines		Veterinary vaccines	
Virus	Cell used for culture	Virus	Cell used for culture
Measles	Chick embryo fibroblasts	Canine distemper	Chick embryo Fibroblasts or dog
Polio	Monkey kidney	Canine hepatitis	Dog
Polio (live)	Monkey kidney or human diploid cells	Foot and mouth disease	Bovine kidney cells
Rabies	Human diploid cells	Rabies	Duck embryo or
Rubella	Rabbit kidney, duck embryo or human diploid cells	Feline panleucopenia	Cat kidney
		Marek's disease	Chick embryo cells



GATE QUESTIONS

GATE 2012

Q.10 Dimethyl sulfoxide (DMSO) is used as a cryopreservant for mammalian cell cultures because

(A) it is an organic solvent
(B) it easily penetrates cells
(C) it protects cells by preventing crystallization of water
(D) it is also utilized as a nutrient

Q.11 Heat inactivation of serum is done to inactivate

(A) Prions
(B) mycoplasma
(C) complement
(D) pathogenic bacteria

Q. 12 Match the entries in **Group I** with the methods of sterilization in **Group II.**

Group I	**Group II**
P. Serum	1. Autoclave
Q. Luria Broth	2. Membrane filtration
R. polypropylene tubes	3. UV irradiation
S. Biological safety cabinets	4. Gamma irradiation
	5. Dry heat

(A) P-5,Q-3,R-1,S-4
(B) P-1,Q-4,R-5,S-3
(C) P-2,Q-1,R-4,S-3
(D) P-4,Q-1,R-3,S-5

Answers:

10-C 11-C 12-C

Career Avenues